A Lab for Latent Actors: Hiding Segments, Probes, Settled Decisions, Injected Records and Sources

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Abstract

Two kinds of patient, invisible in the data, can explain every trial result. The first question is which part of their structure the evidence actually reveals. The second is how a decision maker can force the rest into view, especially when someone with an interest in the outcome chooses what is revealed. This companion to the note *Partial Gluing and Level Two* [8] builds a synthetic laboratory for these questions. It is run as an observe–orient–decide–act loop. Part I proves four theorems.

- *Theorem A (hiding segment).* Pair tables never identify two latent types. They leave a segment of explanations, on which the probability of every event on k variables is a polynomial of degree at most $k-2$ in one unseen parameter. A small subgroup that is very different is indistinguishable from a large one that is mildly different.
- *Theorem B (tempo).* Under pair-only disclosure the weight is never learned (tempo zero). After one informative triple table everything converges at $n^{-1/2}$. Agreement on a single variable is a regular relation. Only the merge of the two types slows to $n^{-1/4}$.
- *Theorem C (stratified value of information).* A probe is worthless when it touches a variable on which the types agree, and collapses the segment otherwise. Its worst-case value is zero whenever the decision has a saddle point on the segment, as a treat-or-not choice does, while its minimax-regret value stays positive whenever the best act changes. Its graded value rises with an explicit signal, and the best probe can be named from the pair tables alone. With verifiable disclosure and a skeptical regulator, sponsors unravel. A robust decision maker can lose from new evidence only when that evidence makes the evidence as a whole vacuous.
- *Theorem D (the decision on the segment).* An act is safe on the whole segment iff it is safe at both walls, and the set-rule choice depends on the pair tables alone, so no undetectable perturbation changes it. Info-gap prefers acts least exposed to the hidden moment, but before the probe its nominal point is a convention. After a probe the decision settles on the oracle act at median sample size $n^* = (z\tau/d)^2$, where d is a computable radius, and never settles when the truth sits on a boundary. In the ring of contingency tables the worst case is a tropical limit of the table count rather than a section, n^* is not a wall-crossing, and the exact conditional probe on the fibre is the efficient one (section 8).

Part II turns to an actor who adds records to tilt the decision, with no relation between injected and genuine records.

- *Theorem E (slopes).* A decision's margin has a price in injected records: the margin divided by the steepest single-record influence, attained by copies of one record.

- *Theorem F (the hidden channel).* Injections that leave every pair table unchanged form a segment of their own. They move the estimate of the unseen parameter by $\varepsilon s/v_T$ while the regulator's segment stays put, so there is always a window in which the decision is taken over and nothing is flagged. The best probe is also the hardest to poison.
- *Theorem G (stealth with equations).* With four variables the tetrads can detect injection, except along their tangent directions. Those directions multiply on the strata, but there the André–Quillen obstruction stops them from being extended.
- *Theorem H (one geometry).* The adversary's segment and the regulator's are the positivity part and the model part of one line, read in units of v_T. The regulator's margin is the distance to the decision boundary and the adversary's room the distance to the detection boundary, so Part II completes the decision lab rather than adding to it.
- *Theorem J (sources and bands).* When the decision maker sees which source and batch each table came from, a trusted source that is adversarial in some batches can move the pooled estimate by at most $c_s\sqrt{\pi/(1-\pi)}$ standard errors without being exposed, at every sample size. Gluing tests on two covers turn every undetected tilt into a bounded delay and localise concentrated attacks without writing the source off.
- *Theorem K (lifting).* Each singularity has a lift: decision chambers and plateaus on the fibre, mixtures of acts, the blow-up of the merge, the adversarial lift whose block size is the poisoning budget, and an orbifold atlas. Around it sit propositions on dilation, on the value of a probe in its size (a Radner–Stiglitz flat start, and a negative Wald value for a fixed rule), on backtracking, and on rectangularity as gluing.
- *Theorem L (one picture).* Settling, the value of a probe, every robustness number, the decision-relevance of a gluing failure and the stealth window are all readings of one chamber arrangement on the fibre. Each robustness number is the same margin to a wall divided by the capacity of a gauge.
- *Theorem M (the descent tower).* The fibre is a fibre of a diagram of local spectra over the contexts. Ambient descent holds algebraically and fails only through positivity (contextuality). For the latent layer: the label switch is the kernel; the sign of Δ_m is a class in $H^1(\Gamma;\mathbb{Z}/2)$; the tetrads are the cokernel of the unsigned incidence matrix; and a hidden channel moves the obstruction up exactly one level, to triple coherence.
- *Theorem N (descent, identification, decision).* Existence of a global law, membership in the two-type model and the decision's sensitivity are three separate conclusions with separate sources (positivity; the $\mathbb{Z}/2$ class and the walls; margin over capacity), and each can fail without the others.

Five pre-registered studies test the quantitative claims, and one example is worked through every step by hand (section 18). In Study 11, twenty-one of the twenty-four registered predictions held, and the three that failed are reported together with what they show. In Study 11b, all four predictions about settling held. In Study 11c, three of four predictions about an exact conditional probe on the lattice of tables held; the one that failed shows that an exact interval's conservativeness can cancel its efficiency gain at n^*. In Study 12, all four predictions about injection held: about 6% of injected records flips a mandated-triple decision without a single replicate being flagged. In Study 13, all five predictions about a sometimes-adversarial source held: pooling is tilted in 88% of replicates, while the gluing filter never settles wrongly more than 2% of the time and is only delayed. The paper is written for three readerships: decision theory under severe uncertainty, Bayesian persuasion and verifiable disclosure, and machine learning with reinforcement learning and adversarial training.

How to read this paper

If you work on	read	because
decisions under severe uncertainty, value of information	sections 2, 3, 5, 6, 9 and 13	the ambiguity set is a stratified fibre rather than a set of priors; for two types on the pair cover the decision on it is closed (existence by two wall checks, invariance on the fibre, a settling size n^*); a probe is worthless or worth the full minimax amount, by stratum; Wald and Savage disagree on that amount; robustness is monotone except for vacuity
persuasion and disclosure	sections 2, 3, 5, 10 and 14	the hiding set is a segment, the regulator's minimal mandate is one named triple, and unravelling and pooling are computed; a sponsor who runs the triple can tilt it through a channel no table check sees
machine learning, RL, GANs	sections 2, 4, 7, 11, 12 and 17	data poisoning solved in a small model (vertex attack, stealth–tilt frontier); tempo as sample complexity, with rates 0, $n^{-1/2}$ and $n^{-1/4}$; a bandit learns the closed-form probe; the regulator–sponsor game rotates like GAN training unless the regulator commits
anyone who wants to follow the computations	sections 18 and 19	one example (the main note's drug example) worked through every theorem, with hand checks, ending in its toric geometry and its descent complex and filtration ($\mathbb{CP}^1$, the Birch point, the hyperdeterminant); then a fourth variable, in stages (a smooth toric 5-fold, induced three-way interactions, the second-triple defence)
data from several sources; robustness, federated learning	sections 20 to 22	a trusted source that is sometimes adversarial: gluing on two covers, bands of fixed width in standard errors, tilt turned into delay (Study 13); lifting the decision maker's singularities, dilation, the value of a probe in its size, backtracking, rectangularity as gluing
anyone who wants the whole paper in one statement	sections 23 and 29	chambers, slices and gauges: one margin over many capacities
algebraic geometry, sheaves and descent	sections 24 and 25	local spectra over the contexts; what each level of the tower fails to separate and what obstructs descent there; what the CSD framework's obstruction classes can and cannot see on these data
categorical probability	section 26	the model as a string diagram, sufficiency on the fibre, de Finetti for batches, Blackwell's order, all worked by hand

Table 1: Reading guide.

1 The lab

A regulator sees trial tables, the pair trials of drugs A, B and C, say, and must decide about a combination nobody has trialled. Patients may come in two latent types: actors that are invisible in the data and shape it [8]. The lab asks what the regulator can learn about the actors, how fast, and what to do about it. It also asks what happens when a sponsor chooses which tables to reveal.

The lab runs as Boyd's observe–orient–decide–act loop [9] (fig. 1).

- *Observe:* the tables revealed so far.
- *Orient:* compute the latent fibre, the set of actor explanations consistent with them.
- *Decide:* choose an act by robust satisficing, and choose a probe.
- *Act:* the probe's outcome is a new table, which cuts the fibre.

Tempo, the speed of the loop, has a precise meaning here: the rate at which each cycle shrinks the fibre.

The decision rules are not new. They are Wald's minimax [11], maxmin expected utility over a set of priors [12], info-gap robustness [13], robust satisficing [14, 15] and the value of information [16], applied under Knightian uncertainty [10]. What is new is the ambiguity set. It is not a convex set of priors chosen by the analyst. It is a fibre of a polynomial map, determined by the evidence and stratified by where the actors agree. Its geometry decides what can be hidden, how fast it is revealed, and what a probe is worth.

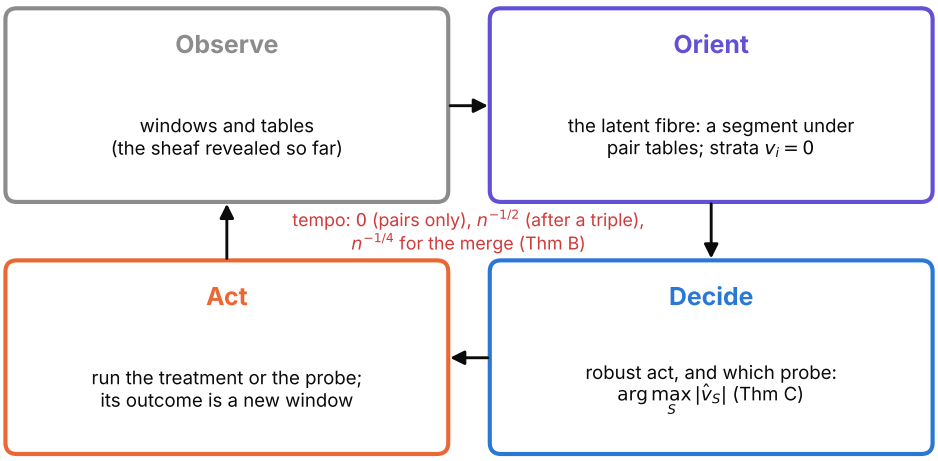


Figure 1: The lab as an OODA loop on the geometry. The probe rule and the tempos are theorems B and C.

2 Setup

Two latent types with weights $w_1 = w$ and $w_2 = 1 - w$ act on m binary variables, such as harms. Type j has harm probabilities $a_j \in [0,1]^m$. The observed law is the mixture of the two product

laws. Write

$$\mu = w_1 a_1 + w_2 a_2, \qquad d = a_1 - a_2, \qquad v = \sqrt{w_1 w_2}\, d, \qquad r = \frac{w_2 - w_1}{\sqrt{w_1 w_2}} = t - \frac{1}{t}, \quad t = \sqrt{w_2/w_1}.$$

A *window* S is a set of variables whose joint table is observed. The central moments of the law are $m_S = \mathbb{E} \prod_{i \in S} (x_i - \mu_i)$, and the law of the variables in a set K is affine in $(\mu, (m_S)_{S \subseteq K})$.

Lemma 1 (moment factorisation [8])**.** *$m_S = v_S\, s_{|S|}(r)$, where $v_S = \prod_{i \in S} v_i$, $s_2 = 1$, $s_3 = r$ and $s_{k+1} = r\, s_k + s_{k-1}$. In particular s_k is a polynomial of degree $k - 2$.*

The *decision problem* follows Section 17 of [8]. An act a has utility u_a on outcomes, with values in $[u_{\min}, u_{\max}]$. Two kinds of cover are used: the *pair cover* (all single and pair tables) and covers with triples added. The *latent fibre* X_C of a cover C is the set of two-type explanations consistent with its tables. The worked act throughout is "treat", with $u = 1 - 1.2\,[E]$ where E is any harm among three variables, against "nothing" with $u = 0$.

3 Theorem A: the hiding segment

Theorem A (hiding segment)**.** *Let $m \geq 3$, and suppose at least three coordinates of v are non-zero.*

(a) The pair cover determines μ, and v up to a common sign. Its latent fibre is, up to relabelling the types, the segment

$$X_{\text{pair}} = \{(w(t),\, \mu + v\, t,\, \mu - v/t) :\ t \in [t_-, t_+]\},$$

where $t_\pm$ are cut out by the box walls. For $v_i > 0$ the constraints are $v_i/\mu_i \leq t \leq (1 - \mu_i)/v_i$. For $v_i < 0$ they are $-v_i/(1 - \mu_i) \leq t \leq \mu_i/(-v_i)$.

(b) Along the segment, the probability of any event on a set K of k variables is a polynomial in r of degree at most $k - 2$. Its coefficient of r^j involves only the products v_S with $S \subseteq K$ and $|S| \geq j + 2$.

(c) Hence robust (worst-case) values over the segment are attained at the box walls $t = t_\pm$, or at no more than $k - 3$ interior critical points. For $k = 3$ only the walls count.

Proof. (a) The pair moments are $m_{ij} = v_i v_j$. With three non-zero coordinates, $v_i^2 = m_{ij} m_{ik}/m_{jk}$ fixes $|v_i|$, and the signs of the products fix v up to a global sign, which relabels the types. Given (μ, v), each $t > 0$ gives $w = 1/(1 + t^2)$, $a_1 = \mu + v\, t$ and $a_2 = \mu - v/t$, and all of them reproduce every single and pair table, since those depend only on μ and v. The box constraints are linear in t and $1/t$, so they cut out an interval.

(b) The law on K is affine in the m_S with $S \subseteq K$, and $m_S = v_S\, s_{|S|}(r)$ with $\deg s_{|S|} = |S| - 2$ (lemma 1).

(c) A polynomial of degree $k - 2$ on an interval has its extremes at the ends or at roots of its derivative. □

The degree bound can drop on a stratum. If $v_i = 0$, every product containing i vanishes, so the terms they carry disappear. Study 11's registered prediction Q2 missed this (section 7). Figure 2(c) shows what the segment means in practice: identical pair tables come from a rare subgroup that differs a lot or from a common one that differs a little. For a regulator these are different drugs.

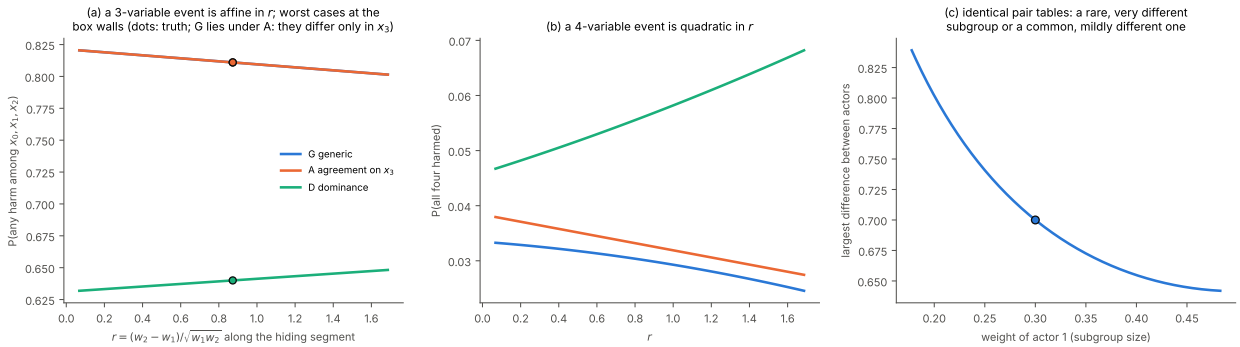


Figure 2: Theorem A on the registered conditions. (a) A three-variable event is affine along the segment, so its worst case sits at a box wall; dots mark the truth. (b) A four-variable event is quadratic. In condition A ($v_3 = 0$) it is affine (the stratum effect). (c) Along the segment for condition G: the subgroup's size against how different it is.

4 Theorem B: three tempos

Theorem B (tempo)**.** *Let each window be observed with n independent samples.*

(a) Pair-only disclosure: tempo zero. *The pair tables are the same at every point of the segment, so no estimator based on them identifies r. The minimax error for r stays at least half the segment's width for every n. The same holds for any prediction whose r-coefficient is non-zero.*

(b) After one triple window S with $v_S \neq 0$. *$r = m_S/v_S$, and every parameter is a smooth function of window moments with non-vanishing Jacobian. So all are estimated at rate $n^{-1/2}$, and the constant for r is proportional to $\sigma_S/|v_S|$ with $\sigma_S^2 = \mathbb{E}[\prod_{i\in S}(x_i - \mu_i)^2]$.*

(c) Relations. *Agreement on one variable ($v_i = 0$ while at least two other coordinates are non-zero) is regular: $v_i = m_{ij}/v_j$ is estimated at $n^{-1/2}$. At the merge ($v = 0$, identical types) the separation $\|v\|$ is estimated only at rate $n^{-1/4}$, which is optimal [26]. Predictions of observed events stay at $n^{-1/2}$ everywhere.*

Proof. (a) is theorem A(a). Two truths at the ends of the segment produce the same distribution of the data, so Le Cam's two-point bound gives minimax risk at least half their distance.

(b) is the delta method.

(c) At $v = 0$ the pair moments are pure noise of size $n^{-1/2}$. A rank-one fit returns $\hat{v}\hat{v}^\top$ of that size, so $\|\hat{v}\| \asymp n^{-1/4}$; Chen [26] shows this rate is optimal for mixtures at coincidence. For agreement, $\hat{v}_i$ is a ratio of moments whose denominator is bounded away from zero. □

In the design discussion we had predicted the slow rate for agreement on one variable. The proof shows it belongs to the merge only. The registration of Study 11 records the correction, which was made before any run (section 7). Another stratum is not identifiable at all: if the types differ on at most two variables, the law depends on v only through one product. The same is true of an absent type ($w_2 = 0$), whose law is exactly that of the merge. So these relations are invisible in any data, not merely slow to estimate (R4 in table 2).

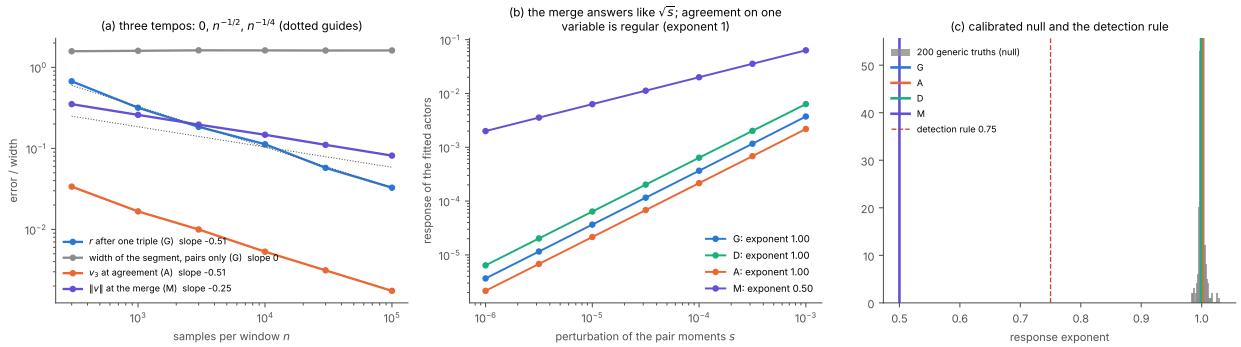


Figure 3: Theorem B. (a) Learning curves: the weight is never learned from pairs (flat). After one triple it converges at $n^{-1/2}$. Agreement on one variable is also $n^{-1/2}$, and the merge is $n^{-1/4}$. (b) The same signatures as response exponents to deterministic perturbations of the pair tables. (c) The null distribution of exponents over 200 generic truths, and the registered detection rule.

5 Theorem C: the value of a probe

For a cover C with fibre X_C, take a probe w' whose exact outcome reveals the post-probe fibre $X_{C\cup w'}(x) \ni x$. Its *minimax value* is

$$\mathrm{VOI}(w') = \min_{x\in X_C} \max_a \min_{x'\in X_{C\cup w'}(x)} U(a,x') \;-\; \max_a \min_{x\in X_C} U(a,x), \qquad U(a,x) = \mathbb{E}_x u_a.$$

This is Wald's minimax applied to information: the gain in the worst case when the adversary picks the truth. It is the minimax analogue of Howard's value of information [16], and by the minimax inequality it is non-negative. Replacing U by minus Savage's regret, $-\big(\max_b U(b,x) - U(a,x)\big)$, gives the *regret value* $\mathrm{VOI}_{\mathrm{reg}}$ [19].

Theorem C (stratified VOI and the mandated window)**.** *Let C be the pair cover and $Z = \{i : v_i = 0\}$ the coincident set. Let w' be a triple window on S.*

(1) On the independence stratum ($v = 0$), $\mathrm{VOI}(w') = 0$ for every probe.

(2) If S meets Z, then $v_S = 0$, the probe does not pin r, and $\mathrm{VOI}(w') = 0$.

(3) If S avoids Z, the probe pins the point, and $\mathrm{VOI}(w') = g$, the minimax gap of perfect information on the segment: $g = \min_x \max_a U - \max_a \min_x U \geq 0$, the same for every such S. It is zero exactly when U has a saddle point on acts $\times$ segment, in particular whenever all acts have their worst case at a common point. For an event on three variables every $U(a,\cdot)$ is affine in r (theorem A), so its worst case is at a wall, and $g > 0$ needs two acts with worst cases at opposite walls. The regret value is $\mathrm{VOI}_{\mathrm{reg}}(w') = \min_a \max_x \big(\max_b U(b,x) - U(a,x)\big)$, which is positive exactly when no act is optimal along the whole segment.

(4) With n samples, the post-probe set is $\{x : |r(x) - \hat{r}| \leq c\sigma_S/(|v_S|\sqrt{n})\}$. So the value is non-decreasing in the signal $|v_S|\sqrt{n}/\sigma_S$, and it vanishes as the truth approaches any plane $\{v_i = 0 : i \in S\}$.

(5) The pair tables estimate v, so the regulator can name the best probe, $\arg\max_S |\hat{v}_S|/\hat{\sigma}_S$, before running it.

Proof. (1) The law is a product, and every point of the fibre gives the same predictions.

(2) and (3) follow from theorem A and $m_S = v_S\, r$. A pinned point gives $\max_a U(a, x)$ for each x, and the adversary minimises over x. Hence $g = 0$ iff $\min_x \max_a U = \max_a \min_x U$, which is the saddle condition; a common worst point x_0 gives $\min_x \max_a U \le \max_a U(a, x_0) = \max_a \min_x U$. After pinning, the regret is 0, so the regret value is the pre-probe minimax regret.

(4) The value is monotone in the post-probe set: a smaller set has a larger worst case. That set shrinks as $|v_S|\sqrt{n}/\sigma_S$ grows, by theorem B(b).

(5) is immediate from (4). □

Corollary 1 (unravelling)**.** *Suppose a sponsor can reveal a triple table verifiably and at no cost, the regulator approves only if the worst case over the revealed fibre meets the requirement, and the sponsor wants approval. Then every sponsor whose true outcome meets the requirement reveals the triple. Approval then coincides with full information.*

Proof. For a safe sponsor, revealing gives approval, because the exact value meets the requirement. Not revealing gives approval only if the worst case over the segment does, and that is never better. For a harmful sponsor neither choice gives approval. This is Milgrom's unravelling argument [22, 23] on the segment. □

The rule borrowed is classical; the structure that makes it apply is the segment. The same structure explains when skepticism fails. If the reported tables come from a different population, for example pooled with a benign one, the skeptic's worst case is taken over the wrong segment (D4).

Proposition 1 (when evidence can hurt)**.** *At level one, with the contamination model of [8], let $C \subseteq C'$. For every act, the robustness on C' is at least the robustness on C, unless the act is vacuous on C' (robustness $-\infty$). So a new window can lower a robust decision maker's robustness only by making the evidence vacuous for that act.*

Proof. $E_\alpha(C') \subseteq E_\alpha(C)$, so the robust value $V_{C'}(a;\alpha) \ge V_C(a;\alpha)$ for every $\alpha \le \lambda^*(C')$, and V is monotone in α. If $V_{C'}(a;\lambda^*(C')) \ge u_0$, the set of admissible contaminations on C' contains that on C intersected with $[1-\lambda^*(C'), 1]$, and it starts no lower. Otherwise the act is vacuous on C'. □

This corrects an earlier statement of ours that new evidence can lower robustness gradually. Under ambiguity aversion, the value of information can be negative [18, 20]. Here that happens only by a jump to vacuity, as when a strongly conflicting trial drives λ^* below the threshold of [8, Prop. 33]. Study 11's registered prediction L8 expected a gradual loss and failed. A post-hoc check confirms the proposition (section 7).

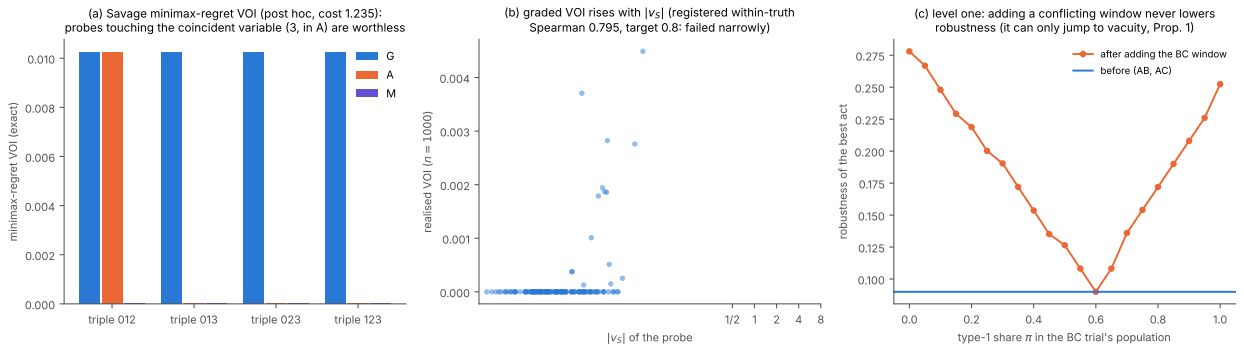


Figure 4: Theorem C. (a) Exact regret VOI per triple (post hoc, harm cost 1.235, at which the best act changes along G's segment). In A, the triples containing the coincident variable are worthless; in M every probe is. The Wald gap g is 0 here at every cost, because both acts have their worst case at the same wall: a saddle point. (b) Graded VOI at $n = 1000$ against $|v_S|$, for random truths where some probe matters. (c) Level one: robustness after adding a conflicting window, over a family of populations. It never falls except by vacuity (proposition 1).

6 Theorem D: the decision on the segment

Theorems A–C say what is hidden, how fast it is revealed and what a probe is worth. They do not say how to choose an act on the segment, whether that choice is stable, or when the evidence settles it. This section does, for two types on the pair cover.

Write $R = [r_-, r_+]$ for the segment in the coordinate r, and $U_a(r)$ for the expected utility of act a at the explanation with parameter r. Let u_a depend on the variables in a set T with $|T| = k$, and let $\chi(x) = \prod_{i \in T}(2x_i - 1)$. For an evidence interval $I \subseteq R$ let $W_a(I) = \min_I U_a$, the act's worst value on I. The *set rule* $\sigma(I)$ chooses, among the acts with $W_a(I) \geq u_0$, one with the largest worst value, and abstains if there is none. The *info-gap rule* [13] needs a nominal point $\hat{r}$ and chooses the act with the largest horizon $\alpha_a(\hat{r}) = \sup\{h : W_a([\hat{r} - h, \hat{r} + h] \cap R) \geq u_0\}$. The *oracle act* is $a^\circ = \sigma(\{r_0\})$ at the true point r_0.

Theorem D (the decision on the segment)**.** *(1)* Existence. *U_a is a polynomial in r of degree at most $k-2$, and $W_a(R)$ is attained at a wall or at one of at most $k-3$ interior critical points. For $k = 3$,*

$$U_a(r) = U_a(\hat{r}) + \beta_a(r - \hat{r}), \qquad \beta_a = v_T \sum_x u_a(x)\chi(x),$$

the top Walsh coefficient of the utility times v_T. So an act meets the requirement on the whole segment iff it does at both walls, and one exists iff $\max_a \min\{U_a(r_-), U_a(r_+)\} \geq u_0$.

(2) Uniqueness, and flat acts. *$\sigma(R)$ is unique unless two admissible acts have equal worst values. For $k = 3$ the horizon is $\alpha_a = (U_a(\hat{r}) - u_0)/|\beta_a|$, or $+\infty$ when the act is admissible on the whole segment. So info-gap prefers acts with a small top Walsh coefficient, the acts least exposed to the hidden triple moment. When two or more acts are admissible on all of R their horizons tie at $+\infty$, and only the nominal value separates them. The info-gap choice and the set-rule choice can differ.*

(3) Invariance and stability. *$\sigma(R)$ is a function of the pair tables alone, so it is the same for every law on the fibre: a perturbation the decision maker cannot detect never changes it. An info-gap choice before the probe is not invariant in this sense, because pair tables carry no*

information about r, so its nominal point is a convention; the midpoint in r and the midpoint in t can give different acts. Under detectable perturbations $\sigma(R)$ changes only where some $W_a(R)$ crosses u_0 or two worst values cross; these are algebraic hypersurfaces in the pair moments, and the distance to them is the decision's stability margin.

(4) Settling *($k = 3$). After a triple window with n draws per window, the post-probe set is $I_n = [\hat{r} - h_n, \hat{r} + h_n] \cap \hat{R}$ with $h_n = z\hat{\tau}/\sqrt{n}$, where $\hat{\tau}$ estimates the asymptotic standard deviation τ of $\sqrt{n}\,\hat{r}$ (theorem B). For an interval $[c-h, c+h]$ inside R, σ chooses a° iff every constraint*

$$g_j(c) - \gamma_j h \geq 0$$

holds, where $g_0 = U_{a^\circ} - u_0$ and $\gamma_0 = |\beta_{a^\circ}|$, and $g_b = U_{a^\circ} - U_b$ and $\gamma_b = |\beta_{a^\circ}| - |\beta_b|$ for every other act b. Hence:

- *(a) The* settling radius *is $d = \min_{\gamma_j > 0} g_j(r_0)/\gamma_j$. Constraints with $\gamma_j \leq 0$, against steeper acts, never bind as h grows.*
- *(b) $P(\sigma(I_n) = a^\circ) \to P\big(g_j(r_0)\sqrt{n} + \xi_j \geq \gamma_j z\tau \text{ for all } j\big)$, with $\xi \sim N(0, \Sigma)$ the delta-method limit of the plug-in constraints. If one constraint binds, the median settling size is*

$$n^* = (z\tau/d)^2,$$

  whatever Σ is.
- *(c) On the stratum $g_j(r_0) = 0$, where the truth sits on a crossing or on the requirement, the settling probability tends to the plateau $\Phi(-\gamma_j z\tau/s_j) < 1$, with $s_j^2 = \Sigma_{jj}$. The decision never settles.*

Proof. (1) For binary variables the law of T has the exact expansion $P_T(x) = \sum_{S \subseteq T} m_S \prod_{i \in S}(2x_i - 1) \prod_{i \in T \setminus S} p_i(x_i)$, with $m_\emptyset = 1$, $m_{\{i\}} = 0$ and $p_i(1) = \mu_i$. By lemma 1, $m_S = v_S s_{|S|}(r)$ has degree $|S| - 2$, so U_a has degree at most $k - 2$. Its minimum on an interval is at an endpoint or at a root of U_a', of which there are at most $k - 3$. For $k = 3$ only $S = T$ depends on r, through $m_T = v_T r$, and its coefficient is $\sum_x u_a(x)\chi(x)$.

(2) For affine U_a, $W_a([\hat{r} - h, \hat{r} + h]) = U_a(\hat{r}) - |\beta_a|h$ until the interval meets the wall on the downhill side; from then on the worst value is the wall value, which is at least u_0 if it is reached first. Solving for h gives the horizon.

(3) R, μ and v are determined by the pair tables (theorem A), and so are the U_a. σ depends on the $W_a(R)$ only through the order of the admissible ones, which changes only at the stated equalities. The non-invariance of a nominal choice is an example (check DS6).

(4) For affine U_a, $W_a([c - h, c + h]) = U_a(c) - |\beta_a|h$. The set rule chooses a° iff $W_{a^\circ} \geq u_0$ and $W_{a^\circ} \geq W_b$ for all b, since an act with $W_b > W_{a^\circ} \geq u_0$ is admissible; these are the constraints. At $c = r_0$ they hold exactly for $h \leq d$, which is (a). The plug-in constraints $\hat{g}_j$ are smooth functions of the window tables, so $\sqrt{n}(\hat{g}_j - g_j) \to N(0, \Sigma)$ by the delta method, and $\hat{\tau} \to \tau$. With one binding constraint the settling probability tends to $\Phi\big((g_j\sqrt{n} - \gamma_j z\tau)/s_j\big)$, which equals $1/2$ exactly when $\sqrt{n} = \gamma_j z\tau/g_j = z\tau/d$; this is (b). If $g_j = 0$ the argument is $-\gamma_j z\tau/s_j$ for every n, which is (c). □

Three remarks fix the scope.

- *Settling is not expiry.* In the main note's time-varying decisions, a decision expired when a new distinction entered the utility, and no radius predicted it [8, Section 17]. Settling is driven by evidence alone, and it is computable.

- *Level one and level two together.* If conflicting windows are added, the contamination model of [8] applies on top of the segment, and its vacuity bound [8, Prop. 36] can make every act inadmissible. A theorem for the combined worst case is open.
- *What is not covered.* Parts (2) and (4) are for utilities on three variables, where U_a is affine. For four variables U_a is quadratic, and settling has to track the interior critical point. More than two types, and other covers, are open.

Checks DS1–DS9 verify the parts exactly or by bisection on random truths (section 7).

7 Studies 11 and 11b

Study 11: Theorems A–C and the disclosure game

The study was pre-registered and frozen with SHA-256 before any registered condition was computed (`engine/study11_lab/PREREG_study11.md`). Three amendments were made before the run, and each is logged in the registration:

- the correction to Theorem B noted above;
- two design fixes to illustrations;
- one change to L6, prompted by a smoke test on an invented truth, which showed that the minimax VOI can be zero everywhere; L6 was reduced to the zero pattern and a common value.

One deviation arose during the run. For some truths the pooled tables of the disclosure game have no two-type explanation at all, and the rule adopted (reject) is logged in `DEVIATIONS_study11.md`. Table 2 lists the predictions.

The three failures say more than the passes.

- *Q2* failed because the registration ignored the stratum effect that theorem A(b) states.
- *L7* failed narrowly because the registered statistic ranked probes by $|v_S|$ alone, while the theorem's signal is $|v_S|/\sigma_S$. Post hoc, on fresh truths, the within-truth correlation is 0.85 with $|v_S|$ and 0.87 with $|v_S|/\sigma_S$.
- *L8* failed because the registered expectation was false: robustness cannot fall gradually (proposition 1). Post hoc, every member of the family satisfies monotonicity, computed by exact bisection. A strongly conflicting window ($\lambda^* = 0.30$) makes every act vacuous, which is the only way down.

Study 11b: when the decision settles

Study 11b tests theorem D(4). It was registered and frozen with SHA-256 before any registered design was simulated (`engine/study11b_settle/PREREG_study11b.md`), after the engine checks and two smoke runs on an invented design. The first smoke run showed that a second constraint close to the binding one moves the median above n^*, as theorem D(4b) allows; the designs were then fixed so that one constraint binds.

There are five designs on two truths. In each, a treatment act ($u = b -$ [any harm], or $b -$ [at least two harmed]) faces a flat conservative act ($u = b -$ [variable 0 harmed], $\beta = 0$), or the

	prediction	outcome
Q1	pair rank $2m$, plus one triple $2m+1$ ($m = 3, 4, 5$)	holds
Q2	a 3-variable event is affine; the 4-variable event is quadratic, not affine (G, A, D)	**fails for A**: affine there, since $v_{0123} = 0$
Q3	extremes of the 3-variable event at the walls	holds
R1–R3	response exponents: G 1.00, D 1.00, A 1.00 (agreement is regular), M 0.50	hold
R4	absent type = merge: identical tables	holds
R5	null band and detection rule	holds
R6	agreement rests exactly on the windows containing the variable	holds
L1–L4	slopes: r after a triple -0.51; v_3 at agreement -0.51; merge -0.25; pair-only width flat	hold
L5	the probe rule matches the oracle; it avoids the coincident variable	holds
L6	exact VOI: zero pattern and common value	holds ($g = 0$: a saddle point)
L7	graded VOI monotone in $\lvert v_S \rvert$ within a truth (mean Spearman ≥ 0.8)	**fails narrowly** (0.795)
L8	some conflicting window lowers robustness (drug family)	**fails**; explained by proposition 1
D1–D7	disclosure game (section 10)	all hold
I1, I3	illustrations: bandit reaches the rule; catch-up is finite	hold
I2	illustration: gradient play does not approach the mixed equilibrium	**fails**: it approaches, then cycles

Table 2: Study 11: registered predictions and outcomes.

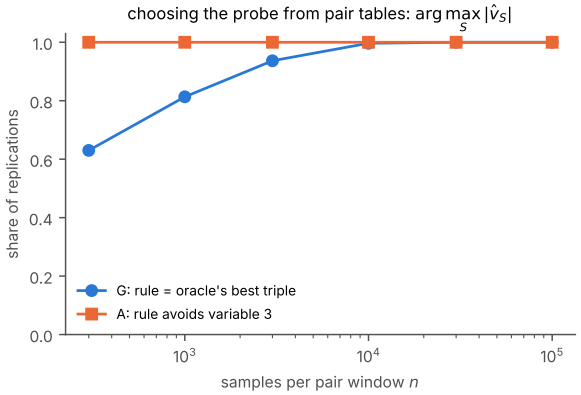


Figure 5: Choosing the probe from pair tables. The rule $\arg\max_S |\hat{v}_S|$ matches the oracle's choice in 99.7% of replications at $n = 10^4$. In condition A it never picks a triple containing the coincident variable.

requirement alone. Before the probe the set rule chooses the conservative act, or abstains; the oracle chooses the treatment. D3 puts the boundary on the other side of r_0, D5 uses a second truth, and D4 puts the truth on the crossing, the stratum $d = 0$. Each cell has 4000 replicates; each replicate samples three pair windows and the triple window independently. The regulator uses the plug-in $\hat{\tau}$ and $z = 1.96$.

	prediction	outcome
S1	settling rate within 0.05 of the delta-method prediction in ≥ 18 of 20 cells, and within 0.08 in all	holds: 20 of 20, largest gap 0.014
S2	empirical median settling size within a factor 1.25 of $n^* = (z\tau/d)^2$	holds: ratios 0.94, 1.08, 1.00, 0.93
S3	D4 stays on the predicted plateau (within 0.05) and does not rise between $n = 3000$ and 10^5	holds: largest gap 0.021; 0.27 at both ends
S4	without the triple, the set rule on the estimated segment picks the population's pre-probe act ($\geq 95\%$)	holds: 98–100%

Table 3: Study 11b: registered predictions and outcomes.

All four predictions held (table 3 and fig. 6). The predicted medians were $n^* = 3270$ (D1, D2), 6410 (D3) and 2001 (D5). The settling curves are shallow: at $8n^*$ between 85% and 99% of replicates had settled. The reason is in theorem D(4b): the plug-in constraints carry noise from the pair tables as well as from $\hat{r}$, and that noise widens the curve without moving its median. On the stratum (D4) the share settled stayed near 0.28 from $n = 10^3$ to 10^5, as predicted. At most 0.75% of replicates had pair tables with no two-type explanation.

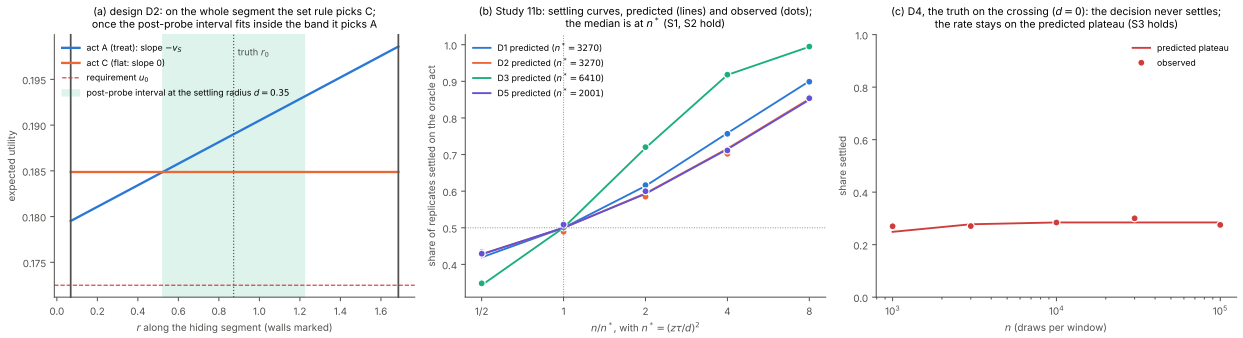


Figure 6: Theorem D and Study 11b. (a) Design D2: the treatment's line and the flat act's line on the segment. On the whole segment the set rule picks the flat act; once the post-probe interval fits within the settling radius it picks the treatment. (b) Settling curves against n/n^*: predicted (lines) and observed (dots); each crosses one half at n^*. (c) The stratum $d = 0$: the share settled stays on the predicted plateau and never rises to one.

8 Theorem D and the ring of contingency tables

The companion [65] puts the geometry of couplings into a graded ring: the Ehrhart ring of a polytope of tables with rational margins. Its degree-k piece has the integer tables with k times the margins as a basis. Its Hilbert function counts them, and the count crosses the walls of the margin space by one binomial term at a time. A reviewer asked two questions about it. Is the decision

functional of theorem D, the worst case of an act over the hiding set, a section of this ring? And is the settling size n^* a wall-crossing? The short answers are no and no. The ring does give two things, though: the correct algebraic form of the worst case, and an exact probe that settles sooner. This section works it out for three binary variables, the case of theorem D(4).

The fibre. Fix the three pair tables of a law P on $\{0,1\}^3$. The laws that share them form a segment $P+\varepsilon\chi$, with $\chi(x)=\prod_i(2x_i-1)$. This is the no-three-way-interaction model. Its toric ideal is principal, generated by the single quartic $p_{000}p_{011}p_{101}p_{110}-p_{001}p_{010}p_{100}p_{111}$ (Macaulay2, check EH1), so χ is its only Markov move. Moving along the fibre moves the triple moment at unit speed, because

$$m_T(P+\varepsilon\chi)-m_T(P)=\varepsilon\sum_x\chi(x)\prod_i(x_i-\mu_i)=\varepsilon\prod_i\big((1-\mu_i)+\mu_i\big)=\varepsilon.$$

So the fibre is an interval of m_T, ended where a cell reaches zero. If P has denominator N, then at sample size $n=kN$ the integer tables with the pair tables of nP are the points $nP+j\chi$, $j\in\mathbb{Z}$, a lattice of step $1/n$ in m_T. The hiding set R of theorem A is the part of this interval where a two-type explanation exists, $m_T\in v_T\,[r_-,r_+]$.

Proposition 2 (the hiding set inside the ring)**.** *For three binary variables and a two-type law P with denominator N:*

(i) The integer tables with the pair tables of kNP are the lattice points of k times a lattice segment of length ℓ. Their number is the Ehrhart polynomial $k\ell+1$, the degree-k Hilbert function of the ring.

(ii) R is a strict sub-segment. Its ends are where some a_{ji} reaches 0 or 1, while every cell of the law is still positive. Those ends are algebraic numbers and generally irrational. So the number of tables in kR is not an Ehrhart polynomial, and not even a quasi-polynomial.

(iii) The expected utility of an act on the three variables is affine on the fibre, $U_a=U_a(P)+c_a\,(m_T-m_T(P))$. Its slope is the Walsh coefficient $c_a=\sum_x u_a(x)\chi(x)$, and $\beta_a=v_Tc_a$ in the coordinate r.

Proof. (i) Every cell of a table with the given pair tables is an integer plus or minus one free integer, so the fibre is $T_0+\mathbb{Z}\chi$, cut by $T_0+j\chi\geq 0$. The ends are integers, and dilating by k multiplies them by k. (ii) The ends of R are the ends of theorem A's box, where a coordinate of $a_1=\mu+vt$ or $a_2=\mu-v/t$ reaches 0 or 1; a cell of the mixture vanishes only if both components vanish there. The count is $\lfloor n\,(R_+-m_0)\rfloor-\lceil n\,(R_--m_0)\rceil+1$. With irrational ends, its second differences at any fixed period do not vanish. (iii) This is the computation above with u_a in place of the moment. □

For the worked truth $w=\frac{2}{5}$, $a_1=(\frac{4}{5},\frac{3}{10},\frac{7}{10})$, $a_2=(\frac{1}{5},\frac{7}{10},\frac{3}{20})$, the denominator is $N=5000$. The fibre is $m_T\in[-0.0671,\,0.0609]$, a lattice segment of 640 steps. The hiding set $R=[-0.0214,\,0.0100]$ is 25% of it (fig. 7(a)).

Proposition 3 (the decision functional and the ring)**.** *(i) Not a section, but a tropical limit. The worst value W_a is a minimum of linear forms, so it is not an element of the ring. It is the limit of a weighted count over the same basis of tables:*

$$\min_T U_a(T)-\frac{\log\#}{s}\;\leq\;-\frac{1}{s}\log\sum_T e^{-sU_a(T)}\;\leq\;\min_T U_a(T),$$

where T runs over the tables in nR (or in the whole fibre) and # is their number. As $s \to \infty$ the middle term tends to the lattice minimum. That minimum is W_a exactly over the fibre, whose ends are lattice points, and within $|c_a|/n$ over R.

(ii) Over the whole fibre the worst case is an integer programme. *Take the quartic, oriented so that its leading term is the costlier side. Its normal form (Conti–Traverso) is the table that minimises U_a. As a function of the pair tables this minimum is piecewise linear. It breaks only where the binding cell changes, and those are the walls where the fibre's length, the leading coefficient of the table count, breaks too. So a skeptic who allows every law with the given pair tables has decision walls that are linear on each chamber of [65]. The set rule on R has algebraic walls instead, because positivity of the explanation, not of the table, ends R.*

(iii) n^* is not a wall-crossing. *Raising n moves the margins along a ray, and chambers are cones. So the table count stays one polynomial, and no wall is crossed. The lattice is still visible at n^*, though. The truth sits $n^* d\,|v_T| = (z\tau)^2|v_T|/d$ lattice steps from the decision wall: about ten in Study 11b's designs. The conditional law of the position has standard deviation $\tau_c|v_T|\sqrt{n^*}$, about five steps.*

(iv) The exact probe is the efficient one. *Given its own pair tables, the triple window is a lattice point on its fibre. Its conditional law is the exponential family $\propto e^{\psi j}/\prod_x (T + j\chi)_x!$ in the three-way log odds ratio $\psi = \sum_x \chi(x) \log p_x$. This is the sample space of Fisher's exact test in [65, Thm. 7.1(iv)], tilted. Mapping ψ to r through the known pair tables gives an estimate with asymptotic standard deviation*

$$\tau_c = \frac{1}{|v_T|\sqrt{\sum_x 1/p_x}} \;\le\; \tau_m,$$

where τ_m is that of the plug-in moment (with v known). The inequality is Cramér–Rao along the fibre, because $\sum_x 1/p_x$ is the Fisher information of the move. So with the pair tables known, the exact probe settles at the smaller size $n_c^* = (z\tau_c/d)^2$.

Proof. (i) Bound the sum between its largest term and # times its largest term. Over the fibre the minimum of an affine function is at an end, and the ends are lattice points (proposition 2(i)). Over R the nearest lattice point to an end is within $1/n$ of it. (ii) The Gröbner basis of a principal ideal is its generator, and reduction by it applies the move against the cost while the table stays nonnegative. The end it reaches is $\min_x$ of a linear form in the margins, taken over the cells with the right sign of χ; both this minimum and the fibre's length change formula exactly where the argmin changes. (iii) A ray kNP has fixed binding cells (check EH5). At n^* the post-probe half-width $z\tau/\sqrt{n^*}$ equals d; in lattice steps that is $n^* d|v_T|$. (iv) The conditional law is classical, with sufficient statistic j. The Fisher information of $P + \varepsilon\chi$ in ε is $\sum_x \chi(x)^2/p_x$. The plug-in moment is a regular estimator of ε on this one-parameter model, so its variance is at least the inverse information. □

Checks EH1–EH7 (`engine/check_ehrhart.py`) verify each statement, as follows.

- EH1: the brute-force counts.
- EH2: the quasi-polynomial test, for periods up to 12.
- EH3: the tropical bounds, on three acts.
- EH4: Conti–Traverso against enumeration and linear programming, and the kinks along lines of margins.

- EH5: the lattice reading of n^*.
- EH6: $\tau_c \le \tau_m$ on 300 random truths, and a simulation of the conditional MLE.
- EH7: the exactness of the conditional interval, coverage at least 0.95 at every ψ.

Study 11c: settling with the exact conditional probe

Proposition 3(iv) makes a prediction that can fail. A regulator who knows the pair tables and uses the exact conditional interval for ψ should settle at n_c^*, earlier than the plug-in regulator of Study 11b. But exact intervals are conservative, and at n^* the conditional law spans only a few lattice steps. Study 11c was registered and frozen with SHA-256 before any registered design was simulated (`engine/study11c_exact/PREREG_study11c.md`), after the checks above and one smoke run on an invented design. It reuses Study 11b's designs D2, D3, D5 and the stratum D4, with the pair tables known. One triple window feeds both probes, and there are 2000 replicates per cell.

	prediction	outcome
E1	exact probe's median settling size within a factor 1.25 of $n_c^* = (z\tau_c/d)^2$	holds: ratios 1.08, 1.06, 1.09
E2	exact probe's median below the plug-in probe's in every design	**fails**: D2 2735 $<$ 2799, D3 5285 $<$ 5612, but D5 1954 $>$ 1860
E3	exact settling rate within 0.06 of the normal theory with τ_c at every n	holds: largest gap 0.046
E4	D4 (stratum): exact rates at $n = 3000$ and 10^5 within 0.05, both below one half	holds: 0.021 and 0.020 (plateau 0.025)

Table 4: Study 11c: registered predictions and outcomes.

Three of the four predictions held, and one failed (table 4, fig. 7(d)). The exact probe settles close to n_c^*, but always a little later. Its medians are 6–9% above n_c^*, and its rate sits below the normal-theory curve at every n up to $4n_c^*$. This is the price of an exact interval: its coverage is at least 95%, never exactly 95% (check EH7), and at n^* only a few lattice steps carry the conditional law.

E2 failed in one design. The efficiency gain is $(\tau_m/\tau_c)^2$ in sample size. That is 12% for truth G, which beats the conservativeness, so in D2 and D3 the exact probe settled first. For truth H the gain is 8%, and conservativeness ate it: in D5 the exact probe settled 5% later than the plug-in. So the exact probe is the efficient one in the limit, but not uniformly better at n^*. A mid-p interval, which is less conservative, is the natural repair. It was not registered, and it is not tested here.

On the stratum D4 neither probe settles. With the pair tables known, the plateau is the one-sided 2.5% of theorem D(4c), not Study 11b's 0.27. That larger value came from the noise in the estimated pair tables.

Gap 3, answered.

- *Is the decision functional a section?* No. The worst case is a tropical object: the $s \to \infty$ limit of a weighted Hilbert function. Over the whole margin fibre it is an integer programme solved by the toric ideal's Gröbner basis, and linear on chambers. Over the hiding set it is not polyhedral, because positivity of the explanation, not of the table, draws its walls.

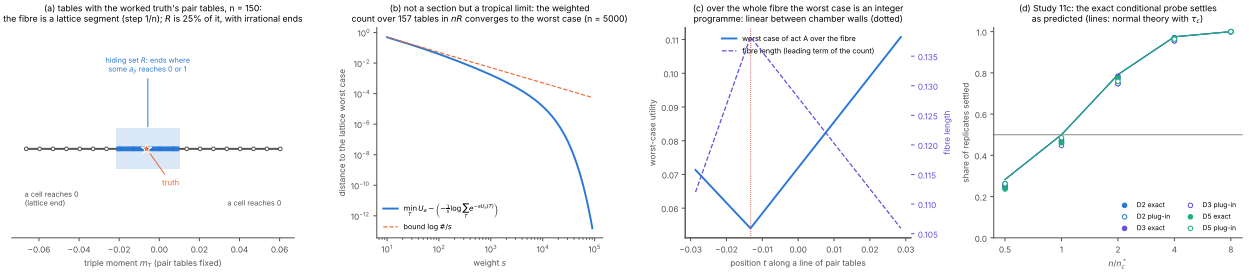


Figure 7: The decision and the ring of contingency tables. (a) The worked truth. The tables with its pair tables form a lattice segment in m_T, shown at $n = 150$ (dots). The hiding set R (blue) is a quarter of it, ended by the explanation's positivity walls rather than by a table's. (b) Act A's worst case over the tables in nR as a tropical limit. The distance from the weighted count to the lattice minimum stays below $\log \#/s$ and vanishes as s grows. (c) Over the whole fibre, the worst case along a line of pair tables is linear between chamber walls (dotted), where the fibre's length (dashed) also breaks. (d) Study 11c: settling with the exact conditional probe (filled) and the plug-in probe (open) against n/n_c^*; the lines are the normal theory with τ_c.

- *Is n^* a wall-crossing?* No. Raising n moves along a ray inside one chamber. What does cross a wall is the truth leaving the stratum $d = 0$, and that is a change of design, not of sample size.
- *What the ring contributes.* The ring supplies the exact sample space of the probe. Using it is the efficient way to settle, $\tau_c \leq \tau_m$, and Study 11c measures what conservativeness costs at the few lattice steps that matter at n^*.

9 For decision theory and the value of information

Knight's distinction [10] between risk and uncertainty has a geometric form here. Under risk the fibre is a point; under pair-only disclosure it is a segment, and nothing in the data puts a probability on it.

- Maxmin expected utility [12] and Wald's minimax [11] supply the rule.
- Info-gap [13] and robust satisficing [14, 15] supply the satisficing form.
- The ambiguity set is neither chosen nor convex in the priors. It is a semialgebraic segment whose length the sponsor controls by choosing windows.

Several consequences are worth stating for this readership.

- *The minimax value of information is stratified.* A probe touching a coincident variable is worth nothing; any other probe is worth the same full amount, g or the minimax regret (theorem C). Howard's value of information [16] is never zero merely because of geometry.
- *Wald and Savage part ways.* With a treat-or-not decision on a three-variable harm, Wald's gap is always zero, since both acts are worst at the same wall (a saddle point). Savage's minimax regret [19] still values the probe whenever the best act changes along the segment (fig. 4a). Which criterion a regulator adopts decides whether the mandated window is worth anything.

- *Information cannot hurt gradually.* Blackwell's ordering [17] has a robust counterpart (proposition 1): more windows never lower robustness, except by vacuity.
- *The best probe can be named in advance,* from the pair tables alone. That turns the value of information into a design rule.
- *Info-gap needs a nominal point the segment does not supply.* Before the probe, pair tables say nothing about r, so an info-gap choice depends on a convention; the set rule does not (theorem D(3)). After the probe the nominal point is identified and info-gap is legitimate; it then prefers acts with a small top Walsh coefficient, those least exposed to the hidden moment.
- *The decision settles at a computable sample size,* $n^* = (z\tau/d)^2$, and never settles on the stratum $d = 0$ (theorem D(4)). Study 11b confirms both.

10 For persuasion and disclosure

In Bayesian persuasion [21], a sender commits to an experiment. In verifiable disclosure [22, 23], a sender chooses what to reveal from what it knows. Drug approval combines both [24], and a sender can also select its sample [25].

The lab gives this setting a concrete geometry. The sponsor's moves change the regulator's fibre:

- revealing only pairs leaves a segment;
- pooling with another population moves the segment;
- running the triple and disclosing it selectively is verifiable disclosure.

A mandate needs a size as well as a window. By theorem D(4) the regulator's decision settles at median size $n^* = (z\tau/d)^2$, so a mandated triple much smaller than n^* leaves the regulator on its conservative act about half the time or more, and a truth on a boundary is never settled at any size.

Registration and results-reporting rules, such as Section 801 of the US Food and Drug Administration Amendments Act of 2007, constrain these moves, and the model treats them as policies.

Study 11 crossed four sponsor strategies with four regulator policies on 2000 random drugs (fig. 8). The results:

- *Pairs only.* A naive regulator (midpoint) falsely approves 0.5% and falsely rejects 0.4%. A skeptic (worst case on the segment) falsely approves none and falsely rejects 2.8%. The hiding is narrow on average (mean width 0.008) but decisive near the threshold (fig. 9).
- *Mandated triple.* Requiring the triple window removes every error.
- *Pooling.* Reporting tables pooled with a benign population makes both naive and skeptical regulators falsely approve 99.9% of harmful drugs: the skeptic's worst case is taken over the wrong segment. Registration of the population restores the skeptic's rates. In 0.05% of cases the pooled tables admit no two-type explanation and are rejected (deviation DV1).
- *Selective disclosure.* Under skepticism every sponsor discloses and there are no errors, as corollary 1 predicts. Under a naive regulator some harmful sponsors hide, and 0.5% are falsely approved.

- *Catch-up.* After a false approval, post-market use reveals the triple in 4.8 rounds on average.

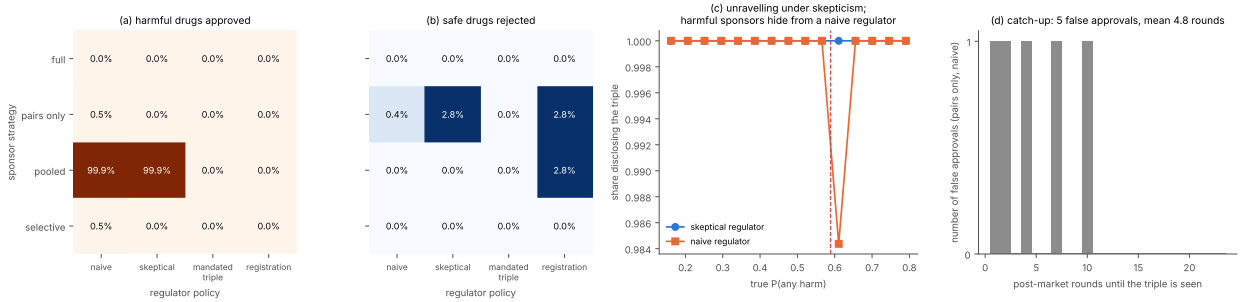


Figure 8: The disclosure game. (a, b) Error rates by sponsor strategy and regulator policy. (c) Unravelling: under a skeptical regulator every sponsor discloses (ties included), while a naive regulator lets harmful sponsors hide. (d) After a false approval, post-market use reveals the triple with probability 0.2 per round.

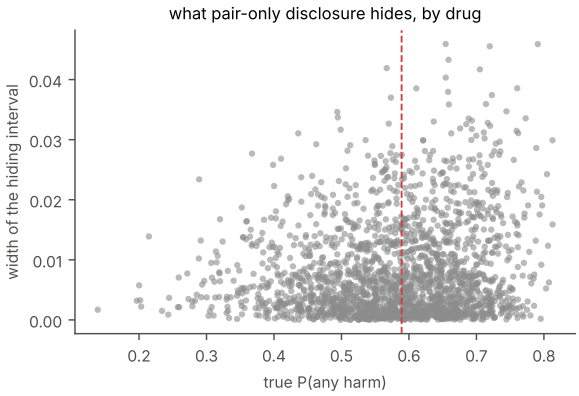


Figure 9: What pair-only disclosure hides, drug by drug: the width of the interval for the triple's harm.

11 For machine learning: RL and adversarial training

- *RL.* The lab is a partially observed decision process [27]. The actors are the hidden state, and the probe is an information-gathering action, as in Bayesian and sequential optimal experimental design [28, 29, 31]. Theorem B gives the sample complexity in closed form: rate 0, $n^{-1/2}$ or $n^{-1/4}$ depending on the stratum. Theorem C(5) gives the optimal probe in closed form, and theorem D(4) gives the sample complexity of the decision itself, $n^* = (z\tau/d)^2$, which is not the same as that of the estimate. The nearest hardness result is for *partial online state information* (POSI) [30]: a learner that can query only some components of the latent state at each step may need $\Omega(A^{H-1}/\varepsilon^2)$ episodes to find an ε-optimal policy (exponential in the horizon H, with A actions), while two structured subclasses are learnable with regret $\tilde{O}(\sqrt{K})$ over K episodes. The lab is a one-step analogue with a different query: the type is never revealed, and a probe returns a marginal table on a window. The segment's geometry

plays the role of their structural assumptions; it is what turns the probe choice into the closed-form rule of theorem C(5).

- *Adversarial training.* The sponsor–regulator interaction is a minimax game, like a GAN's generator and discriminator [32]. It shares their dynamics: simultaneous gradient play rotates around a mixed equilibrium and need not converge [33], while commitment by one side converges. Divergence choice matters as in f-GANs [34]. Here the worst case over the segment plays the role of the discriminator's objective.

Figure 10 shows two illustrations; they illustrate the analogy and are not evidence for the theorems.

- A contextual bandit, given $|\hat{v}_S|$ as a feature, learns to choose probes. After 3000 episodes it reaches 99.7% of the closed-form rule's value, far above random choice.
- In a stylised inspection game between regulator and sponsor, the registered illustration I2 predicted that simultaneous gradient play would not approach the mixed equilibrium (final distance at least the initial one). It failed as registered: the distance falls from 0.83 to 0.33. Post hoc, play then stays on a persistent cycle at distance 0.26–0.34 and never converges, the rotation that [33] describes. With the regulator committing first, as pre-registration does, play converges to honesty.

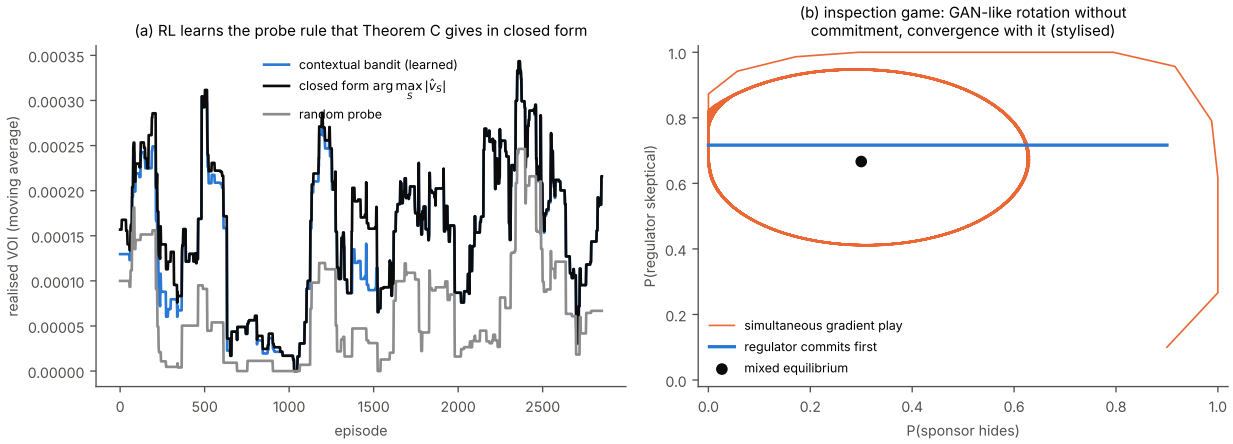


Figure 10: (a) A bandit learns the probe rule that theorem C states in closed form. (b) Inspection-game dynamics: rotation without commitment, convergence with it (stylised).

12 Part II. Injected records: actors who feed the data

Part I asked what a regulator can learn when an interested party chooses which tables to reveal. Part II asks what happens when that party *adds records*: data written to tilt the decision. We study whole-record injection only. The adversary adds complete records, so every window is a marginal of one law.

The model. Genuine records follow the two-type law P. The adversary adds records drawn from a law Q of its choice, a share ε of the data. Within the injected set any dependence is allowed,

since Q is arbitrary; between injected and genuine records there is none. The dataset is therefore a disjoint union, a coproduct of two samples, and every window sees the marginal of

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon Q.$$

This is Huber's contamination model [72] with intent behind it, and the level-one contamination model of [8] read adversarially. It is narrower than editing genuine records, where the reachable set is a whole total-variation ball.

A categorical framing. Provenance is a context the regulator cannot see. Formally it is the environment comonad $W = E \times (-)$ on sets of records, with $E = \{\text{genuine}, \text{injected}\}$. The counit $WX \to X$ forgets provenance, and it is exactly what the regulator observes. The comultiplication copies the label. A W-coalgebra on X is a labelling $X \to E$. "No relation between the two sets" says the dataset is the coproduct $X_g \sqcup X_a$; pushed forward to window tables, a coproduct becomes a mixture. Every procedure the regulator can run factors through the counit. The framing names the situation precisely, but the results below come from three other structures: the polytope of reachable laws, the Walsh characters of $\{0,1\}^m$, and the scheme cut out by the model's equations.

13 Theorem E: slopes and the vertex attack

Let $F = (F_j)$ be decision statistics: smooth functions of the window tables such that the regulator's act is unchanged while every $F_j > 0$. An example is the set-rule constraints of theorem D. Write $\mathrm{IF}_j(x) = \frac{d}{d\varepsilon} F_j\big(P + \varepsilon(\delta_x - P)\big)|_0$, Hampel's influence function [73] of a single injected record x.

Theorem E (slopes and the vertex attack)**.** *(1)* Reachable set. *At budget ε the reachable laws form the polytope $(1-\varepsilon)P+\varepsilon\Delta$, with vertices $(1-\varepsilon)P+\varepsilon\delta_x$. Every window table is a marginal of the one law P_ε, so whole-record injection never makes the windows fail to glue ($\lambda^* = 1$).*

(2) Slopes are linear in the injected law: $\frac{d}{d\varepsilon} F_j(P_\varepsilon)|_0 = \sum_x Q(x)\,\mathrm{IF}_j(x)$.

(3) The vertex attack. *To first order the smallest budget that changes the act is*

$$\varepsilon_1^* = \min_{j,\,x:\ \mathrm{IF}_j(x)<0} \frac{F_j(P)}{-\mathrm{IF}_j(x)},$$

attained by a point mass: copies of one record. The exact smallest budget is $\varepsilon_1^*\big(1 + O(\min_j F_j(P))\big)$.

(4) The injected set is a latent actor. *If Q is a product law, P_ε is a three-type law with weights $(1-\varepsilon)w$, $(1-\varepsilon)(1-w)$ and ε. Its pair covariances are $(1-\varepsilon)vv^\top + \varepsilon(1-\varepsilon)\Delta\Delta^\top$ off the diagonal, with $\Delta = \mu_P - \mu_Q$. They stay rank one exactly when Δ is parallel to v: the injected actor then imitates the latent axis.*

Proof. (1) P_ε is affine in Q, and Q ranges over the simplex. (2) The chain rule, with $\sum_x Q(x)(\delta_x - P) = Q - P$. (3) To first order $F_j(P_\varepsilon) = F_j(P) + \varepsilon \sum_x Q(x)\mathrm{IF}_j(x)$, and a linear function on the simplex is smallest at a vertex. The second-order remainder is $O(\varepsilon^2)$, and $\varepsilon^* = O(\min_j F_j)$, so the relative error is $O(\min_j F_j)$. (4) A mixture's covariance is the weighted within-component covariance plus the between-component term, and a product law has none of its own. □

Part (3) is the first-order energy shift of perturbation theory: the decision's margin divided by the steepest single-record influence. At a tie (the stratum $d = 0$ of theorem D) the margin is zero, and an arbitrarily small injection decides the act. Which act wins is then set by the sign of the influence difference, as in degenerate perturbation theory. Check IN3 shows the exact budget of the best point mass approaching ε_1^* as the margin shrinks, and no random injected law beating it.

14 Theorem F: the hidden channel

Write $\chi_S(x) = \prod_{i \in S}(2x_i - 1)$.

Theorem F (the hidden channel)**.** *Let $m = 3$ and $T = \{0, 1, 2\}$.*

(1) The injected laws that leave every single and pair table unchanged are exactly $Q = P + s\chi_T$, with $s \in [s_-, s_+] = [\max_{\chi_T(x)>0}(-P(x)),\ \min_{\chi_T(x)<0} P(x)]$. They form a segment: the adversary's own hiding segment.

(2) Injecting them leaves μ, v, the regulator's segment R and the pre-probe act unchanged, and moves the estimate of r by exactly

$$\hat{r}(P_\varepsilon) = r_0 + \varepsilon s / v_T.$$

(3) Let r_c be the nearest point beyond which the oracle act changes, on the side $\operatorname{sign}(s/v_T)$, and let $|s|$ be the positivity limit on that side. With exact tables, the act switches at $\varepsilon^* = |r_c - r_0|\,|v_T|/|s|$, and the triple contradicts the pairs ($\hat{r} \notin R$) only at $\varepsilon_{\mathrm{det}} = |w - r_0|\,|v_T|/|s|$, where w is the wall on that side. Since r_c lies inside R, $\varepsilon^* < \varepsilon_{\mathrm{det}}$: there is always a stealth window *in which the decision is taken over and nothing is flagged. With n records per window and one binding constraint, the share choosing the oracle act crosses one half at $\varepsilon_{1/2} = \big(|r_c - r_0| - \kappa z\tau/\sqrt{n}\big)|v_T|/|s|$, where $\kappa = \gamma_j/|g_j'|$ is 1 for a flat competitor.*

(4) For general m and a triple S, $Q = P + s\chi_S$ leaves every single and pair table and every other triple moment unchanged, and moves $\hat{r}_S$ by $2^{m-3}\varepsilon s/v_S$. The budget is therefore proportional to $|v_S|$: the factor that makes S the best probe (theorem C(5)) and sets its tempo (theorem B) also makes it the hardest to poison.

Proof. (1) On $\{0, 1\}^3$ the singles, pairs and total mass are 7 linear functionals of rank 7, so their kernel is one-dimensional. The characters are orthogonal, and χ_T is orthogonal to every character of order at most two, so it spans the kernel. Positivity of Q gives the bounds. (2) The means are unchanged, so central moments are linear in the law, and $m_T(P_\varepsilon) = m_T(P) + \varepsilon s \sum_x \chi_T(x) \prod_{i \in T}(x_i - \mu_i)$. The sum factorises as $\prod_i\big((1 - \mu_i) + \mu_i\big) = 1$. The pair tables give v and R (theorem A), and they are unchanged. (3) follows from (2) and, at finite n, from theorem D(4) with the constraint evaluated at the shifted point. (4) The same factorisation over S, with a factor $\sum_{x_i} 1 = 2$ for each $i \notin S$; for any set S' with $S' \not\supseteq S$ a factor $\sum_{x_i}(2x_i - 1) = 0$ appears. □

Corollary 2 (capacity of the channel)**.** *The hidden channel of a triple S has capacities*

$$\mathrm{cap}_+(S) = s_+ = \min_{\chi_S(x)<0} P(x), \qquad \mathrm{cap}_-(S) = -s_- = \min_{\chi_S(x)>0} P(x),$$

one for each direction: the probability of the rarest outcome that the injection must drain. On the side where the adversary needs $\operatorname{sign}(s/v_S)$, the flip budget is

$$\varepsilon^*(S) = \frac{|r_c - r_0|\,|v_S|}{2^{m-3}\,\mathrm{cap}(S)},$$

and the detection budget is the same expression with the distance to the wall. A probe whose rarest joint outcomes are rarer leaves the adversary less room. Among triples, the regulator should prefer those with large $|v_S|/\mathrm{cap}(S)$*; this is the poisoning counterpart of the probe rule* $\arg\max_S |v_S|/\sigma_S$ *of theorem C(5).*

Proof. $Q = P + s\chi_S$ is a law iff $P(x) + s\chi_S(x) \geq 0$ for all x. For $s > 0$ only the cells with $\chi_S = -1$ can reach zero, which gives $s \leq \min_{\chi_S<0} P(x)$, and symmetrically for $s < 0$. The budgets follow from theorem F(3) and (4), with the shift $2^{m-3}\varepsilon s/v_S$. □

On Study 11's four-variable truth G the two rules agree. The best probe, $\{0,1,2\}$, has the largest $|v_S|/\sigma_S$ (0.097) and the smallest largest shift per unit of injected share (4.61 in r); the order of the four triples is the same under both rules: $\{0,1,2\}$, $\{0,2,3\}$, $\{1,2,3\}$, $\{0,1,3\}$, with shifts 4.61, 5.94, 7.75 and 10.85 (check IN14).

15 Theorem G: stealth when there are equations

With three variables the model has no equations on the pair tables, only inequalities, so any injection into pairs is invisible except through an empty fibre. With four variables there are two, the tetrads $T_1 = M_{01}M_{23} - M_{02}M_{13}$ and $T_2 = M_{01}M_{23} - M_{03}M_{12}$: the off-diagonal pair covariances have rank one [77]. Let $N(Q) = \sum_x Q(x)\nabla T(x)$ be the tetrads' first-order response, the tetrad Jacobian applied to Q.

Theorem G (stealth with equations)**.** *Let* $m = 4$*.*

(1) Tangent stealth. *The tetrads move by* $\varepsilon N(Q) + O(\varepsilon^2)$*. An injection with* $N(Q) = 0$ *is a point of the scheme* $\operatorname{Spec}\mathbb{R}[M]/(T_1, T_2)$ *over the dual numbers* $\mathbb{R}[\varepsilon]/(\varepsilon^2)$*, based at* P*: it is invisible to first order: a straight injection moves the tetrads only at order* ε^2*, so a Wald test of the tetrads has noncentrality of order* $n\varepsilon^4$ *rather than* $n\varepsilon^2$*.*

(2) The stealth–tilt frontier. *For a tilt functional with influence* IF *and whitened Jacobian* N_w*,* $\Phi(\delta) = \max\{\sum_x Q(x)\mathrm{IF}(x) : Q \in \Delta,\ |N_w(Q - P)|_\infty \leq \delta\}$ *is concave and piecewise linear in* δ*. With* $n_x \in \mathbb{R}^2$ *the whitened tetrad response to one injected record* x*, it has the closed form*

$$\Phi(\delta) = \min_{\lambda \in \mathbb{R}^2} \Big[\max_x \big(\mathrm{IF}(x) - \lambda^\top n_x \big) + \delta \, \|\lambda\|_1 \Big],$$

a minimum of finitely many affine functions of δ*. An optimal injection mixes at most three records.* Φ *rises from the best invisible tilt* $\Phi(0)$ *to the vertex tilt* $\max_x \mathrm{IF}(x)$*, which it reaches at* $\bar{\delta} = \|n_{x^*}\|_\infty$ *for the best record* x^**.*

(3) Strata. *The tetrad Jacobian has rank* 2 *at a generic truth and at a single coincidence (*$v_i = 0$*), rank* 1 *when two coordinates of* v *vanish, and rank* 0 *at the merge (*$v = 0$*). So the space of first-order stealthy injections grows on the strata, which are also where Part I's hiding happens.*

(4) Obstruction. *The model's ring* $A = \mathbb{R}[M]/(T_1, T_2)$ *is commutative, so the obstruction to extending a first-order stealthy injection to second order is an André–Quillen class, in* $\mathrm{Ext}^1_A(L_{A/\mathbb{R}}, \kappa(P))$ *[37, 38]. Where the tetrads form a regular sequence it is explicit: for a first-order stealthy direction* A *(*$JA = 0$*, with* J *the tetrads' Jacobian and* H *their second derivative at* P*),*

$$\mathrm{ob}(A) = \big[\tfrac{1}{2} H(A, A)\big] \in \operatorname{coker} J,$$

and A *extends to a second-order stealthy injection iff* $\mathrm{ob}(A) = 0$*. Hence:*

- *Where J has rank 2 (generic truths, single coincidences) the cokernel is* 0*, every A extends, and an adversary that bends its injection, letting Q depend on ε, keeps the tetrads still to third order.*
- *On the two-coincidence stratum, say $v_2 = v_3 = 0$, the cokernel is the line spanned by $T_1 - T_2$ and $\mathrm{ob}(A) = q(A) = dM_{03}\,dM_{12} - dM_{02}\,dM_{13}$ evaluated on A. It vanishes exactly on the quadric cone $\{q = 0\} \cap \ker J$, so a generic stealthy direction is obstructed and moves the tetrads at order ε^2 along every bent injection.*
- *At the merge $J = 0$, the cokernel is all of $\mathbb{R}^2$, and $\mathrm{ob}(A) = \frac{1}{2}H(A, A)$ itself.*

A non-zero cokernel is necessary for an obstruction but not sufficient: the class can vanish on special directions. So the strata give more first-order stealth but obstructed second-order stealth.

Proof. (1) Taylor expansion. A $\mathbb{R}[\varepsilon]/(\varepsilon^2)$-point based at P is a tangent vector annihilated by the Jacobian of the defining equations. The noncentrality is $n\,T(P_\varepsilon)^\top \Sigma_T^{-1} T(P_\varepsilon)$ with $T(P_\varepsilon) = O(\varepsilon^2)$. (2) Since $\sum_x Q(x)(\delta_x - P) = Q - P$, the constraint is $|\sum_x Q(x) n_x|_\infty \le \delta$: a linear programme in Q with one equality ($\sum Q = 1$), four inequalities and 16 non-negative variables. Its dual has a free multiplier $\lambda = \lambda^+ - \lambda^-$ for the four inequalities and one for the equality, and it is the displayed minimum; strong duality holds because $Q = P$ is feasible. The value is a minimum of affine functions of δ, one per dual vertex, so it is concave and piecewise linear. A basic optimal solution has at most as many non-zero coordinates as active constraints, the equality and at most two of the tetrad bounds, hence at most three records. For $\delta \ge \|n_{x^*}\|_\infty$ the point mass at x^* is feasible and optimal. (3) The covariances respond to the law surjectively, so the rank is that of $\partial T/\partial M$. With $M_{ij} = v_i v_j$ and, for example, $v_3 = 0$, the rows are $M_{01}\,dM_{23} - M_{02}\,dM_{13}$ and $M_{01}\,dM_{23} - M_{12}\,dM_{03}$, which are independent. With $v_2 = v_3 = 0$ both rows equal $M_{01}\,dM_{23}$, and $v = 0$ kills both. Check IN8 confirms every single and every pair of coincidences. (4) A bent injection $P + \varepsilon A + \varepsilon^2 B$ with $JA = 0$ keeps the tetrads to third order iff $JB = -\frac{1}{2}H(A, A)$ is solvable, where H is their second derivative; the class of $\frac{1}{2}H(A, A)$ in $\operatorname{coker} J$ is the obstruction. For a presentation by a regular sequence, the cotangent complex is the two-term complex $I/I^2 \to \Omega_{\mathbb{R}[M]} \otimes A$, and $\mathrm{Ext}^1(L_{A/\mathbb{R}}, \kappa(P))$ is that cokernel. If J is surjective the cokernel is zero. On the two-coincidence stratum both rows of J are $M_{01}\,dM_{23}$, so its image is the diagonal of $\mathbb{R}^2$ and the cokernel is detected by $T_1 - T_2$. The quadratic part of $T_1 - T_2 = M_{03}M_{12} - M_{02}M_{13}$ at a point where these four covariances vanish is $dM_{03}\,dM_{12} - dM_{02}\,dM_{13}$, a non-degenerate quadratic form in four of the free directions of $\ker J$; its zero set there is a quadric cone, and off it the class is non-zero (check IN11 finds it non-zero on random stealthy directions). At the merge every covariance vanishes, $J = 0$ and nothing is quotiented. □

On the H4 truth of Study 12 the frontier has three pieces: $\Phi(\delta) = 10.144 + 0.417\,\delta$ up to $\delta = 0.219$, then slope 0.201 up to $\bar{\delta} = 2.744$, where it reaches the vertex tilt 10.743 and stays. The optimal injection mixes three records at $\delta = 0$ (1100, 1101 and 0111), two on the middle piece, and one, the point mass at 1101, from $\bar{\delta}$ on (check IN13).

The ε^2 movement of a straight stealthy injection in (1) is curvature, not obstruction: at smooth points bending removes it. The obstruction proper is (4), and it is André–Quillen rather than Hochschild because the ring is commutative. Hochschild cohomology governs the noncommutative observable algebras of [8, Section 15], and the Hochschild–Kostant–Rosenberg theorem ties the two theories only at smooth points, while the obstruction lives exactly at the singular ones. On random truths a straight stealthy injection moves the tetrads with log–log slope 1.99, against 0.98 for a vertex attack. A bent one moves them with slope 3.0 at a generic truth and 2.0 on the two-coincidence stratum.

16 Theorem H: one geometry for the regulator and the adversary

Fix the pair tables of P. Let Π be the set of all laws with these pair tables, the *pair fibre*, and let $X \subseteq \Pi$ be the regulator's hiding segment, the two-type explanations.

Theorem H (one geometry)**.** *(1)* One line *($m = 3$). $\Pi = \{P + s\chi_T\} \cap \Delta$ is a segment, and X is a sub-segment of it, strictly inside it unless both types meet a wall at once. In the triple-moment coordinate, X is $v_T \cdot R$ and Π is $m_T(P) + [s_-, s_+]$. The regulator reads this line in units of v_T, since $r = m_T/v_T$; the adversary moves along it by εs. The two segments are the model's part and the positivity part of one line, and that is why $1/v_T$ governs the value of a probe (theorem C), its tempo (theorem B) and its vulnerability (theorem F) alike.*

(2) One geometry *($m = 4$). Π has dimension 5. The directions in Π that preserve every equation of the model to first order form a line: the tangent of X. Those equations are the equality of $r_S = m_S/v_S$ across the four triples, and $m_{0123} = v_{0123}(r^2 + 1)$; the tetrads are constant on Π. So an injection invisible to every equation of the model moves the observed law along the regulator's own hiding segment. A single-triple channel χ_S moves r_S alone, and a second triple sees it at first order.*

Proof. (1) Given the singles and pairs, the law of three binary variables is affine in m_T, which is theorem F(1). The two-type laws with these pairs are those with $m_T = v_T r$, $r \in R$. They are laws, so $X \subseteq \Pi$. The two segments end for different reasons. Π ends where some cell of $P + s\chi_T$ reaches 0. X ends at a box wall, where one type's harm probability on one variable reaches 0 or 1. At such a wall, say $a_{1i} = 0$, the first type gives no mass to the cells with $x_i = 1$. But each cell is $P(x) = w_1 \prod_j a_{1j}^{x_j}(1 - a_{1j})^{1-x_j} + w_2 \prod_j a_{2j}^{x_j}(1 - a_{2j})^{1-x_j}$, and the second term is positive whenever $w_2 > 0$ and every a_{2j} lies strictly inside $(0, 1)$. So every cell is still positive, the positivity bound is not reached, and the end of X lies strictly inside Π. Equality is possible only if the second type is at a wall too, so that some cell receives no mass from either type. In the worked example (section 18) the smallest cell is 0.0152 at one wall and 0.0068 at the other, both positive. (2) $\dim \Pi = 16 - 1 - 4 - 6 = 5$. Given the pairs, the model fixes the four triple moments and the quadruple moment up to the single parameter r: four constraints on five coordinates, with Jacobian of rank 4 when no v_i vanishes. The kernel is one-dimensional and contains the tangent of X. A channel χ_S moves m_S alone (theorem F(4)), so it changes r_S and not $r_{S'}$. □

Corollary 3 (the second-triple defence)**.** *Let $m = 4$ and let $S \neq S'$ be triples with $v_S, v_{S'} \neq 0$.*

(a) The single-triple channel $Q = P + s\chi_S$ moves $\hat{r}_S$ by $2\varepsilon s/v_S$ and leaves $\hat{r}_{S'}$ unchanged, so the difference $\hat{r}_S - \hat{r}_{S'}$, which is 0 under the model, moves at first order. Any second triple catches it, with the noise of a difference of two estimates (theorem B).

(b) An injection that no second triple, no tetrad and no quadruple moment can see to first order must move all four r_S together and m_{0123} consistently with them: it moves the observed law along the regulator's hiding segment (theorem H(2)). Its room is then the distance to the segment's walls, as at $m = 3$.

Proof. (a) is theorem F(4) together with $r_S = r_{S'} = r$ under the model. (b) is theorem H(2). □

As a check, take the truth with weights $(0.35, 0.65)$, $a_1 = (0.8, 0.3, 0.7, 0.25)$ and $a_2 = (0.2, 0.7, 0.15, 0.7)$, where all four r_S equal 0.629. Injecting 5% of records from $P + s\chi_{\{0,1,2\}}$, with s half its capacity (0.0137), moves $\hat{r}_{\{0,1,2\}}$ to 0.533, exactly the predicted shift $2\varepsilon s/v_S = -0.096$, while the other three stay at 0.629 (check IN12).

Proposition 4 (two boundaries of one geometry)**.** *On the line of theorem H(1) mark the truth* r_0*, the* decision boundary r_c *where the oracle act changes, and the* detection boundary*, the walls of* R*. The regulator's margin is the distance from* r_0 *to the decision boundary (the post-probe form of theorem D(3)). The adversary's room is the distance from* r_0 *to the detection boundary. A stealthy takeover is possible on a side exactly when the decision boundary is nearer, which holds whenever* $r_c \in R$*. At* $m = 4$ *the detection boundary becomes the stealth–tilt frontier of theorem G(2) together with the model equations of theorem H(2).*

Proof. Theorem F(3), read on the line of theorem H(1). □

The decision lab and the adversarial lab are therefore one lab. Part I measures the distance to the boundary where the act changes; Part II measures the distance to the boundary where the change becomes visible. Both are read on the same stratified fibre, and both scale with the same v_T. Checks IN1–IN14 verify Theorems E–H and their corollaries.

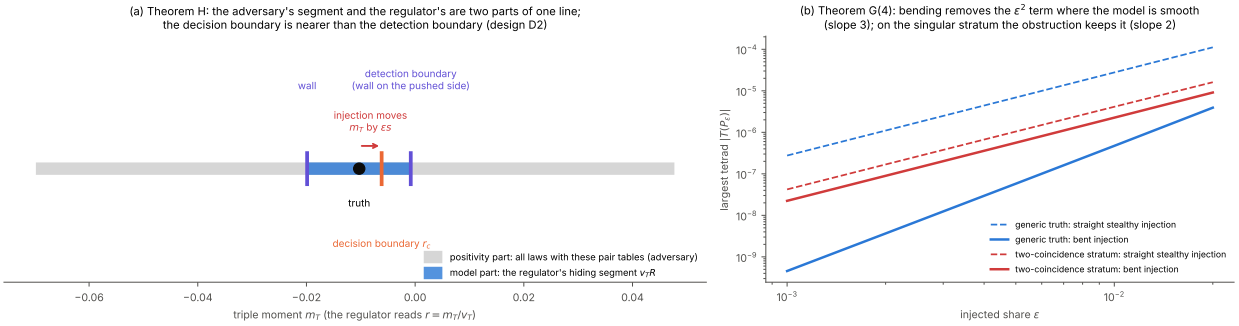


Figure 11: (a) Theorem H and proposition 4 on design D2, in the triple-moment coordinate. All laws with the observed pair tables form the grey segment; the regulator's hiding segment is the blue part of it. The injection walks the truth to the decision boundary well before the detection boundary. (b) Theorem G(4): tetrads against the injected share. A straight stealthy injection moves them at order ε^2. Bending removes that term at a smooth point (slope 3) but not on the singular stratum (slope 2), where the André–Quillen obstruction is non-zero.

17 Study 12: injection in practice

Study 12 was registered and frozen before any registered design was simulated (`engine/study12_inject/PREREG_study12.md`). One irregularity is disclosed in the registration. The smoke run's tetrad part used the $m = 4$ truth then intended for H4. That truth was demoted, and H4 moved to a fresh truth before freezing; the criteria were fixed in code before the smoke run and were not changed.

There are two designs.

- *H1–H3: the hidden channel.* Design D2 of Study 11b, with $Q = P + s\chi_T$ at the positivity limit $s = 0.058$ and $n = 10^5$ records per window. The theory gives $\varepsilon_{1/2} = 0.058$ (closed form; 0.059 from the delta-method curve), $\varepsilon^* = 0.071$ and $\varepsilon_{\text{det}} = 0.164$.
- *H4: first-order stealth at* $m = 4$*.* A Wald test of the tetrads at level 0.05. Two injections have the same first-order tilt of $\hat{r}$ (0.3): the best tetrad-blind law from the linear programme, and a point mass.

	prediction	outcome
H1	share choosing the genuine act within 0.05 of prediction (≥ 9 of 10 cells); half-way point within 10%	holds: 10 of 10, largest gap 0.012; 0.0581 vs 0.0587
H2	flag rate within 0.05 (≥ 6 of 7); half-way point within 5% of ε_{det}	holds: largest gap 0.005; 0.1636 vs 0.1635
H3	at $\varepsilon = 0.07$ the decision is taken over (≤ 0.5) and at most 1% is flagged	holds: 0.31 and 0
H4	tetrad-test rejection within 0.04 of the noncentral-χ^2 prediction (≥ 5 of 6); stealth ≤ 0.10 at every n; vertex ≥ 0.80 at $n = 50000$	holds: largest gap 0.016; stealth 0.045–0.048; vertex 0.84, 1.00, 1.00

Table 5: Study 12: registered predictions and outcomes.

All four held (table 5 and fig. 12). Injecting about 6% of records into the mandated triple, records that leave every pair rate unchanged, takes the decision over half the time. At 7% it is taken over 69% of the time, and not one replicate in 4000 is flagged. Detection begins only near 15%. At $m = 4$ the tetrad-blind injection costs about 6% of the tilt and stays at the test's level at every sample size, while the point mass with the same tilt is rejected 84% of the time already at $n = 2000$.

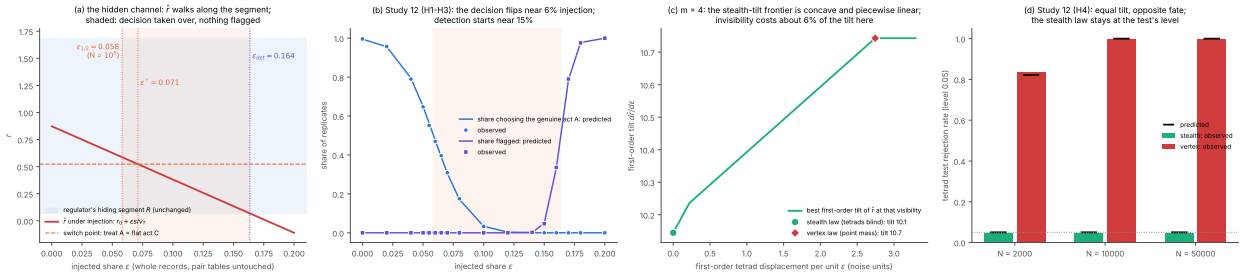


Figure 12: Part II. (a) The hidden channel on design D2: the estimate of r walks along the regulator's segment as the injected share grows, while the segment itself does not move. The decision is taken over between $\varepsilon_{1/2}$ and ε_{det} with nothing flagged (shaded). (b) Study 12, H1–H3: share choosing the genuine act and share flagged, predicted (lines) and observed. (c) The stealth–tilt frontier at $m = 4$ (theorem G(2)); the vertical axis is zoomed. (d) Study 12, H4: tetrad-test rejection for injections of equal tilt.

For the three readerships.

- *Decisions.* A satisficing margin has a price in injected records, ε_1^* (theorem E). Against the hidden channel the price is $|r_c - r_0|\,|v_T|/|s|$ (theorem F), and no consistency check on the tables can raise it.

- *Persuasion and disclosure.* Mandating the triple (corollary 1) opens the channel through which a sponsor who runs it can tilt it. The channel moves joint outcomes without changing any pair rate, so it is invisible to everything Part I's regulator checks. Defences are a second triple, which sees a single channel at first order (theorem H(2)); patient-level audit; or four or more variables, where every model-invisible injection must move along the hiding segment itself.

- *Machine learning.* This is data poisoning [74] in a model small enough to solve. The vertex attack is the optimal single-point poison. ε^* is a certified budget for this pipeline, in the spirit of [75], and the stealth–tilt frontier is the trade-off that robust high-dimensional estimators [76] are built to cut.

18 A worked example, end to end

This section carries one example through every step of the paper, with enough arithmetic that each number can be checked by hand. The truth is population 1 of the main note's drug example [8, Example "conflicting pair trials"]: two patient types with weights $(0.6, 0.4)$ and harm probabilities $a_1 = (0.5, 0.1, 0.4)$ and $a_2 = (0.05, 0.5, 0.1)$ for the harms A, B, C of three drugs. The script `engine/worked_example.py` recomputes every number below and asserts the hand checks (WE0–WE18). Table 6 collects the numbers, and fig. 13 draws them.

Step 0: the law. Each cell is $P(x) = 0.6\prod_i a_{1i}^{x_i}(1-a_{1i})^{1-x_i} + 0.4\prod_i a_{2i}^{x_i}(1-a_{2i})^{1-x_i}$. For example $P(000) = 0.6\cdot 0.5\cdot 0.9\cdot 0.6 + 0.4\cdot 0.95\cdot 0.5\cdot 0.9 = 0.162 + 0.171 = 0.333$. The eight cells, in the order $000, 001, 010, 011, 100, 101, 110, 111$ (harms A, B, C), are

$$0.333,\quad 0.127,\quad 0.189,\quad 0.031,\quad 0.171,\quad 0.109,\quad 0.027,\quad 0.013,$$

exact to three decimals. The triple-combination harm is $1 - P(000) = 0.667$.

Step 1: level one. The regulator sees the pair tables AB, BC, AC. Every law with these pair tables is $P + s\chi$, where $\chi(x) = \prod_i(2x_i - 1)$ is -1 on $000, 011, 101, 110$ and $+1$ on $001, 010, 100, 111$ (theorem F(1)). Positivity bounds s by the rarest cell of each sign: $s \le P(110) = 0.027$ and $s \ge -P(111) = -0.013$. Since $\chi(000) = -1$, the triple harm of $P + s\chi$ is $0.667 + s$, so level one predicts it in [0.654, 0.694], the main note's level-one interval. The tables glue ($\lambda^* = 1$): P itself is a global law.

Step 2: level two. The means are $\mu = (0.32, 0.26, 0.28)$; for instance $\mu_A = 0.6\cdot 0.5 + 0.4\cdot 0.05$. The pair covariances are $m_{AB} = P(11\cdot) - \mu_A\mu_B = 0.040 - 0.0832 = -0.0432$, $m_{AC} = 0.122 - 0.0896 = 0.0324$ and $m_{BC} = 0.044 - 0.0728 = -0.0288$. From $v_i^2 = m_{ij}m_{ik}/m_{jk}$,

$$v = (0.22045,\ -0.19596,\ 0.14697) = \sqrt{0.24}\,(0.45,\ -0.40,\ 0.30),$$

with the sign fixed by $v_A > 0$, the largest coordinate. So $v_T = v_A v_B v_C = -0.006349$. The triple moment is $m_{ABC} = \sum_x P(x)\prod_i(x_i - \mu_i) = 0.002592$, and the unseen coordinate is $r_0 = m_{ABC}/v_T = -0.40825 = (w_2 - w_1)/\sqrt{w_1 w_2}$. The box walls of theorem A cut the segment at $t \in [0.68892, 1.32681]$, that is $r \in [-0.76263, 0.57312]$. At the left wall the second type's A harm is 0 (weights $(0.678, 0.322)$); at the right wall the first type's B harm is 0 (weights $(0.362, 0.638)$). Along the segment the triple harm is affine (theorem A), $0.664408 - 0.006349\,r$, so level two predicts it in [0.66077, 0.66925]. The main note had reported [0.6613, 0.6687] from a grid trace of the fibre. The grid misses the walls, and the exact interval is the one given here; the main note is corrected.

Step 3: the ladder. With $\Delta_m = m_{AB}m_{AC}m_{BC} = 4.031\times 10^{-5} > 0$ and $\mathrm{Det} = m_{ABC}^2 + 4\Delta_m = 1.680\times 10^{-4}$, the main note's inverse formula gives $w_1 w_2 = \Delta_m/\mathrm{Det} = 0.24$, which is $0.6\cdot 0.4$. The law sits on the top rung of the ladder, a genuine mixture.

Step 4: the tower. In the posterior tower [7] the outcome x lies in $X = \{0,1\}^3$ and the law P in $\mathcal{D}X$. The mixture $0.6\,\delta_{\Pi(a_1)} + 0.4\,\delta_{\Pi(a_2)}$ is a point of $\mathcal{D}^2X$, a law over laws. The pair tables pin down only a segment of such points. After a triple with n draws per window, the regulator's uncertainty over the segment is approximately $N(\hat{r}, \tau^2/n)$, a point of $\mathcal{D}^3X$. Which type is called "1" is a label: swapping it sends (r, v) to $(-r, -v)$ and changes no prediction.

Step 5: gluing, and a per-window contrast. The windows AB, BC, AC overlap in single variables, and their nerve is a hollow triangle. The genuine tables agree on the overlaps and glue. For contrast, the main note's conflicting trials replace the BC table by population 2's (weights $(0.25, 0.75)$). The B margin is then 0.26 in one table and 0.40 in the other, the linear gap is 0.28, and $\lambda^* = 0.86$. That is a direct influence on the bottom rung of the ladder, not contextuality. Whole-record injection can never produce it (theorem E(1)).

Step 6: the Stanley–Reisner complex. Over an independent law the fibre is the Stanley–Reisner scheme of the complex on $\{w_1, w_2, d_1, d_2, d_3\}$ with facets $\{w_1, d_1, d_2, d_3\}$, $\{w_2, d_1, d_2, d_3\}$ and $\{w_1, w_2, d_i\}$ [8]. Its ideal, with $w_2 = 1 - w_1$, has the reduced Gröbner basis $\{w_1(1-w_1)d_id_j : i < j\}$, and it equals $(w_1) \cap (w_2) \cap (d_1, d_2) \cap (d_1, d_3) \cap (d_2, d_3)$; the script checks both with SymPy. The truth has $w_1w_2 = 0.24$ and $d = a_1 - a_2 = (0.45, -0.40, 0.30)$, all non-zero, so it lies on no facet. No variable is coincident, and the triple is a probe of positive value (theorem C).

Step 7: the decision. The regulator approves either the triple combination T, with $u_T = 1 - [\text{any harm}]$, or the pair AB, with $u_P = 0.872 - [H_A \vee H_B]$, subject to $u_0 = 0.30$. The pair's harm is observed: $P(H_A \vee H_B) = 1 - P(00\cdot) = 0.540$, so $U_P = 0.332$ on the whole segment. The triple's value is the line $U_T(r) = 0.335592 + 0.006349\,r$.

- *Wall checks* (theorem D(1)). The worst value of T is at the left wall, 0.33075; the worst value of P is 0.332. Both meet u_0, and the set rule picks the pair.
- *The oracle.* At the truth $U_T = 0.333 > 0.332$, so the triple is better. The two are equal at $r_c = -0.5658$, where the triple harm is 0.668: the triple is worth its premium only while its harm stays below 0.668.
- *Info-gap* (theorem D(2)). After the probe, with nominal point r_0, both acts are admissible on the whole segment, so both horizons are infinite: a saturation tie, broken by the nominal value in favour of T.
- *Value of the probe* (theorem C). Wald's gap is 0: $\min_r \max_a U = \max_a \min_r U = 0.332$, a saddle point. Savage's regret is not: T's largest regret is $0.332 - 0.33075 = 0.00125$ at the left wall, and P's is $0.33923 - 0.332 = 0.00723$ at the right wall. So Savage already prefers T before the probe, and the regret value of the probe is 0.00125.

Step 8: the probe and settling. The tempo constant of theorem B is $\tau = 12.71$, so the post-probe half-width $z\tau/\sqrt{n}$ is 0.249 at $n = 10^4$ and 0.079 at $n = 10^5$. The settling radius (theorem D(4)) is the harm gap over the slope, $d = (0.333 - 0.332)/0.006349 = 0.1575$, so the decision settles on the triple at median size $n^* = (1.96 \cdot 12.71/0.1575)^2 \approx 25\,000$.

Step 9: the vertex attack. Before the probe the set rule's margin is only $0.332 - 0.33075 = 0.00125$. By theorem E(3) the cheapest single record to inject is 111, at a first-order budget of 0.186%; the exact budget is 0.187%. Injected 111 records raise $P(H_A \vee H_B)$, lowering U_P to 0.33114, and move the left wall inward from -0.763 to -0.729, raising T's worst value to 0.33114. A tiny injection flips the decision because the decision sits next to a tie.

Step 10: the hidden channel. The pair's sponsor, who wants the triple rejected, injects records from $Q = P + 0.027\chi$, the end of the positivity segment. In cell order, Q is

$$0.306, \quad 0.154, \quad 0.216, \quad 0.004, \quad 0.198, \quad 0.082, \quad 0, \quad 0.040.$$

Every pair table is unchanged, and the apparent triple harm becomes $0.667 + 0.027\,\varepsilon$ (theorem F(2)); in r units the shift is $\varepsilon s/v_T = -4.25\,\varepsilon$. The decision flips when the apparent harm reaches 0.668, at $\varepsilon^* = 0.001/0.027 = 0.037$. The triple contradicts the pairs only when the apparent harm leaves the hiding segment at 0.66925, at $\varepsilon_{\text{det}} = 0.00225/0.027 = 0.083$. With $n = 10^5$ records per window the share choosing the triple crosses one half already at $\varepsilon_{1/2} = (0.1575 - 0.0788) \cdot 0.006349/0.027 = 0.0185$. Bisection on the regulator's pipeline gives the same ε^* and ε_{det}.

Step 11: one line, and the ladder under injection. In harm units the line of theorem H(1) is the interval $[0.654, 0.694]$. Its model part is $[0.66077, 0.66925]$, and on it sit the truth (0.667), the decision boundary (0.668) and the detection boundary (0.66925). The decision boundary is nearer, so the stealth window $[0.037, 0.083)$ exists (proposition 4). The hidden channel leaves the pair covariances, and so $\Delta_m > 0$ and $\text{Det} > 0$, unchanged: the law never leaves the top rung, and the only boundary it crosses is the box. Moving to the signed rung needs a pair covariance through zero; the cheapest single record for that is 011, at 5.4%. With three variables there are no model equations, so theorem G has nothing to test; with a fourth variable a second triple would see the channel (theorem H(2)).

Step 12: what breaks. Past ε_{det} the post-probe fibre is empty: no two-type explanation in the box matches the pairs and the triple. The complex of Step 6 does not change, because it belongs to the model, not to the data. A product injection, in contrast, is a third type (theorem E(4)), and explaining it needs a three-type model with a larger complex; that case is open.

Step 13: the toric reading. The line of Step 1 is also an object of toric geometry, and the companion paper on coupling polytopes [1] supplies the dictionary. Every number below is checked exactly (WE14–WE18); the cells of P are rationals with three decimals, so the polynomials have rational coefficients. Figure 14 draws the picture.

- *The fibre is a Delzant segment, and its manifold is* $\mathbb{CP}^1$. Fixing all three two-way margins of a $2 \times 2 \times 2$ table gives a boundary fibre of shape $2 \times 2 \times 2$ [1, Prop. 10.5(i)]: a capacitated transportation polytope of dimension $(2-1)^3 = 1$. Its tangent lattice is $\mathbb{Z}\chi$, and χ has entries ± 1, so it is primitive. A segment is Delzant [2], and the compact toric symplectic manifold with this moment polytope is $\mathbb{CP}^1$.
- *The capacities are the Kähler data.* The two facets are the endpoints, $\{p_{110} = 0\}$ at $s_+ = 0.027$ and $\{p_{111} = 0\}$ at $s_- = -0.013$. With the truth as base point, the facet values are 0.027 and 0.013, so $[\omega] = 0.027\,[D_+] + 0.013\,[D_-]$ [1, Prop. 2.2]. Its degree, the symplectic area, is the lattice length 0.040. The channel capacities of corollary 2 are exactly these facet values.

- *The moment map is the third cumulant.* The moment map is $\mu([z_0 : z_1]) = -0.013 + 0.040\,|z_1|^2/(|z_0|^2 + |z_1|^2)$. Its fibres are the latitude circles, one for each law with these pair tables, and its two fixed points are the poles. In the coordinate of the line, $m_{ABC}(P + s\chi) = 0.002592 + s$ exactly, so the moment coordinate is the third cumulant up to a constant. Pair data see only second cumulants [1, Props. 9.1–9.2]; the hidden channel is the flow along the circle action, and it moves only the third.

- *The adversary sits at a fixed point.* The tilt of a linear objective is largest at a vertex of the polytope, which is the image of a torus-fixed point [1, Prop. 6.4]. That is why the extremal injection Q of Step 10 has a zero cell ($p_{110} = 0$): it is the north pole.

- *The no-three-way-interaction model meets the line once: the Birch point.* The log-linear model without three-way interaction is the toric quartic $p_{000}p_{011}p_{101}p_{110} = p_{001}p_{010}p_{100}p_{111}$ [3]. On the line the s^4 terms cancel, and the restriction is the cubic

$$-s^3 - \tfrac{27}{1250}s^2 - \tfrac{37971}{5\,000\,000}\,s - \tfrac{143613}{6\,250\,000\,000}.$$

  It has one real root in the segment, $s = -0.003048$, and two complex roots $-0.00928 \pm 0.0863i$; the fourth point of intersection that Bézout's theorem requires is at infinity. By Birch's theorem the real root is the unique maximum-entropy law with these pair tables, the analogue of the gluing filler of [1, Prop. 10.2]. Its triple harm, 0.6640, lies inside the hiding segment and below the decision boundary 0.668. A regulator who adopts it approves the triple, the right act here, and is immune to the hidden channel, which leaves the pair tables and hence the Birch point unchanged. But it gives up the probe: had the truth been on the other side of 0.668, the Birch regulator would not know. At the truth the three-way odds ratio is 0.569.

- *The hyperdeterminant is never reached.* Cayley's hyperdeterminant of the $2 \times 2 \times 2$ table, a quartic [4], restricts to the line as exactly

$$\mathrm{Det}(s) = s^2 + \tfrac{81}{15625}\,s + \tfrac{6561}{39\,062\,500} = m_{ABC}^2 + 4\Delta_m,$$

  the main note's formula, with roots $-0.002592 \pm 0.012698\,i$. So $\mathrm{Det} > 0$ on the whole line, and every law with these pair tables has real tensor rank two: the line meets the rank boundary only at complex points, and the hidden channel never leaves the top rung of the ladder (Step 11). The other two intersections are at infinity.

- *A wall nearby.* At the north pole the cell 011 has only 0.004 left. Pair tables slightly different from these would make p_{110} and p_{011} vanish at the same vertex, a degenerate vertex: the analogue of the walls $f_i = g_j$ of the 2×3 family [1, Thm. 3.1, Prop. 3.3]. Away from such walls the capacities are affine in the pair tables; across one, the binding cell changes and the capacity has a kink.

- *The monotone fibre.* At the midpoint $s = 0.007$ (harm 0.674) both facet values equal 0.020. There the disc potential $W = T^{\lambda_+}z + T^{\lambda_-}z^{-1}$ has its critical points, so by Cho and Oh [5] the equator is the only torus fibre with non-vanishing Floer cohomology. As in [1, Example 6.2(a)], the monotone point is not the maximum-entropy point. We attach no statistical meaning to it.

One caution about vocabulary: "level two" means couplings of random measures in [1] and mixtures of product laws here. The two are related through flattening, but they are not the same object.

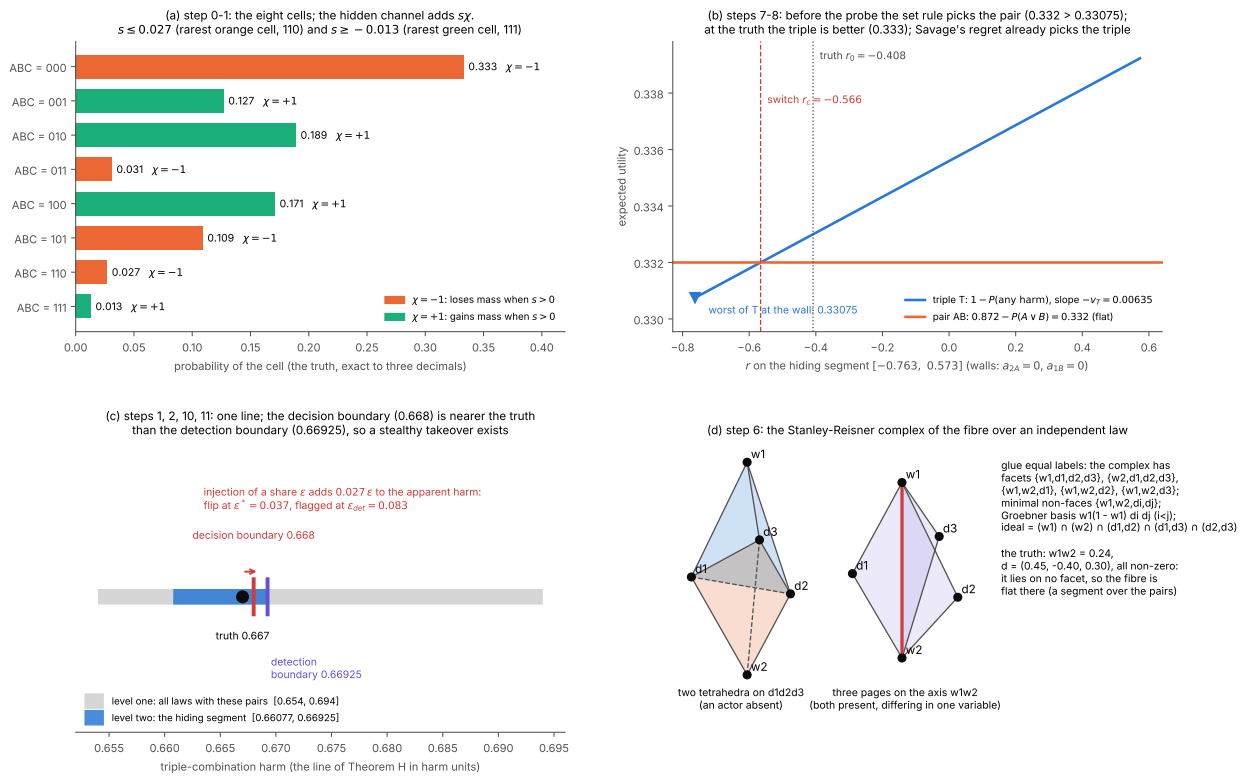


Figure 13: The worked example. (a) The eight cells and the sign of the top Walsh character; the channel's capacity is set by the rarest cell of each sign. (b) The decision on the hiding segment: before the probe the set rule picks the pair, while at the truth the triple is better. (c) The line of theorem H in harm units: level one (grey), level two (blue), the truth, the decision boundary and the detection boundary. (d) The Stanley–Reisner complex of the fibre over an independent law, drawn as two tetrahedra and a three-page book glued along equal labels.

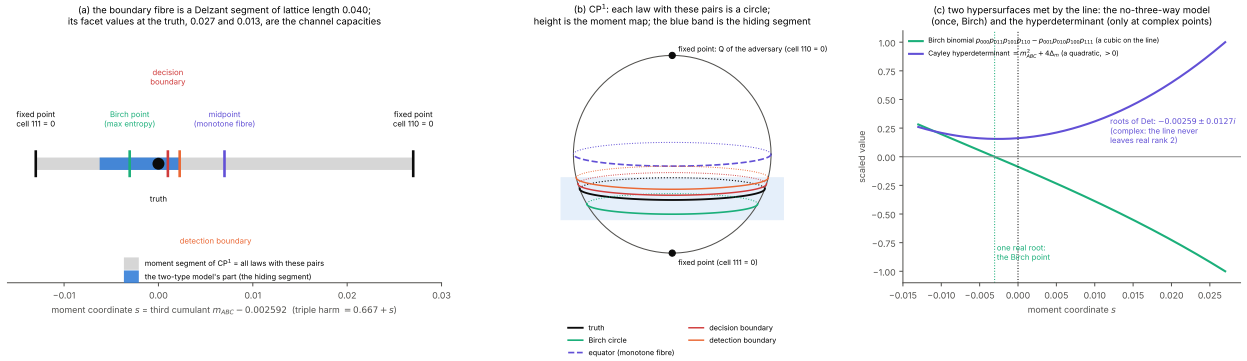


Figure 14: Step 13, the toric reading of the worked example. (a) The moment segment of $\mathbb{CP}^1$ in the coordinate s (the third cumulant, shifted): its two fixed points, the truth, the Birch point, the monotone midpoint, the hiding segment and the two boundaries. (b) The same data on the sphere: each law with these pair tables is a latitude circle. (c) The two hypersurfaces restricted to the line: the no-three-way quartic becomes a cubic with one real root (the Birch point); Cayley's hyperdeterminant becomes $m_{ABC}^2 + 4\Delta_m$, positive throughout.

quantity	formula	value
triple harm	$1 - P(000)$	0.667
channel capacity	$P(110)$, $P(111)$	$s \in [-0.013, 0.027]$
level-one interval	$0.667 + s$	$[0.654, 0.694]$
pair covariances	$P(11) - \mu_i\mu_j$	-0.0432, 0.0324, -0.0288
triple moment, v_T	$\sum_x P(x)\prod(x_i - \mu_i)$; $\prod v_i$	0.002592; -0.006349
unseen coordinate	$r_0 = m_{ABC}/v_T$	-0.408
hiding segment	box walls	$r \in [-0.763, 0.573]$
level-two interval	$0.664408 - 0.006349\,r$	$[0.66077, 0.66925]$
rung	$w_1w_2 = \Delta_m/\mathrm{Det}$	0.24 (top)
decision	worst values; oracle	pair before, triple after
regret value of the probe	$\min_a$ largest regret	0.00125 (Wald: 0)
settling	$d = 0.001/\lvert v_T\rvert$; $(z\tau/d)^2$	0.1575; $n^* \approx 25\,000$
vertex attack	margin / steepest influence	record 111, 0.19%
hidden channel	$(0.668 - 0.667)/0.027$; $(0.66925 - 0.667)/0.027$	$\varepsilon^* = 3.7\%$; $\varepsilon_{\mathrm{det}} = 8.3\%$
rung change	cheapest record	011, 5.4%
toric fibre	boundary fibre $2 \times 2 \times 2$	Delzant segment, $\mathbb{CP}^1$, length 0.040
Birch point	one real root of the cubic	$s = -0.00305$, harm 0.6640
hyperdeterminant	$m_{ABC}^2 + 4\Delta_m$ on the line	roots $-0.0026 \pm 0.0127i$

Table 6: The worked example in one table. Every value is recomputed by `engine/worked_example.py`.

Step 14: the descent complex and the filtration, by hand. Theorem M (section 24) says that the degrees and groups of every obstruction come out of a construction. On the drug example the whole construction fits on a page (fig. 15).

The complex. The contexts are AB, BC, AC, their overlaps B, A, C, and the empty context, whose single cell carries the total mass. The nerve is a hollow triangle: three vertices and three edges, but no 2-simplex, because $AB \cap BC \cap AC = \emptyset$. The Čech complex of signed measures is

$$C^0 = \mathbb{R}^4 \oplus \mathbb{R}^4 \oplus \mathbb{R}^4 \xrightarrow{d^0} C^1 = \mathbb{R}^2 \oplus \mathbb{R}^2 \oplus \mathbb{R}^2 \xrightarrow{d^1} C^2 = \mathbb{R},$$

with dimensions 12, 6 and 1. d^0 subtracts the two restrictions of a family to each overlap (fig. 15(b)). It has rank 5, and d^1 has rank 1. Hence

$$H^0 = 12 - 5 = 7, \qquad H^1 = (6 - 1) - 5 = 0, \qquad H^2 = 1 - 1 = 0.$$

The 7 compatible signed families are 3 single rates, 3 pair covariances and the total mass. The marginal map from the 8 cells of a signed law onto them has rank 7. The one direction it loses is χ_{ABC}, the line s of Step 1. So the fibre of Step 1 is exactly the kernel of descent, and nothing obstructs descent algebraically (check WD1–WD3).

The filtration, stage by stage.

- D_1, *gluing.* The pair tables come from one law, so $\lambda^* = 1$. The liar-cycle inequality $P(A{=}B) + P(B{=}C) + P(A{\neq}C) \le 2$ has value 1.40 here. The liar cycle itself has $\lambda^* = 0$ and value 3.
- D_2, *signs.* The sign cochain on the graph Γ of pairs has cycle product

$$M_{01}M_{02}M_{12} = (-0.0432)(0.0324)(-0.0288) = 4.03 \times 10^{-5} > 0,$$

so the class in $H^1(\Gamma;\mathbb{Z}/2)$ is 0 and sign $v = (+,-,+)$ solves it. The other solution, $(-,+,-)$, is the label switch: the kernel $H^0(\Gamma;\mathbb{Z}/2) = \mathbb{Z}/2$ (fig. 15(c)). The "exactly one harm" law, uniform on 100, 010, 001, has $M_{ij} = -1/9$ on every edge. Its cycle product is $-1/729 < 0$, so it has class 1, and no two-type explanation exists although its tables glue.

- D_3, *moduli.* With three edges and three vertices the unsigned incidence matrix is invertible, so coker $B = 0$ and the moduli are solved, not tested:

$$|v_0| = \sqrt{\frac{M_{01}M_{02}}{M_{12}}} = 0.2205, \qquad |v_1| = 0.1960, \qquad |v_2| = 0.1470,$$

  which is $\sqrt{0.24}\,(0.45, 0.40, 0.30)$. This is why three variables carry no model equation.

- D_4, *walls.* The walls cut $t \in [0.689, 1.327]$, so the segment $R = [-0.763, 0.573]$ is non-empty. A witness for this stage is a joint law ($\mu = (0.382, 0.459, 0.463)$) with both classes zero whose walls cross ($t_{lo} = 1.095 > t_{hi} = 1.075$): signs and moduli solve, but no box of probabilities holds the types.

- D_5, *the triple.* $\hat{r} = r_0 = -0.408$ lies inside R. The hidden channel moves $\hat{r}$ along the kernel of descent. Past $\varepsilon_{\text{det}} = 0.0833$ it leaves R; at $\varepsilon = 0.10$, $\hat{r} = -0.834$, and the triple contradicts the pairs (check WD8).

Labels. For one replicate of Study 13's design A1b ($n = 1000$ per batch), the batches' triple moments form a 0-cochain on a simplex nerve (fig. 15(d)). Its coboundary has energy $24 \times (1.48 \times 10^{-5} + 1.12 \times 10^{-4})$, split into between-source and within-source parts. Source C's statistic is 1.53, under $c_s = 2.39$: the adversary's four shifted batches, 0.0052 each, sit inside the band.

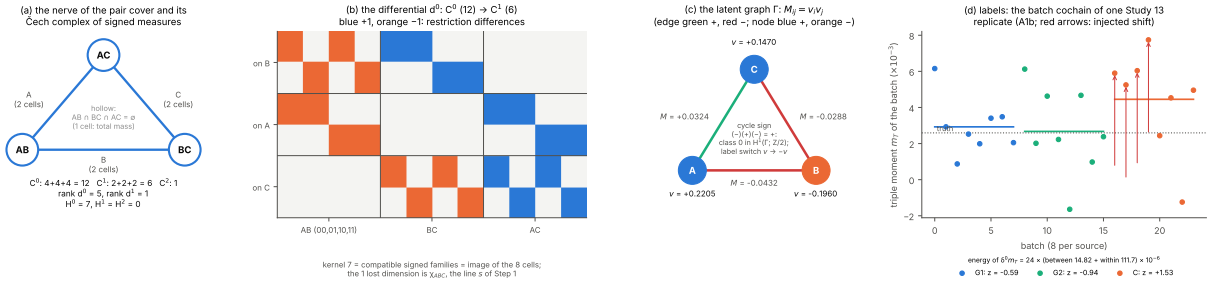


Figure 15: The descent complex of the drug example. (a) The nerve of the pair cover, a hollow triangle, with the dimensions and ranks of its Čech complex. (b) The differential d^0 as a matrix: each row compares the two restrictions to one overlap. (c) The latent graph Γ: pair covariances on the edges and v on the nodes. The cycle's sign is $+$, so $M = v_i v_j$ is solvable, uniquely up to the label switch. (d) The labelled layer: the triple moments of the 24 batches of one Study 13 replicate, with source means. The injected shifts (arrows) stay under the source threshold.

19 The example with four variables, in stages

Add a fourth harm D to the example of section 18, with probabilities 0.2 and 0.7 in the two types; everything else is unchanged. The computation is staged so that each stage adds one layer of structure, and so that the three-variable interactions show up one face at a time. The script `engine/worked_example4.py` recomputes every number and asserts it (checks W4-1 to W4-14); fig. 16 draws the stages.

Stage 1: the law and its four faces. The sixteen cells are exact four-decimal rationals, from $P(0000) = 0.1809$ to $P(1111) = 0.0031$. Each triple of variables is a face: a $2 \times 2 \times 2$ table with its own line, capacities, Birch point and hyperdeterminant, as in Step 13. The ABC face is the three-variable example exactly. All four faces carry the same unseen coordinate, $r_S = m_S/v_S = -0.408$ for every triple S, because there is one pair of types behind them. Each face's hyperdeterminant $m_S^2 + 4\Delta_S$ is positive on its whole line.

face	m_S	v_S	capacities	Birch point s	three-way odds ratio	r_S
ABC	0.002592	−0.006349	0.027, 0.013	−0.00305	0.569	−0.408
ABD	−0.00432	0.010582	0.027, 0.013	0.00233	1.449	−0.408
ACD	0.00324	−0.007936	0.0486, 0.0254	−0.00557	0.522	−0.408
BCD	−0.00288	0.007055	0.0252, 0.0188	0.00087	1.132	−0.408

Stage 2: three-way interactions. Along each face's line the log three-way odds ratio decreases strictly, and it is zero at the face's Birch point. So a face's three-way interaction says on which side of its Birch point the truth lies: an odds ratio below one means a larger triple moment than the Birch point's, as for ABC and ACD (0.569 and 0.522), and above one a smaller, as for ABD and BCD (1.449 and 1.132). The two-type model ties the four together through the single r, but they are not alike. The four-way odds ratio of the truth is 0.507.

Now remove every interaction of order three or more at once. The global maximum-entropy law with the six pair tables, the pairwise log-linear law found by iterative proportional fitting, is the four-variable Birch point. Its faces are *not* their own Birch points: marginalising it induces a small negative three-way interaction on every face, with odds ratios 0.977, 0.985, 0.960 and 0.965. A law with no three-way interaction among four variables still shows three-way interaction among three of them. The three-variable maximum-entropy answer of Step 13 and the four-variable one therefore differ: the ABC triple moment is -0.000456 at the face's Birch point and -0.000336 under the global one.

Stage 3: the pair fibre is a smooth toric 5-fold. All laws on $\{0,1\}^4$ with the six pair tables form a polytope Π of dimension $16 - 1 - 4 - 6 = 5$. Its tangent lattice is the set of integer tables with zero one- and two-way margins. The four triple characters and the four-way character span a sublattice of index 16 in it, because sums such as $(\chi_{ABC} + \chi_{ABD})/2$ are already integral; a saturated basis with entries $0, \pm 1$ exists. In that lattice:

- Π has 20 vertices, found by exact enumeration, and 9 facets. Only nine of the sixteen cells give facets, namely 0011, 0110, 0111, 1001, 1011, 1100, 1101, 1110 and 1111; the other seven can never reach zero with these pair tables.
- Every vertex lies on exactly five facets (simple), and at every vertex the five facet normals form a lattice basis (smooth). So Π is Delzant, and its toric manifold X_Π is a smooth compact toric 5-fold [2] with Euler characteristic 20, the number of torus-fixed points.
- The f-vector is $(20, 50, 54, 31, 9, 1)$, and the h-vector is $(1, 4, 5, 5, 4, 1)$. For a smooth projective toric variety the h-vector of the simple polytope gives the even Betti numbers, so the Poincaré polynomial of X_Π is $1 + 4t^2 + 5t^4 + 5t^6 + 4t^8 + t^{10}$. It is palindromic (Poincaré duality), and $b_2 = 4$ is the number of facets minus the dimension.
- At the truth the facet values are the probabilities of the nine facet cells, so these nine numbers are the Kähler data, $[\omega] = \sum_c P(c)\,[D_c]$ [1, Prop. 2.2]. The lattice volume of Π, which is the symplectic volume of X_Π, is 1.27×10^{-8}.

- Projecting Π to two faces' triple moments is the moment map of a two-dimensional subtorus. For five of the six pairs of faces the image is the product of the two faces' lines, a rectangle: those two three-way structures can be chosen independently. For ACD and BCD, which share the pair CD, the image is a pentagon, a rectangle with one corner cut by 0.002. When both channels are pushed to their extremes at once they compete for the same rare cells, and the two faces' three-way moments are coupled. This is the corner-cutting of the coupling paper's 2×3 family, here appearing inside a projection.

The toric picture does not always survive. Moving the fourth harm (Stage 6) shows smooth chambers ($a_{2D} = 0.5$ and 0.7), a wall where Π is not simple ($a_{2D} = 0.6$: 17 vertices, some on six facets), and, at $a_{1D} = 0.3$, a chamber where Π is simple but some vertices have index 2: the moment polytope of a toric orbifold [6], not of a manifold. In 50 random laws and random two-type truths the fibre was always simple and never smooth (check W4-14). This answers, for the four-variable pair fibre, the question of [1, Prop. 10.5] of where the toric picture stops: it depends on the chamber.

Stage 4: the model is a conic arc. Inside the five coordinates (four triple moments and the four-way moment) the two-type laws with these pair tables satisfy three linear equations, $m_S/v_S = m_{S'}/v_{S'}$, and one quadric, $m_{ABCD} = v_{ABCD}(r^2 + 1)$ with $v_{ABCD} = 0.001555$. So the hiding segment is an arc of a plane conic in the 5-dimensional polytope; on any two faces it projects to a straight segment through the origin. The fourth harm's own box walls allow $t \in [0.408, 1.633]$, which contains the three-variable range $[0.689, 1.327]$. So here the walls are slack, the segment is the same $r \in [-0.763, 0.573]$, and the ABC harm interval stays $[0.66077, 0.66925]$. A more extreme fourth harm would cut it.

Stage 5: the decision and the hidden channel with four variables. The decision of Step 7 is unchanged: the pair before the probe, the triple at the truth. Three numbers change.

- *The channel is narrower.* The ABC channel is now $P + s\chi_{ABC}$ on sixteen cells, and its capacity is the rarest sixteen-cell probability with the needed sign, 0.0099 (cells 1101 and 1110). It moves the face by $2s$ (theorem F(4)), so the capacity in face units is 0.0198, against 0.027 with three variables. The flip needs $\varepsilon^* = 0.001/0.0198 = 5.05\%$ (against 3.7%), and the triple contradicts the pairs at $0.00225/0.0198 = 11.4\%$. The whole fibre still allows m_{ABC} anywhere in the three-variable line $[-0.0104, 0.0296]$, but only by moving other faces too.

- *A second triple catches it.* The channel moves $\hat{r}_{ABC}$ and leaves the other three at -0.408 (corollary 3). With 10^5 records per window, the standard deviation of $\hat{r}_{ABC} - \hat{r}_{ABD}$ is 0.047, so a two-sided test at level 0.05 fires once $3.12\,\varepsilon > 1.96 \cdot 0.047$, at $\varepsilon = 3.0\%$, before the flip at 5.05%.

- *Staying on the model is expensive.* An adversary who keeps every pair table and every model equation must make the observed law the two-type law at some r. The least share that reaches the switch point is exactly $1 - \min_x P_{r_c}(x)/P(x) = 18.3\%$; the walls follow at 42.2%. With three variables the same formula gives 3.7%, the hidden channel itself.

So the fourth variable does not change what is hidden, but it changes what hiding costs: from 3.7% and invisible to 5.05% and caught at 3.0%, or 18.3% to stay invisible.

Stage 6: chambers. The table of Stage 3's last paragraph: $a_{2D} = 0.5$ and 0.7 give smooth fibres with 20 vertices and 9 facets; $a_{2D} = 0.6$ lies on a wall, with 17 vertices and some vertices on six facets; $(a_{1D}, a_{2D}) = (0.3, 0.7)$ gives 28 vertices, 10 facets and vertex index 2.

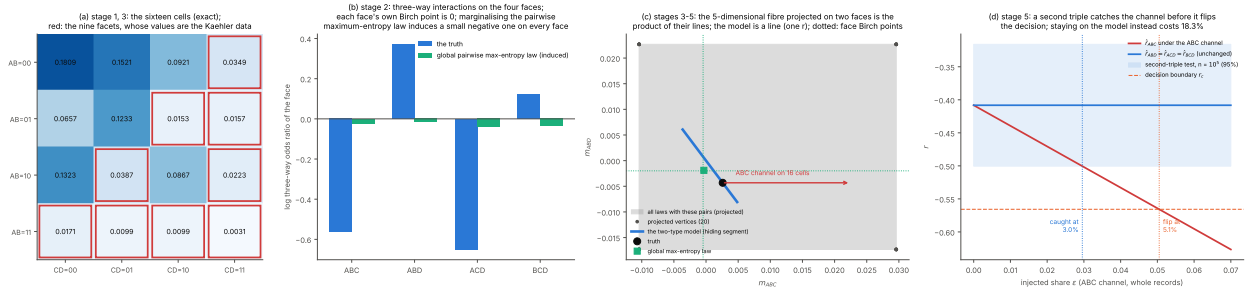


Figure 16: The example with four variables. (a) The sixteen cells; red outlines mark the nine facets of the pair fibre, whose probabilities are the Kähler data. (b) Three-way interactions on the four faces: the truth's, and those induced on the faces by the global pairwise maximum-entropy law. (c) The 5-dimensional fibre projected on the ABC and ABD triple moments (a rectangle, the product of the two faces' lines), the two-type model, the truth, the global maximum-entropy law, and the ABC channel. (d) Under the ABC channel only $\hat{r}_{ABC}$ moves; a second-triple test with 10^5 records per window fires at 3.0%, before the flip at 5.05%.

Stage 7: the filtration with four variables. Two stages that are empty at $m = 3$ become tests at $m = 4$ (fig. 17).

- D_3. coker B now has dimension 2, spanned by the two log tetrad ratios $\log(M_{01}M_{23}/M_{02}M_{13})$ and $\log(M_{02}M_{13}/M_{03}M_{12})$. Both are 0 on the truth. Injecting a product law at $\varepsilon = 0.1$ (theorem E(4)) keeps every cycle sign but moves them to -1.65 and -0.49.
- D_5. The four triple windows give a 0-cochain r_S, and the model requires $\delta r = 0$. On the truth all four equal 0.408. This is r_0 of Step 2 up to sign: the types are relabelled so that $v_A > 0$, which is the label switch of D_2 in action. Half-capacity χ_{ABC} injection at $\varepsilon = 0.05$ moves r_{ABC} to 0.486 and leaves the other three at 0.408. The obstruction moves up exactly one level (check WD10).

Every stage now has a witness that passes the earlier stages and fails there. Each stage also computes a value that the earlier ones did not:

- the gluable fraction;
- the cycle sign;
- the solved moduli, or the tetrads;
- the segment;
- the coherence of r_S;
- the between-source energy.

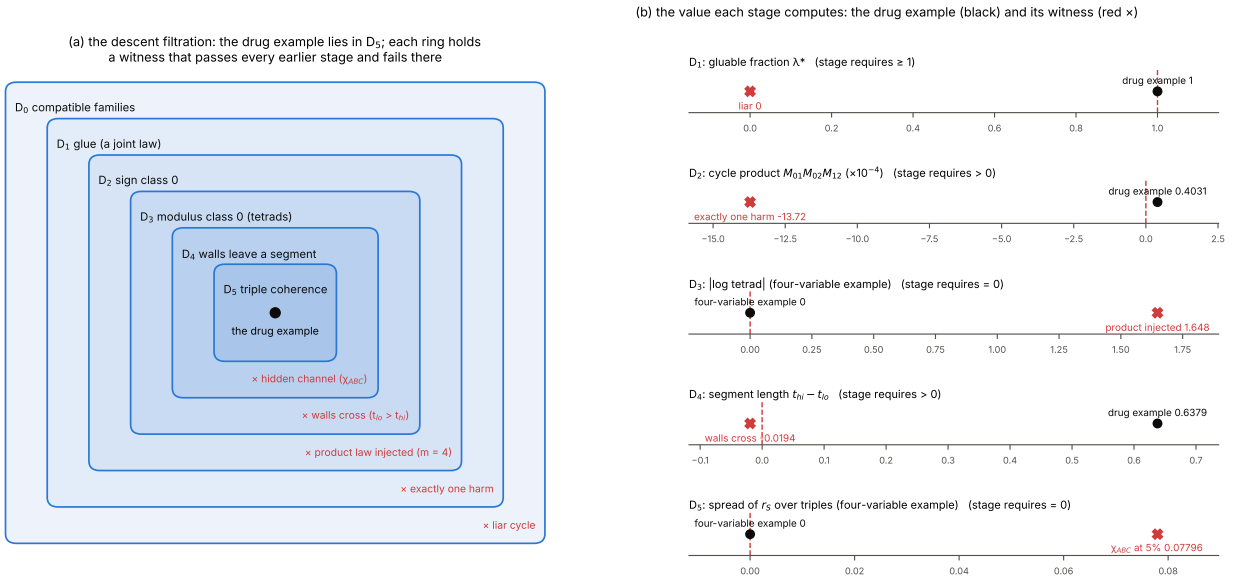


Figure 17: The descent filtration on the worked examples. (a) The nested stages $D_0 \supset \cdots \supset D_5$. The drug example lies in D_5, and each ring holds a witness that passes every earlier stage and fails there. (b) The value each stage computes, for the example (black) and its witness (red): λ^*; the cycle product; the log tetrads (four variables); the segment length; the spread of r_S over the triples (four variables). The dashed line is the stage's requirement.

20 Part II, continued. Sources: a trusted source that is sometimes adversarial

Theorems E–H let the adversary inject into the whole pool, and the regulator sees only the pooled tables. In practice data arrive from *sources*, in *batches*, and the decision maker knows where each table came from and nothing else about provenance. The dangerous case is not a hostile source, which can be dropped once it is identified. It is a *trusted* source that is genuine most of the time and adversarial in some batches, and that adds just enough to tilt the decision or to delay it. Writing such a source off throws away its genuine data. Trusting it lets the adversary in. This section shows what the labels buy, using the worked example of section 18 as the truth.

Sources, batches and sections. There are three sources, G_1, G_2 and C, each sending K_C batches. A batch is the triple window (the probe) with n records and the three pair windows with ρn records each, since pair tables come from large registries. Every table is a *section* over its context, labelled by (source, batch). All sections come from one population, so they must glue. Source C is genuine in most batches. In some of them a share ε_b of the records is drawn from the hidden channel $Q = P + s\chi_T$ of theorem F. That leaves every single and pair table unchanged and moves the triple moment by exactly

$$m_T\big((1-\varepsilon_b)P + \varepsilon_b Q\big) = m_T(P) + \varepsilon_b s,$$

because the coefficient of χ_T in the centred triple moment is $\prod_i \sum_{x_i} (2x_i - 1)(x_i - \mu_i) = 1$. The pair sections therefore always glue, and the triple sections can fail to glue only along one coordinate, m_T. The gluing test is a test on that coordinate.

Monad and comonad. The provenance comonad of section 12 now has observable labels: $W = L \times (-)$ with $L = \{\text{sources}\} \times \{\text{batches}\}$. The family of batch laws is an element $\Pi = \frac{1}{K}\sum_b \delta_{P_b}$ of $D(DX)$, where D is the distribution monad. *Pooling is the monad's multiplication* $\mu : D(DX) \to DX$, which averages the batch laws. The data come from one population exactly when Π lies in the image of the unit $\eta : DX \to D(DX)$, that is, when all batch laws are equal: this is the gluing condition. Injection in some batches moves Π off the image of η. Since $\mu \circ \eta = \mathrm{id}$, pooling cannot tell Π from $\eta(\mu\Pi)$: the adversary's footprint lies in the fibre of μ. A rule that pools factors through the counit of W, which forgets the labels. A rule that tests gluing is a coKleisli arrow $WX \to A$ that does not factor through the counit. The comonad supplies the labels that make the monad's inner layer visible. As in section 12, this framing names the situation; the results come from the following statistics.

The gluing filter. Write $\hat{m}_b$ for batch b's triple moment, σ_m^2 for the variance of its influence function (per record), and $\sigma_b = \sigma_m/\sqrt{n}$. The filter tests two covers, the fine one first.

1. *Batches.* $z_b = (\hat{m}_b - \tilde{m})/(\sigma_b\sqrt{1 + 1/K_C})$, where $\tilde{m}$ is the median of the three source means. Triple sections with $|z_b| > c_b$ are dropped.

2. *Sources.* For each source j, Z_j compares its pooled triple moment (kept batches) with that of the other two. If the largest $|Z_j|$ exceeds c_s, that source's remaining triple sections are dropped.

Pair sections are always kept, and the decision is taken on the colimit of the sections that remain. The median reference matters. With one compromised source, the median lies between the two genuine source means, so a strongly shifted source cannot pull genuine batches out of the pool. A leave-one-out reference can: it drops genuine batches on the far side (swamping), and that biases the pool toward the adversary. The Gaussian model showed this before the filter was fixed.

Theorem J (sources and bands)**.** *Let source C hold a share π of the triple records, and let the other sources be genuine.*

(1) Pooling. *A rule that pools faces a whole-pool injection of share $\pi\bar{\varepsilon}_C$, where $\bar{\varepsilon}_C$ is C's average share. All of theorem F's budgets apply, divided by π. For two triple sections with shares ε and ε', the gluable fraction is $\lambda^* = 1 - |\varepsilon - \varepsilon'|\,\mathrm{TV}(P, Q)$. Hence $1 - \lambda^*$ is a lower bound on the difference in contamination (Tier 1).*

(2) The band. *If C's source statistic has mean a, then C moves the pooled triple moment by exactly*

$$a\sqrt{\pi/(1-\pi)} \quad \textit{pooled standard errors},$$

whatever K_C, K and n are. An adversary that keeps its exposure risk at most β has $a \le c_s - \Phi^{-1}(1-\beta)$. It therefore moves the pooled estimate by at most

$$B = \big(c_s - \Phi^{-1}(1-\beta)\big)\sqrt{\pi/(1-\pi)}$$

standard errors, at every sample size. *Its per-batch share shrinks like $1/\sqrt{n}$: the adversary rides inside the noise band.*

(3) No tilt. *Let d be the settling radius (theorem D) and $\kappa = \sigma_m/(|v_T|\tau) \le 1$. Under the filter, such an adversary settles the decision on the wrong act with probability at most $\Phi\big(\kappa B - z - d\sqrt{N}/\tau\big) \le \Phi(B - z)$, up to the genuine rate. This tends to 0 as N grows. Under pooling, a fixed share of C with $\pi\bar{\varepsilon}_C \in (\varepsilon^*, \varepsilon_{\mathrm{det}})$ settles on the wrong act with probability tending to one.*

(4) Delay. *The sample size at which the decision settles half the time moves from* $(z\tau/d)^2$ *to at most*

$$N_{1/2} \le \big((z\tau + B\sigma_m/|v_T|)/d\big)^2 \approx \big((z+B)/z\big)^2 (z\tau/d)^2.$$

(5) Localisation. *A gluing failure can be localised down to the finest cover whose sections are less noisy than the shift. A shift* Δ *in one batch is seen at the batch level once* $\Delta \gtrsim c_b\sigma_b$*. The same total shift spread over* k *batches is invisible there when* $\Delta/k < c_b\sigma_b$*, and only the source level sees it. With two sources a failure cannot be localised, since* Z_1 *and* Z_2 *are opposite in sign and proportional in size. With three, the median makes the batch test immune to swamping by a single compromised source.*

Proof. (1) μ is linear, so the pooled law is $(1-\pi\bar{\varepsilon}_C)P + \pi\bar{\varepsilon}_C Q$. For two laws on one context, $\lambda^* = \sum_x \min(P_\varepsilon(x), P_{\varepsilon'}(x)) = 1 - \mathrm{TV}(P_\varepsilon, P_{\varepsilon'})$, and $P_\varepsilon - P_{\varepsilon'} = (\varepsilon - \varepsilon')(Q - P)$. (2) Let D be C's mean shift. Its statistic has mean $a = D/\big(\sigma_m\sqrt{1/(\pi N) + 1/((1-\pi)N)}\big)$. The pooled shift is πD, and the pooled standard error is $\sigma_m/\sqrt{N}$, so the ratio is $a\,\pi\sqrt{1/\pi + 1/(1-\pi)} = a\sqrt{\pi/(1-\pi)}$. The exposure probability is $\Phi(a - c_s) + \Phi(-a - c_s) \ge \Phi(a - c_s)$. (3) The triple part of the decision statistic $\hat{r} - \hat{r}_c$ is $(\hat{m}_T - m_T)/v_T$, so the variance of its triple part is σ_m^2/v_T^2 (check SR5). Settling on the wrong act needs the shift to exceed $d + z\tau/\sqrt{N}$. When the filter keeps C, the statistic is conditioned on $|Z_C| \le c_s$, which only lowers the shift, because the pooled estimate and Z_C are positively correlated. When the filter drops C, the pool is genuine. Under pooling with a fixed share, the shift $\pi\bar{\varepsilon}_C s/|v_T|$ exceeds d by a fixed amount while the noise shrinks. (4) The median of the settling statistic crosses $z\tau/\sqrt{N}$ when $d - B\sigma_m/(|v_T|\sqrt{N}) = z\tau/\sqrt{N}$. Exposure only helps the decision maker. (5) The batch statistic of a single shifted batch has mean Δ/σ_b up to the factor $\sqrt{1 + 1/K_C}$. With two sources, $\hat{m}_1 - \hat{m}_{12}$ and $\hat{m}_2 - \hat{m}_{12}$ are proportional with opposite signs. The median of three numbers, one of them arbitrary, lies between the other two. □

The theorem turns the adversary's "just enough" into geometry. Gluing is tested on a single coordinate, the fibre coordinate m_T of theorem H. The adversary's operating region is a band of fixed width, B standard errors, around the genuine estimate. Because the standard error shrinks, the band narrows with N and eventually clears the switch. So the filter turns every undetected tilt into a delay, and the delay is bounded. With $c_s = 2.394$ (Bonferroni over three sources) and $\pi = 1/3$: $B = 1.69$ at exposure risk one half, and $B = 1.10$ at exposure risk 0.2. Both are below $z = 1.96$, so an adversary that stays hidden cannot even make a wrong settlement likely. The delay factor is at most $((z+B)/z)^2 = 3.5$ and 2.4. The adversary pays a vanishing share for it: on the worked example's truth, 4.4% of the records in half of C's batches at $N = 192{,}000$ (table 7).

Tier 2. If every source were compromised alike ($\pi \to 1$), the band B would be unbounded. No gluing test sees a shift common to all sections, $\lambda^* = 1$, and only the geometry is left: the walls (theorem F(3)), the model's equations at $m \ge 4$ (theorems G and H) and a second triple. The labels help exactly as far as the sources disagree.

21 Study 13: a trusted source that is sometimes adversarial

Study 13 was registered and frozen before any registered design was simulated (`engine/study13_sources/PREREG_study13.md`). The checks SR1–SR10 and the smoke runs used random truths and an invented truth; on the registered truth only the Gaussian model was used. Before freezing, the filter's order was changed on the strength of that model, and the change is disclosed in the registration (swamping, above).

Design. The truth and the decision are those of section 18: approve the triple (T) or the pair (P), with the switch at triple harm 0.668 against a truth of 0.667. There are three sources with $K_C = 8$ batches each, triple windows of n records, and pair windows of $25n$. The hidden channel uses $s = 0.027$. The thresholds are $c_b = 3.078$ and $c_s = 2.394$, Bonferroni at family-wise 0.05, and $z = 1.96$. The adversary knows P, n, K and the thresholds, but not the noise. Five adversaries are compared (table 7):

- none (A0);
- *sometimes, just enough*: 4 of C's 8 batches, at exposure risk 0.2 (A1) or 0.5 (A1b);
- *concentrated*: A1's source shift in one batch (A3);
- *tuned to pooling*: a share of 0.18 in every batch of C (A2), so a pooled share of 0.06, in the middle of the stealth window $(\varepsilon^*, \varepsilon_{\text{det}}) = (0.037, 0.083)$.

Four rules see the same data:

- naive pooling;
- the gluing filter;
- an oracle that knows C and drops its triple sections;
- writing C off entirely.

There are $N = 24n$ triple records in all, from 12,000 to 192,000, with 2000 replicates per cell.

n per batch	N	A1 (4 batches)	A1b (4 batches)	A3 (1 batch)	A2 (8 batches)
500	12,000	0.177	0.274	0.710	0.18
1000	24,000	0.125	0.193	0.502	0.18
2000	48,000	0.089	0.137	0.355	0.18
4000	96,000	0.063	0.097	0.251	0.18
8000	192,000	0.044	0.068	0.177	0.18

Table 7: Study 13: the injected share in each adversarial batch of source C. "Just enough" shrinks like $1/\sqrt{n}$; the fixed share does not.

All five predictions held (table 8 and fig. 18). Without labels the adversary takes the decision over. Under pooling, a share of 0.18 in source C's batches settles the wrong act in 88% of replicates at $N = 192{,}000$, and nothing is flagged. With labels, the same attack is caught by its own growth: C's gluing statistic grows like $\sqrt{N}$ (fig. 18(c)), and the filter matches the oracle that knows C to within 0.006. The adversary that stays hidden only delays. Its statistic rides flat under c_s, and the filter's half-way point moves from 24.7k to 52.8k records (A1) and 55.7k (A1b), as Theorem J(4) bounds (59k and 85k for an adversary that is never exposed; exposure, which drops C's triples, is what keeps A1b close to A1). At no sample size does the filter settle on the wrong act in more than 2% of replicates. A concentrated attack is localised: the poisoned batch is removed 97% of the time, C's seven genuine batches stay in, and the filter then beats both the oracle and writing C off, because it keeps C's genuine triples. On genuine data the filter costs at most 0.012 in right-rate.

The price of not knowing which source is compromised is the gap between the filter and the oracle under the tuned adversary, about 0.16 in right-rate at 48k. No test on these tables can close it. The adversary sits inside the noise band by construction, and only information beyond the

	prediction	outcome
H1	right-rate within 0.05 of the Gaussian model in $\geq 90\%$ of the 100 cells, within 0.08 in all	holds: 100 of 100; largest gap 0.031
H2	filter's wrong-rate ≤ 0.05 everywhere; pooling under A2 wrong ≥ 0.75 at 192k; filter within 0.05 of the oracle under A2 for $N \geq 48$k	holds: ≤ 0.020; 0.877; gap ≤ 0.006
H3	half-way points within 15% of prediction for A0, A1, A1b; oracle $\geq$ filter under A1b for $N \geq 24$k	holds: 24.7k, 52.8k, 55.7k against 24.6k, 53.0k, 56.6k; yes
H4	A3's batch removed ≥ 0.90, C dropped ≤ 0.10, filter within 0.05 of A0; under A0 a source dropped ≤ 0.08 and filter within 0.04 of pooling	holds: 0.96–0.97; 0.011–0.017; ≤ 0.025; ≤ 0.044; ≤ 0.012
H5	at 192k the worst right-rate over adversaries is ≥ 0.85 for the filter and ≤ 0.05 for pooling	holds: 0.918 and 0

Table 8: Study 13: registered predictions and outcomes (2000 replicates per cell).

tables (an audit of C, a second triple) can. Note also where pooling looks good: under A1 at 192k pooling (0.976) beats the filter (0.950), because the tuned shift is small at that size and the filter sometimes drops C's genuine data. That is why the comparison that matters is the worst case over adversaries (H5), not any single attack.

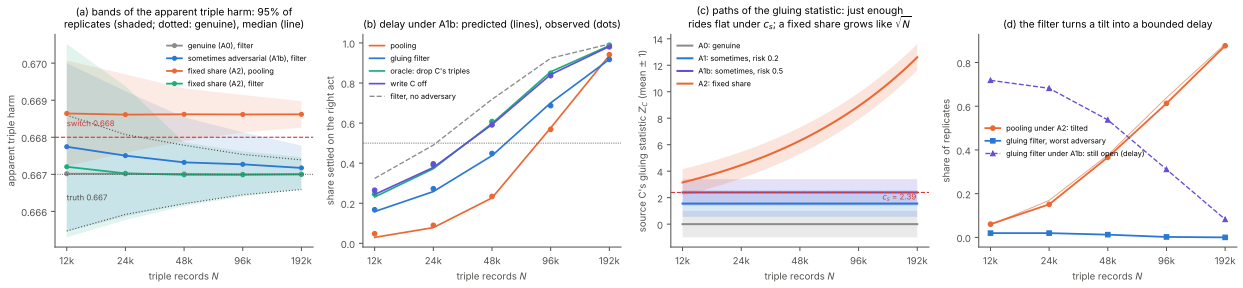


Figure 18: Study 13. (a) Overlaid bands of the apparent triple harm (95% of replicates). Pooling under a fixed share sits above the switch and never comes back. The filter under the sometimes-adversarial source is shifted toward the switch by a band of fixed width in standard errors, which shrinks and clears the switch. (b) Settling curves under A1b: the adversary delays the filter (blue) but cannot stop it. The oracle (green) and writing C off (violet) settle earlier, and pooling (orange) later. (c) Paths of C's gluing statistic (Gaussian model, mean ± 1): just-enough adversaries stay flat under c_s at every N, while a fixed share grows like $\sqrt{N}$. (d) Pooling is tilted with probability rising to 0.88; the filter's wrong-rate under its worst adversary stays below 0.02, and what is left is the delay (open decisions under A1b).

Tearing. The filter's decision is a discontinuous function of the data. When $|Z_C|$ crosses c_s, C's triple sections leave the pool and the estimate jumps by πD. This is a tear in the decision map, along the hypersurface $|Z_C| = c_s$ in table space, and an adversary sitting just below it gains most from it. Lifting smooths the tear, as section 22 shows for acts: replace the 0–1 inclusion of a source by a weight $w(Z_C)$ that falls continuously from 1 to 0 across the threshold.

For the three readerships. *Decisions*: labels turn an unbounded tilt into a delay bounded by $((z+B)/z)^2$, and the bound is a design parameter through c_s. *Persuasion*: a trusted source can buy delay, not a different decision, and the delay it buys is priced in standard errors, not in records. *Machine learning*: this is federated learning with one Byzantine client [40]. Theorem J(2) is a bound on the "a little is enough" attack [41], exact for one coordinate. The median of sources is the coordinate-wise median aggregator [42].

22 Lifting: the decision maker's patchwork

The decision maker meets singularities wherever a map it relies on stops being smooth:

- the decision map at a tie (theorem D, $d = 0$);
- the parametrisation at the merge ($v = 0$, theorem B);
- identification on a coincidence stratum (theorem C);
- the filter at its threshold (the tear of section 21).

Each has a lift. The lift moves to a space one level up, where the map is regular, computes there, and pushes the answer down. The space one level up may be laws on acts, a blow-up, a larger cover, or a labelled total space. This section assembles the lifts into one picture. The total space is hypotheses $\times$ contexts, and each context carries its provenance label. A *vertical block* is a piece $U \times V$: part of the hypotheses over part of the contexts. A patchwork of blocks is a cover of the total space. Lifting means choosing a decision on each block and checking that the choices agree on overlaps: the gluing of the parent note, one level up.

Theorem K (lifting)**.** *Let F be the polytope of laws with the observed pair tables, and $F \cap \mathcal{M}$ its model part (the hiding segment at $m = 3$).*

(1) The decision arrangement. *Expected utility is linear in the law, so every boundary the decision maker cares about, $\{U_a = U_b\}$ and $\{U_a = u_0\}$, is an affine hyperplane. These hyperplanes cut F into* decision chambers*. The* indecision locus *is where they meet $F \cap \mathcal{M}$. The decision is settled before the probe exactly when $F \cap \mathcal{M}$ lies in one chamber. A probe with slices $\{F \cap \mathcal{M} \cap \pi^{-1}(y)\}$ can settle the decision only on slices that lie in a single chamber. If every slice meets every chamber that $F \cap \mathcal{M}$ meets, the probe is on a* plateau*: no outcome settles it. In the worked example $U_T - U_P = 0.128 - P(001)$ is a coordinate hyperplane of the simplex. On the pair line it sits at $s = 0.001$, inside the hiding segment, so the decision is unsettled before the probe. With four variables the decision's gradient along the five-dimensional pair fibre is a multiple of the χ_{ABC} coordinate. The chambers are then* slabs*, unconstrained in the other four directions: literally vertical blocks.*

(2) The act lift. *Replace the acts A by the laws on acts $D(A)$, the tower's own monad applied to actions (a Kleisli lift, i.e. fractional allocation). The lifted utility is bilinear, so the minimax theorem gives a saddle point. On an evidence interval I with worst regrets $a(I)$ of T and $b(I)$ of P, the lifted rule puts weight $\alpha^* = b/(a+b)$ on T. Its worst regret is $ab/(a+b) < \min(a, b)$, and α^* is continuous in the evidence. Before the probe in the worked example:* 85% *on the triple and worst regret* 0.00107*, against* 0.00125 *for the best pure act. The pure rule jumps by a whole act at the switch, while the lifted rule moves by at most* 0.001 *per step of a fine sweep (check LF2).*

(3) The hypothesis lift: blowing up the merge. *Write $v = \rho e$ with $\rho \geq 0$ and $e \in S^{m-1}/\pm$, so $M_{ij} = \rho^2 e_i e_j$. In the coordinates $(\rho^2, [e])$ the pair map is monomial (normal crossings). Hence ρ^2 is estimated at the regular rate $n^{-1/2}$, and $|v| = \rho$ inherits $n^{-1/4}$ as a square root. The exceptional divisor is the space of directions $[e]$, "which way the types would differ if they existed". It is not identified at $\rho = 0$. Every slope $\beta_a = v_T \cdot$ (Walsh coefficient) is $O(\rho^3)$, so near the merge the decision is flat, and ρ^2 says how flat. At $v = 0$ the fitted sampling exponents are -0.498 for ρ^2 and -0.245 for ρ; away from the merge both are -0.51 (check LF7).*

(4) The adversarial lift. *Lift each hypothesis to a triple (P', Q, ε) that projects to the observed law, $O = (1-\varepsilon)P' + \varepsilon Q$. The block of explanations of size ε_0 is $B_{\varepsilon_0}(O) = \{P' \in \mathcal{M} : (1-\varepsilon)P' \leq O \text{ for some } \varepsilon \leq \varepsilon_0\}$. Its projection to the genuine law is the Horowitz–Manski identified set [39]. The decision is* confident at level ε_0 *if it is constant on $B_{\varepsilon_0}(O)$. The confidence radius is*

$$\varepsilon_c(O) = \min_{P' \in \mathcal{M} \text{ in another chamber}} \Big(1 - \min_x \frac{O(x)}{P'(x)}\Big),$$

the smallest poisoning budget of any genuine law that the data cannot rule out. Value of information and robustness to poisoning are two readings of one block structure. In the worked example the chamber wall is $P(001) = 0.128$ against $O(001) = 0.127$. So $\varepsilon_c = 1 - 0.127/0.128 = 1/128$, attained by a genuine two-type law (check LF3). If the injection must also keep the pair tables, the block shrinks to the line and the radius grows to $1/14$. Read forward from the truth, the hidden channel needs 0.037 (theorem F). The observer's block and the adversary's budget are not the same number: the observer must also allow for genuine laws it cannot distinguish.

(5) The cover lift and the orbifold atlas. *Moving up the tower of covers by adding windows (a second triple, corollary 3) or labels (sources, theorem J) routes around a coincidence or a hidden channel, but cannot undo a coincidence that is a fact about the truth. At a vertex of the four-variable fibre with index k (section 19, Stage 6), the toric chart is $\mathbb{C}^5/G$ with $|G| = k$. An orbifold* is *a patchwork of smooth charts divided by finite groups. The decision maker computes on the k-fold cover, where the fibre is smooth, and pushes the answer down. No global toric resolution is needed.*

Proof. (1) Expected utility is linear in the law, and each chamber is an intersection of half-spaces with F. A slice settles the set rule only if its worst values put one act above every other everywhere on it, and that is a single-chamber condition. On the worked example, $U_T = P(000)$ and $U_P = P(000) + P(001) - 0.128$, and along the line $P(001)$ moves by $s\chi(001) = s$. At $m = 4$ the decision depends on the ABC table only, and within the pair fibre the ABC table moves only along its own χ_{ABC} coefficient (theorem F(4)). (2) Von Neumann's minimax theorem for the bilinear form $\alpha \cdot$ (regret) on $[0,1] \times I$. With two acts and affine regrets, the worst case of the mixture is $\max(\alpha a, (1-\alpha)b)$, which is minimised at α^*. Continuity holds because $a(I)$ and $b(I)$ are continuous in the endpoints of I. (3) $M_{ij} = \rho^2 e_i e_j$ is a monomial map in (ρ^2, e). A nonzero linear functional of $\hat{M}$ estimates ρ^2 at $n^{-1/2}$ (delta method), and the square root of an $O_p(n^{-1/2})$ quantity is $O_p(n^{-1/4})$. Finally $v_T = \rho^3 e_0 e_1 e_2$. (4) $Q = (O - (1-\varepsilon)P')/\varepsilon$ is a law exactly when $(1-\varepsilon)P' \leq O$. Hence the smallest ε for a given P' is $1 - \min_x O(x)/P'(x)$; minimise over the other chamber. On the worked example the chamber is $\{P'(001) \geq 0.128\}$, and the bound $1 - O(001)/P'(001)$ is attained, because a two-type law with $P'(001) = 0.128$ and $P' \leq O/(1-\varepsilon)$ elsewhere exists (found by SLSQP; certificate in LF3). (5) This is the standard description of a simplicial toric variety:

at a simple vertex the cone is generated by a basis of a sublattice of index k, and the chart is the quotient of affine space by the finite group the two lattices define [2, 6]. □

Remark 1 (stability of the lift)**.** Fournier-Facio and Willett show that, for groups with the local lifting property and property RFD, every approximate representation is close to a corner of an honest one [46]. We borrow the shape of the statement, not its proof. Here the analogue is exact. A family of batch laws with common pair tables and triple moments on the line is *exactly* the compression (forget the label) of an honest labelled law on $\{\text{genuine}, \text{injected}\} \times X$, with shares $\varepsilon_b = (m_b - m)/s$ (check LF10). An approximate gluing is a corner of an honest gluing one level up. The size of the corner is the injected share, and that share is what Theorems F and J price.

Remark 2 (smoothness deceives)**.** The adversary's goal is to make the injected data as representable as the original: stay on the fibre, keep the pair map a submersion, keep the classical (first-order) picture smooth. Li shows that a classical representation scheme can be smooth while the derived one is not, and that coefficients beyond matrices detect it [47]. The lab has two exact counterparts, offered as analogies.

- *The tangent space versus the obstruction.* The first-order tangent space of the tetrad scheme shows a stealthy direction, but on the two-coincidence stratum the André–Quillen class (theorem G(4)) says it does not extend.
- *Pooling versus labels.* The pooled law sits on the smooth line $\mathbb{CP}^1$. The labelled family is an element of $D(DX)$ off the image of η, and it is visible only with the label coefficients that the comonad supplies (theorem J).

In both cases what stands out is geometric: a class in a cokernel, or a point off a diagonal. A smooth-looking shadow does not make it go away.

Proposition 5 (dilation)**.** *Pair tables pin every single and pair probability, but they dilate every individual-level forecast. With only pair tables disclosed, the forecast $P(C \mid A = a, B = b)$ for a patient with observed (a, b) ranges over an interval of width*

$$\Delta s / P(a, b),$$

where Δs is the length of the hiding segment in the line coordinate s. Every outcome therefore turns a point into an interval [43], and the rarest profiles are dilated most. The triple pins s and removes the dilation. The hidden channel moves the interval but cannot widen it, since the width is set by the pair tables.

Proof. $P(a, b)$ is a pair probability, so it is fixed. $P(a, b, 1)$ moves along the line by $s\chi(a, b, 1)$, with $|\chi| = 1$. So $P(C \mid a, b)$ is affine in s with slope $\pm 1/P(a, b)$. $P(C)$ is a single probability, so it is fixed. □

Proposition 6 (the value of a probe in its size)**.** *In the worked example the value of a probe of n triple records has three readings.*

(1) Bayes *(uniform prior on the hiding segment).* EVSI(n) *is zero to 10^{-12} for $n \le 100$, and all its derivatives vanish at 0: the Radner–Stiglitz flat start [44]. It is convex up to an inflection near $n \approx 1700$, then saturates at* $\text{EVPI} = 9.24 \times 10^{-5}$*. At a cost of 10^{-9} per record the optimal trial has about* 18,000 *records (net value 5.4×10^{-5}). Yet the marginal value at $n = 0$ is below the cost, so a decision maker who reasons at the margin never starts.*

observed (A, B)	$P(C \mid A, B)$ over the hiding segment	width $= \Delta s / P(a, b)$	with the triple
$0, 0$	$[0.263, 0.281]$	0.018	0.276
$0, 1$	$[0.131, 0.169]$	0.039	0.141
$1, 0$	$[0.381, 0.412]$	0.030	0.389
$1, 1$	$[0.169, 0.381]$	0.212	0.325
none	$P(C) = 0.28$ exactly	0	0.28

Table 9: Dilation in the worked example ($\Delta s = 0.00848$; check LF4). The population is pinned and the individual is not.

(2) Savage. *The set rule with the Savage default lowers the worst regret from* 0.00125 *toward* 0.

(3) Wald. *At a saddle point (theorem C) the probe cannot raise the worst case. With sampling noise, the Wald value of the paper's set rule (with the Wald default) is* negative *at every n and tends to* 0 *from below. This is a property of that fixed rule. An ex-ante maxmin-optimal rule can always ignore the probe, so its value is never negative; here it is exactly* 0*, as Blackwell's order requires (proposition 13(4)).*

Proof. (1) The posterior mean of $U_T - U_P$ is negative only when $\hat{r}$ falls far enough below r_c to outweigh the prior. For small n that event has probability below any power of n. Saturation follows by dominated convergence. The numbers are computed by quadrature (check LF5). (2) and (3): the probabilities of settling are those of theorem D(4). Under the Wald default, the rule picks T only when it settles, and with noise that can happen where T is worse. The worst case of P alone is already the pre-probe value (check LF6). □

The decision maker is not led blindly into the flat start, because the curve can be computed in advance. Theorem C(5) names the best probe from the pair tables alone, theorem D(4) gives its settling size, and proposition 6 prices the whole trial. Planning the probe as a whole rather than at the margin is the formal sense of "seeing the landscape".

Backtracking. Gluing failures are often visible only late. A batch section that glues with a weak early reference can fail to glue once more genuine data arrive. A decision maker who treats accepted sections as final lets an adversary that is caught late keep what it injected early. The remedy is to recompute the colimit over the whole history whenever a failure appears, revoking the failed source's earlier sections. This is a dynamic programme over the history. Each step re-solves the gluing linear programme of the parent note with new constraints, and its dual certificate (the witness of [8]) names the sections to revoke.

Proposition 7 (backtracking)**.** *Consider eight rounds, one batch per source per round, with the filter of theorem J run after each round (Gaussian model, check LF8). Over constant doses and durations, the worst adversary moves the final pool by* 1.05 *standard errors when accepted sections are final, although it is flagged in* 74% *of runs. With backtracking the worst adversary moves the pool by* 0.57*.*

Remark 3 (rectangularity as gluing)**.** Epstein–Schneider rectangularity [45] says that the ex-ante set of beliefs C is the set of all pastings of its one-step conditional sets along the information tree. In sheaf language, the conditional sets restrict contravariantly along the tree, and the set of their pastings $\mathrm{rect}(C)$ is the set of global sections of the sheaf they generate, the *sheafification* of C. C is rectangular exactly when it is already a sheaf. Backward induction (dynamic programming)

computes the maxmin over $\mathrm{rect}(C)$, so dynamic programming is exact exactly when beliefs glue. Otherwise the global problem is a linear programme over the joint law, which does not decompose along the tree.

Proposition 8 (the price of not gluing)**.** *For any set of beliefs C on a finite tree and any utility u, the value of observing the tree's information under prior-by-prior updating and dynamic choice satisfies*

$$\mathrm{VOI} \;\geq\; -\operatorname{osc}(u)\, d_H\big(C, \mathrm{rect}(C)\big),$$

where d_H is the Hausdorff distance in total variation. On a two-stage tree with correlated ambiguity ($q_1 = 0.2 + 0.6\alpha$, $q_2 = 0.8 - 0.6\alpha$), a bet at price 0.4 is accepted ex ante (value 0.1) and rejected in every branch. Hence $\mathrm{VOI} = -0.1$, within the bound -0.3 (check LF9).

Proof. Dynamic choice with full Bayesian updating of C maximises the recursive value, which equals $V_{\mathrm{rect}}(f) = \min_{\mathrm{rect}(C)} E[u(f)]$ [45]. Since $C \subseteq \mathrm{rect}(C)$, $V_C \geq V_{\mathrm{rect}}$, and since every element of $\mathrm{rect}(C)$ is within d_H of C, $V_{\mathrm{rect}} \geq V_C - \operatorname{osc}(u) d_H$. So $V_C(f_{\mathrm{dyn}}) \geq V_{\mathrm{rect}}(f_{\mathrm{dyn}}) \geq V_{\mathrm{rect}}(f_0) \geq V_C(f_0) - \operatorname{osc}(u) d_H$, where f_0 is the best plan that ignores the information. □

Conjecture (sharpness). $\sup_u(-\mathrm{VOI})/\operatorname{osc}(u) = d_H(C, \mathrm{rect}(C))$. Moreover $1 - d_H = \min_{\pi \in \mathrm{rect}(C)} \max_{\mu \in C} \sum \min(\pi, \mu)$ is a gluable fraction of the tree, computable with the linear programmes of the parent note. Testing this on families of small trees is an empirical claim and would need its own pre-registration. We have not done it.

Tears. The precision the geometry adds is that it says *where* each map tears:

- on the decision hyperplanes (theorem K(1)), where the pure rule jumps;
- on $|Z_C| = c_s$, where the filter drops a source;
- on the coincidence strata, where identification drops rank;
- at the merge, where the rate drops to $n^{-1/4}$.

Statistics sees each of these only as a change in a rate or in a test's power. Each lift glues one tear: mixtures of acts glue the first, soft weights the second, larger covers the third, and the blow-up the fourth.

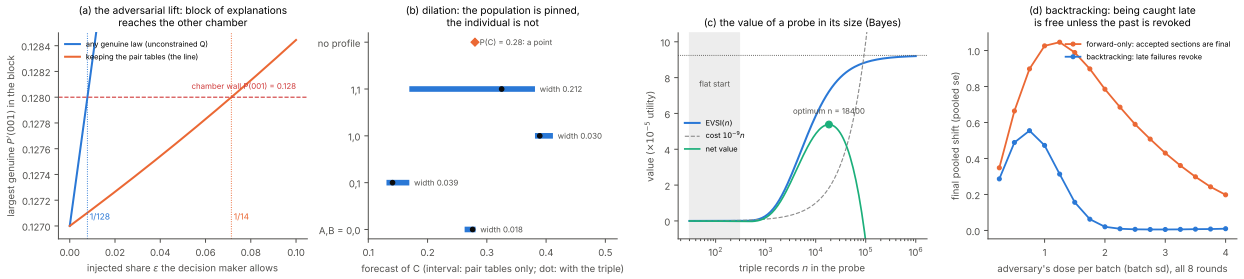


Figure 19: Lifting. (a) The adversarial lift on the worked example: the largest $P'(001)$ among genuine laws the data cannot exclude, as the injected share grows, without constraint and keeping the pair tables. The block reaches the chamber wall 0.128 at 1/128 and at 1/14. (b) Dilation: pair tables pin $P(C)$ (a point), while the individual forecasts are intervals; the triple's values are marked. (c) The value of a probe in its size: the flat start, the inflection and the optimal size at a cost of 10^{-9} per record. (d) Backtracking: the final pooled shift against the adversary's dose, for forward-only acceptance and for backtracking.

23 One picture: chambers, slices and gauges

Theorems A–K were proved one at a time. This section states what they have in common. Every quantity in the paper is a question about how a cover's slices, and a gauge's capacities, meet a chamber arrangement on the fibre.

The objects.

- The *fibre* F is the polytope of laws with the observed pair tables (the pair cover). Its *model part* $\mathcal{M} \subseteq F$ is the set of two-type laws in F, which is the hiding segment at $m = 3$ (theorem A).
- A *decision* is a finite set of linear functionals ℓ_j of the law, with walls $\{\ell_j = c_j\}$: $\ell = U_a - U_b$ with $c = 0$, or $\ell = U_a$ with $c = u_0$. The walls cut F into *chambers*.
- A *probe* refines the cover. With exact data it cuts $\mathcal{M}$ into *slices* S_y. A plateau probe has $S_y = \mathcal{M}$, and the triple has point slices.
- A *gauge* is a convex set $\mathcal{Q}$ of admissible injected laws, with support function $h_{\mathcal{Q}}(\ell) = \sup_{Q \in \mathcal{Q}} \ell(Q)$:
  - the simplex: any record;
  - F itself: pair tables kept;
  - the laws that pass the gluing tests of a labelled cover: sources and batches.
- The *margin* of a law P to a wall is $\delta = c - \ell(P)$, taken positive on the side being approached.

Theorem L (one picture)**.** *Two latent types, the pair cover. Parts (1), (2) and (5) are for $m = 3$, where F is a line. Parts (3) and (4) hold for every m.*

(1) Settling. *The pre-probe set rule chooses the oracle act at every truth in $\mathcal{M}$ if and only if $\mathcal{M}$ lies in one chamber.*

(2) Value of information is slice–chamber incidence. *With exact data, the Savage value of a probe is* $R(\mathcal{M}) - \max_y R(S_y)$*, where* $R(S) = \min_a \max_S \text{regret}_a$*. So it equals* $R(\mathcal{M})$*, which is positive exactly when* $\mathcal{M}$ *meets two chambers, if every slice lies in one chamber, and it is* 0 *on a plateau. The Wald value is* $\min_y W(S_y) - W(\mathcal{M}) \geq 0$*, where* $W(S) = \max_a \min_S U_a$*. For point slices it is positive* if and only if *a wall meets the interior of* $\mathcal{M}$ *and the two acts' slopes along* F *are strictly opposite. This is theorem C's saddle-point condition read off the chambers.*

(3) Robustness is margin over capacity.

(a) Forward *(from the truth, gauge* $\mathcal{Q}$*): the smallest injected share that crosses the wall is*

$$\varepsilon = \delta / \big(h_{\mathcal{Q}}(\ell) - \ell(P)\big).$$

For the simplex the capacity is $\max_x \ell(\delta_x) - \ell(P)$*: the vertex attack, exact for linear* ℓ *(theorem E(3)). For* F *it is a linear programme, and at* $m = 3$ *it is* $s_{\pm}|\ell(\chi_T)|$*: the hidden channel (theorem F).*

(b) Observer *(from the data): the radius of the block of explanations (theorem K(4)) is a water-filling. It removes the observed mass from the cells of lowest* ℓ *first, and it equals* $\delta/(c - \min_x \ell(\delta_x))$ *when the lowest cell holds the removed mass. Requiring the genuine law to lie in* $\mathcal{M}$ *can only raise it. On the line it is* $|s'|/(\text{cap}(O)+|s'|)$*, where* $s' = \delta/\ell(\chi_T)$*.*

(c) Sampling*: the decision settles half the time at* $n_{1/2} = (z\sigma_\ell/\delta)^2$*, where* σ_ℓ *is the per-record standard deviation of the estimate of* ℓ*. This is theorem D(4) with* $d = \delta/|\partial\ell/\partial r|$*.*

(d) Labelled cover*: a source of share* π *that stays unexposed can move* $\hat{\ell}$ *by at most* $B\sigma_\ell/\sqrt{N}$*, with* $B = (c_s - q_\beta)\sqrt{\pi/(1-\pi)}$*. Its capacity is measured in standard errors (theorem J).*

(4) Contextuality matters exactly along wall normals. *Whole-record injection keeps the unlabelled cover gluable (*$\lambda^* = 1$*, theorem E(1)) and moves the law inside* F*. On the labelled cover a contaminated section fails to glue, with* $1-\lambda^* \geq |\Delta\varepsilon|\,\text{TV}$ *(theorem J(1)). A failure in direction* ξ *changes wall* j *at rate* $\ell_j(\xi)$*. It is* decision-relevant if and only if $\ell_j(\xi) \neq 0$*; otherwise no injection along* ξ*, at any share, changes the act.*

(5) Two boundaries. *On* F *the decision walls and the detection boundary* $\partial\mathcal{M}$ *lie on one line (theorem H). A stealth window* $\varepsilon^* < \varepsilon_{\text{det}}$ *exists if and only if a wall meets the interior of* $\mathcal{M}$*, exactly the condition in (1) under which the decision is unsettled.*

Proof. (1) If $\mathcal{M}$ lies in one chamber, some act a dominates on $\mathcal{M}$, so its worst value is the largest and the set rule picks it, as the oracle does at every point. If $\mathcal{M}$ meets two chambers, the oracle act is not constant on $\mathcal{M}$, while the set rule's act is. (2) With exact data the truth lies in its slice and the rule acts on the slice. The worst case over truths is therefore the worst case over slices of the slice's value, which gives both formulas. For two affine acts f, g on an interval, $W = \max(\min f, \min g)$ and $W_{\text{post}} = \min\max(f, g)$. If the slopes have the same sign, or one is zero, the upper envelope's minimum is at the same endpoint as the dominating act's minimum, so the gap is 0. If they are strictly opposite and cross inside, the upper envelope is V-shaped with its minimum at the crossing, which is strictly above both acts' minima. (3a) $\ell\big((1-\varepsilon)P + \varepsilon Q\big) = \ell(P) + \varepsilon(\ell(Q) - \ell(P))$ reaches c first at the Q maximising $\ell(Q)$. On the simplex that Q is a vertex. On the $m = 3$ line, $\ell(P + s\chi) = \ell(P) + s\,\ell(\chi)$ with s up to the positivity limit. (3b) $O = (1-\varepsilon)P' + \varepsilon Q$ means that $y = (1-\varepsilon)P'$ is a sub-measure of O of mass $1-\varepsilon$ with $\ell(y) \geq c\,(1-\varepsilon)$. Maximising the mass is a fractional knapsack, solved by removing mass in increasing order of ℓ. On the line both P' and

Q move along χ, and $Q \geq 0$ bounds $|s_Q|$ by the smallest cell of the relevant sign of $P' = O + s'\chi$. (3c) The delta method, as in theorem D(4). (3d) Theorem J(2), applied to $\hat{\ell}$. (4) ℓ_j is linear. A failure along ξ moves the law by multiples of ξ, and the gluable fraction is that of theorem J(1). (5) Moving along χ, the wall is crossed before the end of $\mathcal{M}$ exactly when it lies inside $\mathcal{M}$. □

Checks CL1–CL9 (`engine/check_capstone.py`) test every part on 150 random truths and random two-act decisions with utilities on the eight cells. They also cover the extension of (3a) to $m = 4$. There, keeping the pair tables, the support function of the five-dimensional fibre gives the adversary no more room than at $m = 3$, because the fibre's ABC projection lies in the three-variable fibre (3.70% both ways). It gives more room than the single χ_{ABC} channel (5.05%).

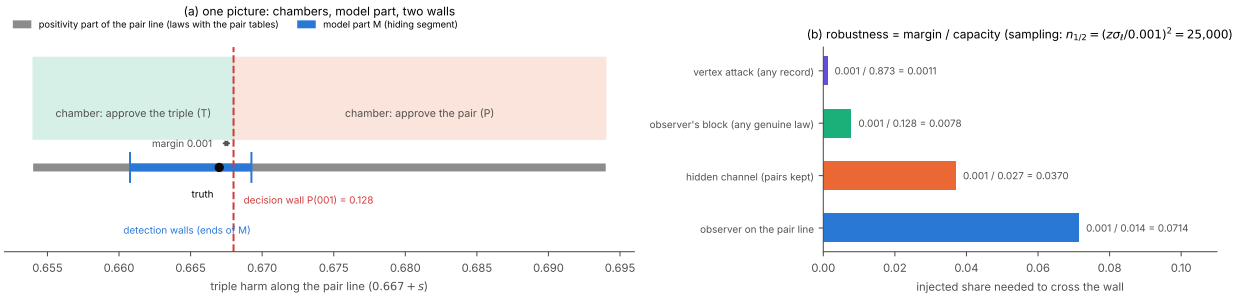


Figure 20: Theorem L on the worked example. (a) The pair line: the positivity part, the model part $\mathcal{M}$ with its two detection walls, and the decision wall $P(001) = 0.128$, 0.001 from the truth. (b) One margin, four capacities: any record, any genuine law, pairs kept, and the observer on the line. Sampling is the same ratio with a standard deviation as capacity.

The worked example in one line. The margin is 0.001 in every reading, and only the capacity changes:

- any record: 0.873, so 0.11%;
- any genuine law the data cannot exclude: 0.128, so 1/128;
- pairs kept: 0.027, so 3.7% (theorem F);
- the observer on the line: 0.014, so 1/14;
- sampling: σ_ℓ, so $n_{1/2} = 25{,}000$ (theorem D);
- a labelled source: B standard errors, so the delay of Study 13.

The value of information, the poisoning budgets, the settling size and the bands of Part II are one geometric fact. A wall sits 0.001 from the truth, inside the hiding segment, and each question measures that distance in its own gauge.

How far this goes. Theorem L is proved for two types on the pair cover, parts (1), (2) and (5) at $m = 3$.

- Parts (3) and (4) use only linearity and hold on any fibre.

- With more types, $\mathcal{M}$ becomes a higher secant variety. It is no longer convex, so (1) needs "connected component of a chamber", and (2)'s slices are no longer intervals.
- With other covers, F is a different fibre polytope, but (3) and (4) are unchanged.
- Non-linear criteria (a risk measure in place of expected utility) curve the walls, and (3a) becomes a support-function inequality rather than an equality.

The evidence is two worked truths, a few designed ones, and random decisions on three variables. The reach beyond that is open.

24 The descent tower

Theorem L reads every quantity off one fibre. This section builds the object the fibre is a fibre *of*. It is a diagram of local spectra over the contexts, together with a descent complex at each level of the tower, and it says which level carries which obstruction. We stay with ordinary spaces and complexes. Categorification is deferred until this layer is explicit, and the end of the section names the first place it is forced.

Contexts and local spectra. Let $V = \{1, \dots, m\}$ be binary variables. The *context category* $\mathcal{C}$ has the observed windows $S \subseteq V$ and all their intersections (the empty context included) as objects, with an arrow $S \to T$ when $T \subseteq S$. For each context let

$$R_S = \mathbb{R}[p_x^S : x \in \{0,1\}^S]/(\textstyle\sum_x p_x^S - 1), \qquad X_S^{\mathrm{alg}} = \operatorname{Spec} R_S,$$

the affine space of signed laws on S, and let $X_S^{\geq 0}$ be its real points with $p \geq 0$, the probability simplex. Marginalisation $\rho_{S,T}$ is a linear map, so it induces ring maps $R_T \to R_S$, and $\rho_{T,U} \circ \rho_{S,T} = \rho_{S,U}$. This gives a diagram $S \mapsto X_S$ over $\mathcal{C}$. Its limit is Spec of the colimit of the rings: families of local laws whose marginals agree on every overlap. The model layer adds the two-type equations on top. Real probability points and the full algebraic spectrum are kept apart throughout, because they behave differently.

Proposition 9 (spectrum and descent)**.** *For the pair cover with $m = 3$ or 4, and for the triple cover with $m = 4$:*

(1) Algebraic descent holds up to a kernel. *The marginal map $X_V^{\mathrm{alg}} \to \lim_{\mathcal{C}} X_S^{\mathrm{alg}}$ is surjective. Its kernel is spanned exactly by the Walsh characters χ_S of the sets S contained in no window: dimension 1 for the pair cover at $m = 3$, 5 at $m = 4$, and 1 for the triple cover. So the presheaf of signed laws satisfies gluing but is not separated, and its fibres are torsors under this kernel. The ambient fibre F of theorem L is such a torsor cut by positivity.*

(2) The linear descent complex is acyclic. *For the Čech complex of signed measures on these covers, $H^q = 0$ for $q \geq 1$. Nothing obstructs descent algebraically.*

(3) Real descent fails only through positivity. *On real points the limit is the local polytope (consistent pseudo-marginals), and the image of the simplex is the marginal polytope. When the nerve has a cycle these differ, and the difference is the contextual families. Example: $A = B$, $B = C$, $A \neq C$ with uniform margins. It has zero Čech energy, extends to a signed law with least negative mass $1/2$, and has $\lambda^* = 0$: it is strongly contextual. The certificate is the dual witness of the gluing linear programme, a Bell-type inequality, and it is not a cohomology class.*

Proof. (1) A signed law on V has the marginals of a consistent signed family whenever the family is consistent. Build it from the Efron–Stein components of the windows, as `signed_extension` in [8] does. Hence the map is onto $\ker d^0$. A vector is killed by every window marginal exactly when all its Walsh coefficients on covered sets vanish. (2) and (3) are computed in DC1–DC3. That the real limit is the local polytope and the image is the marginal polytope is the definition of each. □

Remark 4 (three cases, and contextuality)**.** The descent picture separates three situations, and only the middle one is contextuality in the sheaf-theoretic sense of [48]: local tables that agree on every overlap but come from no joint law.

1. *Incongruence.* The tables disagree on an overlap, and the Čech energy (a coboundary) is non-zero. This is inconsistent data, not contextuality. The worked example's conflicting trials (section 18, Step 5) are of this kind: energy 0.061, B margins 0.26 against 0.40, and $\lambda^* = 0.86$.

2. *Contextuality.* The tables agree on every overlap but do not glue. A signed joint law always exists, so linear algebra cannot see this case; only positivity fails.

3. *Gluing.* A joint law exists, as for every whole-record injection in Part II ($\lambda^* = 1$, theorem E(1)).

The family $A = B$, $B = C$, $A \neq C$ (the three-variable liar cycle) is the extreme of case 2.

- It is *strongly* contextual: $\lambda^* = 0$, so no fraction of it is explained by a joint law, and the contextual fraction $1 - \lambda^*$ is 1.

- Any assignment of A, B, C satisfies at most two of the events $A = B$, $B = C$, $A \neq C$, since the first two force $A = C$. So every joint law obeys

$$P(A = B) + P(B = C) + P(A \neq C) \leq 2.$$

  The liar cycle gives 3, the largest possible violation. The dual witness of the gluing linear programme is an inequality of this kind.

- The least negative mass of a signed extension is 1/2: the "negative probability" reading of the same contextuality.

- It needs a cycle in the cover. On a tree-shaped cover (for example AB, BC), consistent tables always glue.

For the lab, this means pooled-data adversaries cannot produce contextuality. Injection of whole records keeps $\lambda^* = 1$, so contextual data points to separate populations or to window-by-window manipulation, not to a hidden channel. Contextuality is the level-0 obstruction and needs no model. The sign of Δ_m and the tetrads are obstructions only relative to the two-type model (theorem M). All of this is checked in DC3 and DC14.

So the proposition the attached note proposed, "global compatible sections coincide with the joint laws", holds only on the algebraic spectrum modulo the kernel. On probability points it fails by exactly the contextuality the parent note measures. That failure is positivity, not cohomology, which is why the parent note needed λ^* and not only Čech classes.

The latent layer. The two-type model writes the pair covariances as $M_{ij} = v_i v_j$. On the graph Γ of observed pairs this is a *multiplicative* 1-cochain, with the unsigned coboundary

$$\delta^+ : C^0(\Gamma; \mathbb{R}^\times) \to C^1(\Gamma; \mathbb{R}^\times), \qquad (\delta^+ v)_{ij} = v_i v_j.$$

Split $\mathbb{R}^\times = \{\pm 1\} \times \mathbb{R}_{>0}$. On signs, written additively in $\mathbb{Z}/2$, δ^+ is the ordinary coboundary, since $+1 = -1$. On moduli, after taking logarithms, it is the unsigned incidence matrix B of Γ.

Theorem M (the descent tower)**.** *Two latent types.*

(1) Level 1, kernel: label switching is non-separatedness. $\ker(\delta^+)$ *on signs is* $H^0(\Gamma; \mathbb{Z}/2) = \mathbb{Z}/2$ *for connected* Γ*: the switch* $v \mapsto -v$*, which with* $r \mapsto -r$ *exchanges the types. On moduli the kernel is* 0 *if* Γ *is connected and not bipartite (the triangle,* K_4*). It is* 1*-dimensional if* Γ *is bipartite (the path* AB, BC*): there the scale of* v *is not pinned, which is the blow-up direction of theorem K(3).*

(2) Level 1, obstruction. *Pair tables have the algebraic form* $M = \delta^+ v$ *if and only if two classes vanish:*

- *the* sign class *in* $H^1(\Gamma; \mathbb{Z}/2)$*, whose cycle parities are the signs of* $\prod_{cycle} M_{ij}$*;*
- *the* modulus class *in* $\operatorname{coker} B$*.*

On the triangle the sign class is $\operatorname{sign}(M_{01}M_{02}M_{12}) = \operatorname{sign}(\Delta_m)$*, the ladder's rung, and* $\operatorname{coker} B = 0$*, so three variables carry no equation. On* K_4 *the sign classes form* $H^1(K_4; \mathbb{Z}/2) = (\mathbb{Z}/2)^3$*, and* $\operatorname{coker} B$ *is* 2*-dimensional and spanned by the two log tetrad ratios.* The tetrads are the modulus class. *Algebraically, the ideal of the image of* $v \mapsto (v_i v_j)$ *is the toric ideal of* B*: zero on the triangle and generated by the two tetrads on* K_4*. The number of sign cosets left unreached by real* v *is* $|H^1(\Gamma; \mathbb{Z}/2)|$*.*

(3) Level 1, walls. *Beyond both classes, a two-type explanation needs the box walls to leave a non-empty segment. This is a semialgebraic condition, not a cohomological one. With three variables, pair tables are two-type realizable if and only if the sign class vanishes and the walls leave a segment.*

(4) Level 2, triple coherence. *On the triple cover,* $r_S = m_S / v_S$ *is a 0-cochain, and the model requires it to be constant:* $\delta r = 0$*, with the quadruple window adding* $m_{0123} = v_{0123}(r^2 + 1)$*. A single-triple channel* χ_S *leaves levels 0 and 1 unchanged, because it has the same pair tables, and makes* $\delta r \neq 0$ *at first order. The obstruction moves up exactly one level, which is why a second triple detects what no pair check can (corollary 3).*

(5) The fibre. *At* $m = 3$ *the level-0 torsor over the data (the line* $P + s\chi_T$*) meets the level-2 condition (*$\hat{r}$ *inside the walls) exactly in the hiding segment.*

(6) Labels: a degree-0 obstruction. *On the labelled cover the batches pairwise overlap in the whole triple context, so the nerve is a simplex and* $H^q = 0$ *for* $q \geq 1$*. The shared statistic* m_T *is a 0-cochain, and one population means it lies in* $\ker \delta^0$ *(constants). The obstruction to a common realization is therefore the coboundary* $\delta^0 m_T \in B^1 \cong \mathbb{R}^K/\mathbb{R}$ *of dimension* $K - 1$*, in the group* $\mathbb{R}_{\chi_T}$ *of the hidden channel. It is a failure of overlap compatibility,* not *an* H^1 *class. Its energy splits orthogonally as*

$$\|\delta^0 m_T\|^2 = K\big(\textit{between-source} + \textit{within-source}\big),$$

and these are the coarse and the fine covers of the gluing filter. Theorem J's source statistic is the normalised between-source contrast. Its decision-relevant part is $\ell(\chi_T)\,\delta^0 m_T$ *(theorem L(4)). Pooling is the monad multiplication, which forgets* δ^0*, and the band of theorem J is how much of* $\delta^0 m_T$ *the noise can hide.*

(7) What the realization diagram remembers. *Every* 2×2 *table is a mixture of two product laws on its own, so the diagram of* images *(locally two-type laws) has the same limit as level 0. On the triangle, a law with* $\Delta_m < 0$ *has pair tables that glue and are each two-type, yet admit no global two-type realization. Only the diagram of* realizations $S \mapsto \{(v_i)_{i\in S}\}$ *sees this. Its limit is the set of global realizations, and its emptiness is measured by*

$$H^1_+(\Gamma;\mathbb{R}^\times) = C^1/\delta^+C^0 \cong H^1(\Gamma;\mathbb{Z}/2) \oplus \operatorname{coker} B,$$

which is $\mathbb{Z}/2$ *on the triangle and* $(\mathbb{Z}/2)^3 \oplus \mathbb{R}^2$ *on* K_4*. The ladder's rung and the tetrads are the two summands. The gluing of images is ordinary; realizability is not.*

Proof. (1) Over $\mathbb{Z}/2$, δ^+ is the graph coboundary, whose kernel is the locally constant functions. Over $\mathbb{R}$, the unsigned incidence matrix of a connected graph has rank m if the graph is not bipartite, and $m-1$ if it is. (2) $M = \delta^+ v$ splits into $\operatorname{sign} M = \delta(\operatorname{sign} v)$ and $\log|M| = B\log|v|$, and each is solvable exactly when the right side is in the image. On a cycle the sum of the sign cochain is the parity of $\prod M_{ij} < 0$. The toric ideal of B cuts out the Zariski closure of the image (DC7). (3) Two types exist when v exists and some t makes (w, a_1, a_2) valid, which is the segment condition (DC9). (4) χ_S changes no pair table and moves m_S alone, by $2^{m-3}\varepsilon s$ (theorem F(4)). (5) Check DC11. (6) The nerve of a cover in which every finite intersection is non-empty is a full simplex, which is contractible. $\sum_{b<c}(m_c - m_b)^2 = K\sum_b (m_b - \bar{m})^2$, and the total sum of squares splits into between- and within-source parts (DC12). (7) A 2×2 table is the mixture of its two rows, each a product law; the counterexample is DC13, and the isomorphism is (2). □

Corollary 4 (the descent filtration)**.** *On pair data, together with triple windows when they are observed, set*

- D_0*: compatible families;*
- $D_1 \subseteq D_0$*: families that glue (the marginal polytope);*
- $D_2 \subseteq D_1$*: sign class* 0*;*
- $D_3 \subseteq D_2$*: modulus class* 0*;*
- $D_4 \subseteq D_3$*: the walls leave a segment;*
- $D_5 \subseteq D_4$*: triple coherence* $\delta r = 0$*.*

Each inclusion is strict. The witnesses are:

- $D_0 \setminus D_1$*: the liar cycle;*
- $D_1 \setminus D_2$*: a joint law with* $\Delta_m < 0$*;*
- $D_2 \setminus D_3$*: a joint law on four variables with trivial signs and non-zero tetrads;*
- $D_3 \setminus D_4$*: a joint law with both classes zero and no segment;*
- $D_4 \setminus D_5$*: a two-type law with a* χ_S *channel.*

These are checked in DC3, DC5, DC10 and DC16. The order of D_2, D_3 and D_4, which puts algebraic before semialgebraic and signs before moduli, is a convention. The stages are not. What the filtration predicts beyond the moment equations of a given cover is a general count. For any *connected pattern of observed pairs with E edges on m variables:*

- *the two-type model imposes exactly $E - \operatorname{rank} B$ equations on the pair moments: $E - m$ when the graph is not bipartite, $E - m + 1$ when it is;*
- *there are $b_1 = E - m + 1$ independent sign conditions;*
- *exactly 2^{-b_1} of the sign patterns are reachable.*

This is checked against Jacobian ranks on 57 *random graphs (DC15).*

level	kernel (non-separatedness)	obstruction	certificate
0: ambient laws	Walsh characters of uncovered sets (the fibre's directions)	none algebraically; positivity on real points	LP witness (Bell-type inequality), λ^*
1: pair layer, signs	$\mathbb{Z}/2$: the label switch	$H^1(\Gamma; \mathbb{Z}/2)$	$\operatorname{sign} \Delta_m$ on each cycle
1: pair layer, moduli	0 (non-bipartite) or scale (bipartite)	$\operatorname{coker} B$	tetrads ($m \geq 4$)
1: walls	—	positivity	the segment is empty
2: triple layer	—	$\delta r \neq 0$	a second triple; the quadruple equation
labels (degree 0)	—	$\delta^0 m_T \neq 0$ across batches	the gluing tests of theorem J

Table 10: The descent tower: what each level fails to separate, what obstructs descent there, and what certifies the obstruction (checks DC1–DC16).

Where categorification is forced. The first genuinely categorical step is already in (1)–(2). Two local realizations on an overlap agree only up to the label switch. So the realizations form a groupoid, the quotient stack $[\Theta_S/(\mathbb{Z}/2)]$, not a set, and the sign class is the class of the gluing data in Čech H^1 with coefficients in the automorphism group $\mathbb{Z}/2$. A global realization exists when this $\mathbb{Z}/2$-torsor on Γ is trivial. This is the precise sense in which the latent layer needs realization *spaces* rather than sets. Higher stacks, hypercovers and operadic realization would enter only with more types: there the automorphisms permute k types, the coefficient group becomes the symmetric group S_k, and the cohomology is non-abelian. We leave that open.

Two comonads. Two different comonads appear in this paper, and they should not be confused.

- The *provenance comonad* $W = L \times (-)$ (sections 12 and 20) is the environment comonad of a set L of labels (source, batch). Its counit $\varepsilon : L \times X \to X$ forgets the label, and its comultiplication copies it. Its co-Kleisli maps $L \times X \to A$ are the *label-aware decision rules*. A rule that pools factors through ε; the gluing filter of theorem J does not. The counit of a comonad always satisfies the comonad laws, so "detection" is never a failure of the counit. It is the gap between a co-Kleisli rule and its factorisation through ε.

- The *cover comonad* belongs to a cover $f : \coprod_i U_i \to X$. Write f^* for restriction to the pieces and f_* for its right adjoint, so $f^* \dashv f_*$. The adjunction has a unit $\mathrm{id} \Rightarrow f_* f^*$ on the side of X and a counit $f^* f_* \Rightarrow \mathrm{id}$ on the side of the cover. The comonad is therefore $G = f^* f_*$, and it acts on presheaves over the *cover* $\coprod_i U_i$, not over X. Its counit is the counit of the adjunction, and its comultiplication is $f^*(\mathrm{unit}) f_*$. Its coalgebras are the descent data of the cover [51, 53]. The monad $f_* f^*$ on the side of X is the other half of the same adjunction, and its cosimplicial resolution is the Čech complex of (i) below. This is the comonad of the descent tower.

The two meet in the labelled cover of theorem J. There the batches are the pieces of the cover, so the label set L indexes the cover's pieces. Pooling is the multiplication $\mu : D(DX) \to DX$ of the distribution monad D [59], and the gluing condition is that the family of batch laws lies in the image of the unit η (section 20).

Proposition 10 (the comonadic form of the descent tower)**.** *Let $\mathcal{U} = \{U_i\}$ be the cover by observed windows, and let K send a joint law (signed, or a probability law) to its family of window marginals.*

(i) Comonadic realization of descent. *The cosimplicial resolution of the cover comonad has n-th term $\prod F(U_{i_0} \cap \cdots \cap U_{i_n})$ over all ordered $(n+1)$-tuples, repetitions included. Its unnormalised cochain complex has the same cohomology as the alternating Čech complex of proposition 9, because the degenerate part is acyclic. On the pair cover the unnormalised complex has dimensions $12, 24, 54, \ldots$ and cohomology $7, 0, 0$, as the alternating one does (check DC17). So the lab's Čech complex* is *the normalised classical comonadic resolution; no new comparison construction is involved. For a single presheaf F, the descent data of $\mathcal{U}$ are the equalizer $\ker d^0$ of this resolution, and $K : F(X) \to \ker d^0$ is the comparison map at F. Descent at F (the sheaf condition: separated and gluing) means that K is a bijection.*

(ii) Three distinct failures of K.

(a) Injectivity fails. $\ker K$ *is spanned by the Walsh characters of the uncovered sets, which is the hiding fibre and the hidden channel. This is a loss of identifiability. By itself it is not contextuality.*

(b) Surjectivity fails, through positivity only. *On signed laws K is onto $\ker d^0$ and the resolution is acyclic in degrees ≥ 1. This is proved for the lab's covers, not assumed. On probability laws, K misses exactly the contextual families. That failure is decided by a feasibility linear programme and its dual witness, and not by a cohomology class of either natural linearisation (proposition 11(1)).*

(c) Latent descent fails with automorphisms. *Two-type realizations form a groupoid whose automorphisms are the label switch $\mathbb{Z}/2$. Descent for groupoid-valued data is governed by non-abelian Čech H^1 with coefficients in the automorphisms, and here the obstruction is a class in $H^1(\Gamma; \mathbb{Z}/2)$, the sign of Δ_m (theorem M(2)). It is distinct from (a) and (b): a law can glue and still carry this class (the "exactly one harm" law of Step 14).*

(iii) The source-labelled discrepancy. *On the labelled cover the triple-moment discrepancy is the degree-0 coboundary $\delta^0 m_T$ on a simplex nerve (theorem M(6)); it is not an H^1 class. On the unlabelled cover, whole-record injection stays in the image of K ($\lambda^* = 1$), so the discrepancy is invisible there. On the labelled cover, assume two things: every batch's tables arise from an actual batch law (a probability law on all m variables), which is the case under whole-record injection; and δ^0 vanishes on every window, so that all batches agree on every table. Then the*

common compatible family is realised by any one batch law. It is therefore the image under K of a genuine probability law, not merely of a signed solution: positivity comes for free under these two assumptions, and only under them. At $m = 3$, in addition, the batch laws themselves coincide. On three binary variables the 8 cells have 7 free coordinates, and the singles, the pairs and m_T supply exactly 7 that determine the law (the moment normal form of [8]). Vanishing does not place the common law in the two-type model; that needs stages D_2–D_5 (corollary 4). In data $\delta^0 m_T$ is never exactly zero, and "vanishing" means passing the tests of theorem J.

Proof. (i) The resolution of G applied to F is the Čech nerve of the cover evaluated in F [56, 53]. Normalisation, which quotients by degeneracies, preserves cohomology [57], and the ordered and alternating Čech complexes have the same cohomology [56]. The numbers are DC17. The equalizer of the first two coface maps consists of the families that agree on overlaps. (ii)(a), (b): proposition 9 and proposition 11. (ii)(c): theorem M. The "exactly one harm" law is a probability law, so it glues, and its class is DC5 and WD5. (iii) If all batches agree on every window, any one batch's law realises the common family. At $m = 3$ the law is a polynomial function of singles, pairs and m_T (the moment normal form of [8]). □

How each classical theorem applies.

- *Eilenberg–Moore* [54]. Every adjunction $f^* \dashv f_*$ gives a comonad $G = f^* f_*$ on presheaves over the cover and a monad $f_* f^*$ on the side of X, each with its category of (co)algebras and a comparison functor from presheaves on X. In the lab, f^* restricts a presheaf on all contexts to the windows, and the coalgebras are families of window data with restriction-compatible structure.

- *Bénabou–Roubaud* [51]. For a bifibration satisfying the Beck–Chevalley condition, the descent data along f are exactly the (co)algebras of the induced (co)monad. This is what licenses calling the families that agree on overlaps "coalgebras": the lab's restriction maps are marginalisations, which compose and commute with restriction to sub-contexts. The lab uses only its object-level consequence, that descent data are the equalizer of the resolution.

- *Beck's comonadicity theorem* [52, 55]. f^* is comonadic, so that the comparison functor is an equivalence and f is an effective descent morphism [53], if and only if f^* reflects isomorphisms and preserves the relevant equalizers. The lab works with one presheaf at a time, so it sees the object-level counterparts of Beck's two conditions. (ii)(a) is a failure of separation: K identifies laws that differ by χ_S, which is the object-level shadow of not reflecting isomorphisms. (ii)(b) shows that, even where the linear comparison is onto, the probability comparison is not. At the level of these presheaves, then, the window cover does not give effective descent, and (ii) names the two reasons. We do not claim a statement about the comparison functor on all presheaves.

- *Godement* [56]. The cosimplicial resolution of a comonad computes Čech (and derived) cohomology, and the ordered, alternating and normalised versions agree. This is (i), checked numerically in DC17.

- *Normalisation and Dold–Kan* [57]. For cosimplicial abelian groups, the normalised complex is chain-homotopy equivalent to the unnormalised one. This is why the paper's alternating Čech complex can stand for the comonadic resolution.

- *Non-abelian H^1* [58]. Gluing groupoid-valued data up to their automorphisms is classified by H^1 with coefficients in the automorphism sheaf. In the lab this is the $\mathbb{Z}/2$ of the label switch, giving (ii)(c). With k types the coefficients become the symmetric group S_k.

The lab's contribution is therefore not a new equivalence between comonadic descent and Čech cohomology. It is the explicit separation and calculation, inside one realization problem, of four phenomena: the kernel, positivity, provenance, and the latent automorphisms. It also pins down the degree and the meaning of the source-labelled discrepancy.

25 Relation to the CSD framework

The Categorified Spectral Duality (CSD) programme [49] builds, for an admissible operator-semantic system $A = (\mathcal{O}_A, \mathcal{C}_A, \tau_A, \rho_A)$, a spectral stack $\mathrm{Spec}(A)$ by operadic syntactic realization, a left Kan extension and hypersheafification. It then defines an obstruction cohomology $\mathrm{Obs}^\bullet(A) = H^\bullet(\mathrm{Spec}(A), F_A^{\mathrm{lin}})$ from a descent filtration, and assigns contextuality to degrees through $\mathrm{CtxSpec}(A) = \{n : \mathrm{Obs}^n(A) \neq 0\}$. Several of its results are stated under hypotheses (geometricity, perfectness, reconstruction, hyperdescent, full faithfulness). Some computations are deferred to future work, including the linearised sheaf F_A^{lin} for its examples.

We do not import that machinery. The lab is a classical, commutative setting with no operad, so $\mathrm{Spec}(A)$ is not constructed here. What the lab does supply are the data the two frameworks share:

- a context category: the windows and their intersections;
- a covering: by the windows;
- a presheaf of local realizations: local laws, or local two-type realizations up to the label switch.

On these data we can prove what any Čech-type obstruction theory can and cannot do. This is the connection we can actually establish.

Proposition 11 (the lab inside CSD: what an obstruction class can see)**.** *On the lab's covers (the pair cover at $m = 3, 4$):*

(1) No linear or support-based class characterises gluing of probability laws. *Let θ be an invariant of compatible families that factors through either*

(i) the linear Čech cohomology of signed families, or

(ii) the support presheaf (which local outcomes have positive probability).

Then $\theta = 0$ cannot be equivalent to "the family glues". For (i), linear Čech cohomology vanishes in every degree ≥ 1 (DC2), yet the liar cycle is compatible and does not glue. For (ii), the noisy liar $(1-q)\,\mathrm{liar} + q\,(\text{independent uniform})$ has full support for every $q > 0$. It is contextual for $0 < q < 2/3$: $\lambda^* < 1$, and the Bell value $3 - \frac{3}{2}q$ exceeds 2. It glues from $q = 2/3$. So a support-based invariant cannot tell $q = 0.3$ from $q = 1$. For probability laws, gluing is decided by positivity, a linear programme and its dual witness (remark 4), not by the vanishing of such a class. The CSD statement "$\mathrm{Obs}_{\mathrm{CSD}}(A) = 0$ iff every compatible family glues" can therefore hold for probabilistic models only with a linearisation of neither kind. The CSD paper does not construct one, and we do not know whether one exists.

(2) Where Čech cohomology on Γ does detect, the degree is 1. *For all-versus-nothing families the supports are affine parity conditions. The liar cycle's are* $a+b=0$, $b+c=0$, $a+c=1$. *Its edge-parity cochain* $(0,0,1)$ *on the graph* Γ *of windows is not a coboundary, which gives a non-zero class in* $H^1(\Gamma;\mathbb{Z}/2)$ *[50]. This is the same group as the latent sign class of theorem M(2). The nerve of the pair cover is a hollow triangle at* $m=3$ *(reduced Betti numbers* $b_1=1$, $b_2=0$*), and at* $m=4$ *it has* $b_1=3$, $b_2=b_3=0$. *So Čech* H^2 *vanishes on these covers for every coefficient group, and no class of degree* ≥ 2 *arises. Computed on these data, the "contextuality polynomial" of the liar cycle is* t *with* $\mathbb{Z}/2$ *support coefficients and* 0 *with real linear coefficients.*

(3) The lab's degree-two obstruction is a deformation obstruction, not a gluing obstruction. *The class that* does *sit in degree two in the lab is the André–Quillen class* $\mathrm{ob}(A)=[\frac{1}{2}H(A,A)]\in \operatorname{coker} J$ *of theorem G(4). It is the obstruction to extending a first-order deformation along the tetrad scheme. This matches CSD's reading of* H^2 *of a tangent complex as "primary obstruction classes" in deformation theory. In the lab the two obstructions are distinct objects, computed separately: gluing in Čech degree* 1*, deformation in cotangent degree* 2.

(4) What transfers. *CSD's realizations form* ∞*-groupoids. The lab's first genuinely groupoid-valued case is the two-type realization stack* $[\Theta_S/(\mathbb{Z}/2)]$ *(theorem M), whose gluing obstruction is in* $H^1(\Gamma;\mathbb{Z}/2)$ *and whose automorphisms are the label switch. With* k *types the automorphism group is the symmetric group* S_k *and the obstruction becomes non-abelian. That is the first place where higher coherence of the CSD kind could appear in the lab. We have not computed it.*

Proof. (1)(i) The invariant takes the same value on the liar cycle and on any gluing family, because both have zero class in every degree ≥ 1. (ii) The noisy liar at $q=0.3$ and $q=1$ have the same supports, all four cells in every window, yet only the second glues. The numbers are computed in CX4. (2) The parity system is solvable exactly when the cochain is a coboundary (CX3). The Betti numbers are computed over $\mathbb{Z}/2$ (CX2). (3) Theorem G(4) and check IN11. (4) Theorem M. □

CSD notion	lab instance	status
context category $\mathcal{C}_A$, topology τ_A	windows and intersections; covering by windows	defined (section 24)
sheaf of local realizations F_A	local laws; two-type realizations up to label switch	defined; descent computed (proposition 9, theorem M)
$\mathrm{Spec}(A)$ via operad, Lan, hypersheafification	not constructed (no operad in a commutative classical setting)	not instantiated
$\mathrm{Obs}_{\mathrm{CSD}}=0 \iff$ gluing	fails for linear and support linearisations of probability laws	proved (proposition 11(1))
primary obstruction in degree 2	gluing obstruction in Čech degree 1; Čech $H^2=0$ on these covers	computed (CX2, CX3)
H^2 of tangent complex as obstructions	André–Quillen class of theorem G(4)	matches
realizations as ∞-groupoids	the stack $[\Theta_S/(\mathbb{Z}/2)]$	first case; S_k case open
Obs^n, $n\geq 3$ (higher contextuality)	none on these covers	vanish (nerve dimension)

Table 11: The CSD framework on the lab's data (checks CX1–CX5).

Summary. On classical probability tables the descent filtration does distinguish real levels (corollary 4, table 10), but the obstruction that decides gluing is positivity. It is not a cohomology class of the linear or the support linearisation. Where a Čech class does decide (the all-versus-nothing families and the latent sign), it sits in degree 1, in the group $H^1(\Gamma;\mathbb{Z}/2)$. Higher Čech degrees do not arise on these covers, and in the lab degree two is the deformation degree. Degrees are not absolute: they depend on the complex. Roumen's order cohomology places the complete extension obstruction in *relative* degree 2, via a connecting map out of degree 1, and that convention agrees with CSD's (section 26). A proof that CSD's Spec restricts to the lab's diagram would need its construction for commutative classical systems. If such a construction is given, the lab supplies computed test cases for every invariant above.

Relation to Ikeda's entanglement obstruction. Ikeda [60] treats entanglement as a cohomological obstruction in the quantum marginal problem. The setting is a presheaf S of density operators on subsystems, with restriction by partial trace, and the restriction map $j_{\mathcal{U}} : S(I) \to H^0(\mathcal{U}, S)$. The paper names two failure modes: (A) *gluing failure*, when $j_{\mathcal{U}}$ is not surjective, and (B) *non-uniqueness*, when it is not injective. These are exactly the lab's failures (ii)(b) and (ii)(a) of K in proposition 10. The correspondence is more than an analogy: the lab is the commutative part of Ikeda's problem.

Proposition 12 (the lab inside the quantum marginal problem)**.** *Identify a law on $\{0,1\}^m$ with the diagonal density operator in the product basis.*

(1) Dephasing is a retraction. *Full dephasing Δ in the product basis commutes with every partial trace. So if a quantum state ρ has diagonal window marginals, then $\Delta(\rho)$ is a classical law with the same marginals. Hence a diagonal compatible family glues quantum-mechanically if and only if it glues classically. The lab's contextual families, the liar cycle among them, are gluing failures (A) in Ikeda's sense too.*

(2) Non-uniqueness grows. *On the three-qubit pair cover, the kernel of the classical K is spanned by χ_{ABC}, which is $-Z\otimes Z\otimes Z$ on the diagonal. The kernel of the quantum restriction on Hermitian operators has dimension 27, spanned by the weight-3 Pauli strings. Coherences add 26 directions that no two-qubit window sees, and the lab's hidden channel is the single commuting one.*

(3) Witnesses. *Ikeda's separability witnesses ($\mathrm{Tr}(W\sigma) \geq 0$ for all separable σ, with $\mathrm{Tr}(W\rho) < 0$ detecting entanglement) and the lab's dual witnesses of the gluing linear programme are both supporting hyperplanes of a convex set. They separate a point from the set of realizable states, and this is the positivity side of the obstruction in both settings.*

Proof. (1) Δ keeps the diagonal and Tr_C sums diagonal entries over C, so $\Delta \circ \mathrm{Tr}_C = \mathrm{Tr}_C \circ \Delta$ (check QU1). If ρ realises the family, so does $\Delta(\rho)$ (QU2). (2) A Pauli string acting non-trivially on all three qubits has zero partial trace onto any two of them. The 64-dimensional space of Hermitian operators has $1 + 9 + 27 + 27$ strings by weight; the windows see weights 0 to 2, and the kernel is the 27 weight-3 strings (QU3). (3) Hahn–Banach separation in finite dimensions. □

Ikeda's Theorem 3.1 [60] is a different kind of result, and it should be read as a comparison. For a smooth family of full-rank states over a parameter manifold X, it takes the Uhlmann connection A on the amplitude bundle and a separability-witness field W_{sep} that is covariantly constant ($D_A W_{\mathrm{sep}} = 0$). The Bott–Shulman–Stasheff–Dupont construction then gives a Čech–de Rham cocycle whose class is $\left[(2\pi i)^{-k}\,\mathrm{Tr}(W_{\mathrm{sep}}F^k)\right]_{\mathrm{dR}} \in H^{2k}_{\mathrm{dR}}(X)$. For $W_{\mathrm{sep}} = 1$ the form $(2\pi i)^{-k}\,\mathrm{Tr}\,F^k$

is $(-1)^k k!$ times the k-th Chern-character form; it agrees with a Chern class only after Newton's identities. Its degree is even and positive, and it lives on a parameter space. The lab's obstructions are different: positivity, degree 0 ($\delta^0 m_T$) and degree 1 ($H^1(\Gamma;\mathbb{Z}/2)$). The lab's closest object is the Kähler class of the toric fibre $\mathbb{CP}^1$ (Step 13). Its area over 2π is the lattice length $s_+ + |s_-| = 0.040$, which is the adversary's total capacity on the line. The hidden channel moves the point but leaves that length unchanged (QU4). So the curvature-type invariant measures the size of the fibre, while the descent and positivity tests decide realizability and where the truth sits. The two are complementary, and the lab shows it on one example.

26 Markov categories: factorisation, sufficiency, de Finetti and Blackwell

Markov categories [61] give probability a compositional, diagrammatic foundation. Morphisms are Markov kernels; every object has a copy map; and conditional independence, sufficiency and almost-sure equality are defined by diagrams. Three theorems of that programme apply to the lab directly, and a fourth idea, tangent structure, applies at one precise point. Each is shown below on a small example worked by hand (fig. 21, checks MK1–MK6).

Proposition 13 (the lab in a Markov category)**.** *Work in* FinStoch*, finite sets and stochastic matrices.*

(1) The two-type model is a factorisation. *A law P on $\{0,1\}^3$ is two-type exactly when it factors as*

$$I \xrightarrow{w} Z \xrightarrow{\text{copy}} Z \otimes Z \otimes Z \xrightarrow{K_A \otimes K_B \otimes K_C} X_A \otimes X_B \otimes X_C,$$

with Z a two-point object. This says that A, B and C are conditionally independent given Z. The hiding segment is the set of such factorisations with fixed pair marginals.

(2) Sufficiency on the line. *For the family $P_s = P + s\chi_{ABC}$, every pair window is ancillary for s: its Fisher information is 0. The triple window carries $I(s) = \sum_x \chi(x)^2/P(x) = \sum_x 1/P(x)$ at $s = 0$. The pair tables are therefore not sufficient for the hidden coordinate, which is Fritz's synthetic sufficiency [61] read on the fibre.*

(3) De Finetti for batches. *The batch laws define a mixing measure $\pi = \frac{1}{K}\sum_b \delta_{P_b} \in D(DX)$. Records are exchangeable across all batches if and only if π is a Dirac, and only partially exchangeable (within batches) otherwise; this is the de Finetti picture of [62, 63]. Pooling is the multiplication μ, and $\mu(\pi)$ can equal a whole-pool injection exactly. The population version of theorem J's test statistic is $\mathrm{Var}_\pi(m_T)$, which is zero if and only if π is a Dirac.*

(4) Blackwell. *If an experiment E_2 is a garbling of E_1 ($E_2 = E_1 G$ for a stochastic G), then both the Bayes value and the ex-ante maxmin value (over randomized rules) satisfy $V(E_1) \geq V(E_2)$ for every decision problem [64]. A* fixed *rule can lose value under garbling, and so can prior-by-prior updating (proposition 8). These are the only sources of negative value of information in the lab.*

Proof. (1) The composite has cells $\sum_z w_z \prod_i K_i(z, x_i)$, which is the definition of a two-type law. (2) $\partial P_s/\partial s = \chi$, and χ has zero marginal on every proper window. (3) Two records from batches b and c have joint law $P_b \otimes P_c$. Swapping them gives $P_c \otimes P_b$, and the two agree exactly when $P_b = P_c$. Moreover $\mathrm{Var}_\pi(m_T) = 0$ exactly when m_T is π-a.s. constant, and on the line that means all batch laws are equal. (4) Any rule for E_2 composed with G is a rule for E_1 with the same

value against every state. The Bayes case is the classical Blackwell–Sherman–Stein theorem, in its categorical form [64]. □

Worked example 1: the string diagram. For the drug example, $w = (0.6, 0.4)$, and the kernels (rows $z = 1, 2$; columns $x = 0, 1$) are

$$K_A = \begin{pmatrix} 0.50 & 0.50 \\ 0.95 & 0.05 \end{pmatrix}, \quad K_B = \begin{pmatrix} 0.90 & 0.10 \\ 0.50 & 0.50 \end{pmatrix}, \quad K_C = \begin{pmatrix} 0.60 & 0.40 \\ 0.90 & 0.10 \end{pmatrix}.$$

One cell by hand: $P(000) = 0.6 \cdot 0.5 \cdot 0.9 \cdot 0.6 + 0.4 \cdot 0.95 \cdot 0.5 \cdot 0.9 = 0.162 + 0.171 = 0.333$, as in Step 0. The two walls of the hiding segment are two other factorisations. They have the same pair tables and triple harms 0.66925 and 0.66077 (MK1).

Worked example 2: Fisher information per window. The cells are 0.333, 0.127, 0.189, 0.031, 0.171, 0.109, 0.027 and 0.013, so

$$\begin{aligned} I(s) &= \tfrac{1}{0.333} + \tfrac{1}{0.127} + \tfrac{1}{0.189} + \tfrac{1}{0.031} + \tfrac{1}{0.171} + \tfrac{1}{0.109} + \tfrac{1}{0.027} + \tfrac{1}{0.013} \\ &= 3.00 + 7.87 + 5.29 + 32.26 + 5.85 + 9.17 + 37.04 + 76.92 = 177.41. \end{aligned}$$

The rare cells dominate: 011, 110 and 111 supply 82% of it. Each pair window gives 0. The efficient standard deviation of $\hat{s}$ is $1/\sqrt{177.41} = 0.0751$ per record, a little below the moment estimator's $\sigma_m = 0.0797$ used in theorem J (MK2).

Worked example 3: four batches. Three batches are genuine, and one has a share 0.2 of the hidden channel ($s = 0.027$).

- The batch laws sit on the line at $s = 0, 0, 0$ and 0.0054. So the mixing measure is $\pi = \frac{3}{4}\delta_P + \frac{1}{4}\delta_{P_{0.2}}$, which is not a Dirac.
- Pooling gives $\mu(\pi) = P_{0.05}$ exactly, at $s = 0.00135$. That is past the switch (0.001) and short of the detection wall (0.00225). Every pooling rule therefore sees theorem F at share 0.05, the decision is tilted, and nothing is flagged ($\hat{r} = -0.621$, against $r_c = -0.566$).
- Records drawn from a genuine batch and from the adversarial batch are not exchangeable: $\max |P_b(x)P_c(y) - P_c(x)P_b(y)| = 0.0028$. Records from two genuine batches are exchangeable.
- The Dirac test is $\mathrm{Var}_\pi(m_T) = \frac{3}{4} \cdot \frac{1}{4} \cdot (0.0054)^2 = 5.47 \times 10^{-6}$, the quantity that the source statistic of theorem J estimates (MK3, MK4).

Worked example 4: Blackwell on a three-point slice. Take three states, the two walls and the truth: $r = -0.763$, $r = -0.408$ and $r = 0.573$. The utilities are $U_T = (0.33075, 0.333, 0.33923)$ and $U_P = 0.332$. There are three experiments: exact, $E_1 = I_3$; noisy,

$$E_2 = \begin{pmatrix} 0.9 & 0.1 \\ 0.6 & 0.4 \\ 0.1 & 0.9 \end{pmatrix} \quad \text{(signals "low" and "high")},$$

which is $E_1 G$ with $G = E_2$; and none, $E_0 = E_2 \mathbf{1}$.

- *Bayes, uniform prior.*

- Without a probe, $\bar{U}_T = 0.334327 > 0.332$, so the choice is T.
- With the noisy probe, the posterior after "low" is $\propto (0.9, 0.6, 0.1)$, so $E[U_T \mid \text{low}] = 0.332124 > 0.332$, and the choice is still T. The probe never changes the act and is worth 0: it sits on a plateau at this prior.
- With the exact probe, the value is $\frac{1}{3}(0.332 + 0.333 + 0.33923) = 0.334744$.

So $V_0 = V_2 \leq V_1$.

- *Ex-ante maxmin.* At $r = -0.763$ no act exceeds 0.332, so every rule's worst case is at most 0.332, and "always P" attains it. So $V_0 = V_2 = V_1 = 0.332$: monotone, and constant at the saddle point.
- *A fixed rule.* "T if the signal is high" has worst case $0.9{\cdot}0.332{+}0.1{\cdot}0.33075 = 0.331875 < 0.332$: negative value, from the rule, not from the information.

On 300 random decision problems and random garblings, both Bayes and ex-ante maxmin values are monotone (MK5, MK6).

Tangent structure, built. Tangent categories [67] formalise differentiation through an endofunctor T with additive bundle structure. The companion [66] builds the structure for the lab and checks every axiom on Δ_3, Δ_4 and a non-simplex polytope (checks TG1–TG13, TG6b). On polyhedra with affine maps (on simplices, the stochastic matrices), the feasible-direction functor satisfies all the equational axioms, and an injection direction $Q - P$ is exactly a tangent vector. Its additive bundles are commutative monoids without negatives at the boundary. Universality of the vertical lift holds exactly at relative-interior points: at an edge of Δ_3, $(x, u, 0, -u)$ lies over the zero section but not in the image of the lift. On the open simplex the slopes of theorem E are $T(F)(P, Q-P)$, the stealth direction spans the vertical bundle of the marginal map, and the value function fails to preserve $+$ on a wall by exactly $|\ell(\xi)|$ (theorem L(4)). On the tetrad scheme the Weil structure has a universal lift everywhere, but $T^2V \to T_2V$ is not onto on the two-coincidence stratum. The obstruction is $[H(u, v)] \in \operatorname{coker} J$, whose diagonal is twice the André–Quillen class of theorem G(4). Positivity breaks universality; the algebra breaks completion. Used on decision boundaries, the structure gives three results. A smooth map of simplices is a second-order morphism at a corner iff each vanishing output cell has a bilinearly non-negative Hessian on a cone of null directions; Bayesian conditioning qualifies. The post-probe boundary $\{U_T(\hat{r}) = U_P\}$ is curved both ways, so two data sets that each choose T can, pooled, choose P, which no affine decision allows. And 2-jet budgets can reverse the first-order verdict of theorem E(3). At corners the companion keeps layered maps with restriction criteria (strata, resolutions, multi-maps). Each open face is a tangent category, and the corner's universality failure is exactly a change of layer. Blowing up resolves the coincidence and the merge. The explanation multi-map has corners at the edge of its domain and branches that end at positivity walls. The partial maps form a restriction category with joins (axioms checked), the tangent structure on a hiding sheet is the sheet's tangent cone, and pooling along any chord is decided exactly by two polynomials of degree at most six: globally, pooled data fail mostly by acquiring the triangle's negative sign class, not by flipping.

Roumen's order cohomology. Roumen [68] studies classical realizability as the extension of a state σ on an effect algebra E along a weak injective morphism $i : E \hookrightarrow F$. Here E is generated by the windows' outcome events, $F = \mathcal{P}(X)$ with $X = \{0, 1\}^3$, and i sends an event to its cylinder set. His two constructions match the two sides of proposition 11(1) exactly.

- *Cyclic cohomology is the real-linear case.* If σ extends, the connecting class $\partial(j(\sigma))$ vanishes (his Theorem 9.1), but "the converse does not hold, so false positives may arise." The reason, given at the start of his Section 10, is that "the coboundary of a state is zero if and only if it extends to a signed state." This is our no-go (i), reached independently. On the noisy liar the signed test passes for every q (check RO1).

- *Order cohomology is complete because the coefficients carry the cone.* The coefficients are ordered abelian groups, and kernels are replaced by *precones* $\mathrm{prec}(f) = f^{-1}(B^+)$, with $H^n_{\leq} = \mathrm{prec}(\delta)/\mathrm{im}(\delta)$. These are commutative monoids, not groups, and the long sequence is not exact. Degree 1 is the monoid of positive additive maps: $H^1_{\leq}(E) \cong \{\varphi : E \to \mathbb{R}_{\geq 0} \text{ additive}\}$. His Theorem 10.5 says that σ extends if and only if $\sigma \in \mathrm{prec}\big(\partial : H^1_{\leq}(E) \to H^2_{\leq}(F, E)\big)$, and Proposition 10.4 identifies that precone with the image of restriction from $H^1_{\leq}(F)$. With $F = \mathcal{P}(X)$, the criterion asks whether σ is the restriction of a positive measure on X. That is the marginal-problem linear programme, so the criterion is $\lambda^* = 1$.

We computed the test literally: an additive map $\mathcal{P}(X) \to \mathbb{R}_{\geq 0}$, constrained on all 6049 disjoint pairs, whose restriction along i is σ. On the noisy liar it fails exactly for $q < 2/3$ and holds from $q = 2/3$, where the Bell value $3 - \frac{3}{2}q$ reaches 2 (RO2). It agrees with the linear programme on 200 random compatible families (RO3). So order cohomology does detect full-support contextuality, but it is not a linearisation that escapes proposition 11: it succeeds because positivity is built into its coefficients, and computing it is the lab's linear programme. It is the precise sense in which the positivity obstruction becomes cohomological once the coefficients are ordered. Its obstruction lives in relative degree 2, which shows again that the degree of an obstruction is a property of the complex chosen, not of the phenomenon.

Injection as a soft intervention. Categorical causal models [69] and partial Markov categories [70] give a language for interventions and updates. In the lab the correspondence is exact once the causal model is named. Take the graph $E \to X$, where the provenance node $E \in \{\text{genuine}, \text{injected}\}$ is the label of the provenance comonad, $X \mid E{=}\text{genuine} \sim P$ (generated through the latent type Z), and $X \mid E{=}\text{injected} \sim Q$. Whole-record injection at share ε is the soft intervention that replaces E's mechanism δ_{genuine} by Bernoulli(ε). The observed law is then exactly P_ε on every window (CA1). Per-window injection is not a single causal model: it needs a different intervention in each window. That is exactly the case that can break gluing; injecting into the AB window alone gives $\lambda^* = 0.78$, while one intervention always keeps $\lambda^* = 1$ (CA2). An additive perturbation of a joint law is not automatically a Pearl-style intervention; it becomes one only relative to a named mechanism, here E. No lab theorem needs the categorical do-calculus yet, so we use it as language only.

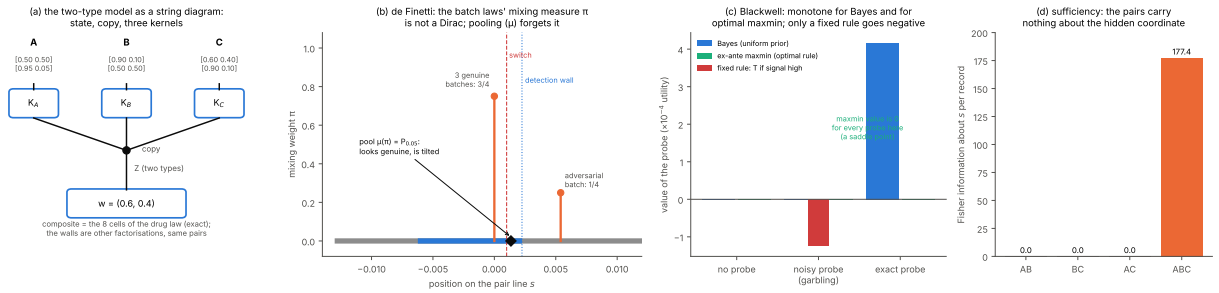


Figure 21: Markov categories on the worked examples. (a) The two-type model as a string diagram: the state w, the copy map, and three kernels (their matrices above). (b) De Finetti for four batches: the batch laws on the pair line, with their mixing weights. The pool $\mu(\pi)$ (diamond) lands past the switch and inside the detection wall, so it is tilted and unflagged. (c) Blackwell on a three-point slice: value of the probe for Bayes (uniform prior) and for ex-ante maxmin, for no, noisy and exact probes. Only a fixed rule goes negative. (d) Fisher information about the hidden coordinate per record: zero in every pair window.

27 Assembly: descent, identification and decision

The theorems of the paper answer three separate questions about observed window tables O: does a global law exist, does a latent-model law exist, and how does a decision respond? The answers are not one obstruction. This section states them together, with three separate conclusions. It is an assembly of results proved above, not a new theorem; its value is that it shows where each conclusion comes from.

Theorem N (descent, identification, decision)**.** *Two latent types, the pair cover on three binary variables, and window tables O (with source and batch labels where stated).*

(i) Existence. *The global laws consistent with O form the polytope $F(O) = (P + \ker K) \cap \Delta$, a translate of the kernel $\ker K = \mathbb{R}\chi_{ABC}$ cut by positivity. $F(O)$ is non-empty if and only if O is compatible on overlaps (zero Čech energy) and passes the positivity programme ($\lambda^* = 1$). Equivalently, σ_O lies in the precone of Roumen's connecting map. Compatibility alone gives only a signed law. For labelled batches whose tables come from batch laws, a common law exists if and only if δ^0 vanishes on every window. (propositions 9 to 11; [68], Theorem 10.5.)*

(ii) Latent-model membership. *$F(O)$ contains a two-type law if and only if the sign class in $H^1(\Gamma; \mathbb{Z}/2)$ vanishes and the walls leave a segment. The two-type laws in $F(O)$ then form the hiding segment, and a triple window decides whether the law lies on it (stage D_5). The label switch is the kernel of this layer. (theorem A; theorem M(1)–(5); corollary 4.)*

(iii) Decision sensitivity. *For a decision with walls $\{\ell_j = c_j\}$:*

- *it is settled before a probe if and only if the hiding segment lies in one chamber;*
- *a probe's Savage value is $R(\mathcal{M}) - \max_y R(S_y)$, and its Wald value is positive if and only if a wall crosses the segment with strictly opposite slopes;*
- *every robustness number is the margin to a wall over a capacity: forward and observer injection budgets, settling size, and the labelled band of $B = c_s\sqrt{\pi/(1-\pi)}$ standard errors. Under labels, an unexposed source can delay the decision, by a factor of at most $((z+B)/z)^2$, but cannot tilt it when $B < z$.*

(theorems D, J and L; proposition 6.)

Proof. Each item is the cited result restricted to the stated setting. (i) The kernel and the signed surjectivity are proposition 9. The positivity criterion is the marginal-problem linear programme, which equals Roumen's precone test by his Proposition 10.4 (check RO3). The labelled case is proposition 10(iii). (ii) Theorem M(2), (3) and (5) and the strictness witnesses of corollary 4. (iii) Theorem L(1)–(3) and theorem J(2)–(4). □

The three conclusions can fail independently. Each combination occurs in the lab:

- a family can glue and fail (ii): the "exactly one harm" law;
- it can pass (i) and (ii) and still be decision-fragile: the drug example's margin is 0.001, and the observer's block is 1/128;
- it can be tilted while passing every test: the hidden channel inside the stealth window, which only labels or a second triple expose.

The theorem is stated for two types, the pair cover and three variables. Parts of (i) and (iii) hold on any fibre, because they use only linearity and positivity. Its extension to more types, where the model part is a non-convex secant variety and the label switch becomes the symmetric group S_k, and to other covers, is open.

28 Relation to the geometry, and open problems

Theorem A rests on the moment normal form and the Stanley–Reisner structure of the latent fibre in [8]. Its strata, where types agree on variables or merge, are the hiding places the theorems measure.

At level one, gluing tables along observed separators is Sullivant's toric fibre product [35, 36]. The latent model needs gluing along a hidden separator, because the type is in every window. A rule for that case would say which covers identify the actors and would extend theorem A beyond the pair cover. We do not know one.

Related work. Deciding over the set of parameters that the data cannot distinguish is the programme of partial identification [78, 79, 80, 81, 71]. Identified sets of discrete models with unobserved variables are fibres of polynomial maps, and [82] compute them by polynomial programming. That literature treats the identified set as a generic set, often an interval. What the lab adds is the use of its algebraic structure (degree, strata, moment factorisation) to obtain decision-theoretic results: the two wall checks, the stratified value of a probe, the settling size, the hidden channel, and the nesting of the adversary's segment and the regulator's on one line. Singular learning theory [83, 84] uses the same singular geometry for estimation and model choice, not for decisions. Horowitz and Manski [39] identify a distribution from contaminated data; their identified set is the regulator's side of the reachable polytope of theorem E(1). Part II sits in robust statistics [72, 73] and the study of data poisoning [74, 75, 76].

Further open problems:

- decisions that combine the segment with level-one contamination, and settling for utilities on four or more variables (theorem D covers the two-type pair cover and, for settling, three variables);

- more than two types, where the segment becomes a higher-dimensional cell;
- a real test on published aggregate trial tables;
- injection into separate windows, which can break gluing ($\lambda^* < 1$) and be caught at level one, and editing of genuine records (a total-variation ball);
- an adversary uncertain about P, and obstructions beyond second order on the strata;
- choosing windows to minimise the hidden channel's capacity.

29 Scope of each result

Several results hold well beyond the setting in which the paper develops them, and others hold only in that setting. To prevent misreading, table 12 records for each result the number of latent types, the cover and the number of variables for which it is proved, why it extends when it does, and what is open. "Any" means the proof uses only linearity of expected utility in the law and positivity, or a statement about an arbitrary graph or cover. "Computed" means verified for the listed cases by a check, with no general proof.

Table 12: Scope by result. "Types" is the number of latent types; "Var." the number of binary variables.

result	types	cover	var.	why it extends, and what is open
Part I: the hiding segment and the decision				
A: hiding segment	2	pair	$m \geq 3$	needs three non-zero v_i. Open: $k \geq 3$ (a higher-dimensional, non-convex hiding set); partial covers
B: tempo	2	pair, +triple	any m	open: $k \geq 3$; bipartite covers, where $\lvert v \rvert$ is not pinned
C: value of a probe	2	pair + one triple	any m	open: several probes jointly; other covers; $k \geq 3$
D: decision on the segment	2	pair	any m; settling $m = 3$	settling needs affine utilities in r (three variables). Open: settling for $m \geq 4$
Part II: injected records				
E: slopes, vertex attack	any	any	any	linear in the injected law; exact for linear decision functionals
F: hidden channel	2	pair	$m = 3$; (4) any m	(4) covers one triple S at any m. Open: several channels moved jointly
G: stealth with equations	2	pair	$m = 4$	open: $m \geq 5$, where more model equations appear
H: one geometry	2	pair	$m = 3, 4$	open: $m \geq 5$
J: sources and bands	any	labelled, one scalar statistic	any	band formula for any scalar statistic in the Gaussian limit. Open: several statistics; adaptive adversaries over rounds
Study 13	2	pair + triple, 3 sources	3	one registered truth

result	types	cover	var.	why it extends, and what is open
Lifting and the one picture				
K(1), (2), (4)	any	any	any	linear in the law; act lift by the minimax theorem
K(3): blow-up of the merge	2	pair	any m	open: blow-ups for $k \geq 3$
K(5): orbifold atlas	2	pair	$m = 4$	standard toric geometry; the Markov-basis reading is conjectural
L(3), (4): margin over capacity, relevance	any	any fibre	any	support function of the gauge; linearity
L(1), (2), (5)	2	pair	$m = 3$	open: $m \geq 4$, where the model part is a curve in a higher-dimensional fibre
dilation; value in probe size; backtracking	2	pair	$m = 3$	worked example; dilation width $\Delta s / P(a, b)$ holds on any pair line
price of not gluing (rectangularity)	—	finite trees	—	proved; sharpness conjectural
Descent and categorical structure				
spectrum and descent	any	any cover	any	kernel and signed surjectivity in general; acyclicity computed for the pair covers ($m = 3, 4$) and the triple cover ($m = 4$)
M(1)–(3): sign and modulus classes	2	any graph of pairs	any	$H^1(\Gamma; \mathbb{Z}/2) \oplus \operatorname{coker} B$ for any graph. Open: S_k coefficients for $k \geq 3$ (non-abelian)
M(4), (5): triple layer, fibre	2	triple cover; pair	$m = 4$; $m = 3$	open: $m \geq 5$
M(6), (7): labels; realization diagram	any; 2	labelled; pair	any	degree-0 statement for any labelled cover
descent filtration (corollary)	2	any connected pair graph	any	strictness witnesses computed at $m = 3, 4$; the counts $E - \operatorname{rank} B$ and b_1 are general
comonadic form	any	any cover	any	classical theorems; the lab uses their object-level consequences
CSD no-go	any	covers with a cycle	any	counterexamples on the triangle; degrees computed for the pair covers
quantum marginal problem	—	qubit windows	any	dephasing retraction general; the kernel count (27) is for three qubits
Markov categories	any	any	any	factorisation for k types is a k-point object; Blackwell and de Finetti as cited
Ring of contingency tables (section 8)	2	pair	$m = 3$	fibre and Gröbner statements hold for any table model; the hiding set, τ_c and Study 11c are for three binary variables
Tangent structure [66]	any	polyhedra	any	cone structure on any polyhedron with affine maps; universality iff relative interior; jets proved on the $m = 4$ tetrad scheme

result	types	cover	var.	why it extends, and what is open
Roumen's order cohomology	any	any	any	his Theorem 10.5; equals the marginal-problem LP
N: assembly	2	pair	$m = 3$	as its parts

The pattern is consistent. Results that rest on linearity and positivity hold on any fibre and any cover. These are the slopes, margin over capacity, the relevance criterion, the source band, the descent kernel and the positivity criterion. Results that rest on the geometry of the two-type model need its parametrisation. These are the hiding segment, the tempos, settling, the closed forms of the hidden channel, and realizability. They are proved for two types, mostly on the pair cover, and some only for three or four variables. Two companion projects take up the narrow results. One builds the tangent structure [66]; it is now written. The other develops the two-type theory on arbitrary covers, with a three-type computation as a proof of concept for S_k descent.

30 Limits

- All data are synthetic.
- Part II treats whole-record injection by an adversary who knows P; the hidden channel is for $m = 3$ and the detection test for $m = 4$ uses only the tetrads.
- There are two types and binary variables.
- Theorem D's settling part covers utilities on three variables; Study 11b tests it on five designs and two truths.
- The decision in Study 11 is one stylised treatment choice. Its Wald gap g is 0 by a saddle point; the regret illustration uses a post-hoc cost.
- The disclosure game assumes the sponsor knows the truth.
- The regulator's naive policy uses the segment's midpoint.
- The illustrations are stylised.
- In Study 11, three registered predictions and one illustration failed, as reported.
- The settling predictions use the delta method and the plug-in $\hat{\tau}$; they were tested from $n^*/2$ upward, not at small n. Study 11c's exact probe assumes the pair tables are known.

A Reproducibility

- `engine/pce/lab.py` implements the model, the segment and sampling.
- Study 11 is in `engine/study11_lab/`:
  - the registration and its hash manifest: `PREREG_study11.md`, `FROZEN_PREREG_study11.txt`;
  - the frozen script as a text copy: `run_study11_as_frozen.py.txt`;

  - the script that was run, with the logged deviation: `run_study11.py`, `DEVIATIONS_study11.md`;
  - results and post-hoc analyses: `results_study11.json`, `posthoc_study11.py`, `posthoc_study11.json` (PH1–PH4).

- `engine/pce/segment_decision.py` implements theorem D: the utility polynomials, the set and info-gap rules, the settling constraints and the $m = 3$ estimator; `engine/check_decision_segment.py` runs checks DS1–DS9.

- Study 11b is in `engine/study11b_settle/`: `PREREG_study11b.md`, `FROZEN_PREREG_study11b.txt`, `design11b.py`, `run_study11b.py` (with a frozen text copy), `results_study11b.json`.

- Study 11c is in `engine/study11c_exact/`: `PREREG_study11c.md`, `FROZEN_PREREG_study11c.txt`, `run_study11c.py` (with a frozen text copy), `results_study11c.json`; the ring-of-tables checks are `engine/check_ehrhart.py` (EH1–EH7) with `engine/pce/ehrhart.py`, and the figure is `lab/fig_ehrhart.py`.

- `engine/pce/injection.py` implements Part II: the injected law, the hidden channel, influence functions, budgets, the tetrads and the stealth–tilt programme; `engine/pce/duality.py` implements theorem H and the obstruction of theorem G(4); `engine/check_injection.py` runs checks IN1–IN14.

- Study 12 is in `engine/study12_inject/`: `PREREG_study12.md`, `FROZEN_PREREG_study12.txt`, `design12.py`, `run_study12.py` (with a frozen text copy), `results_study12.json`.

- `engine/worked_example.py` recomputes every number of section 18 and checks it (WE0–WE18), writing `lab/worked_example.json`.

- `engine/worked_example4.py` recomputes the four-variable example of section 19 in stages and checks it (W4-1 to W4-14), writing `lab/worked_example4.json`.

- `lab/fig_lab.py` draws every figure.

- The pairwise proposition is check SC7 of [8].

References

[1] V. K. Awasthi, Coupling polytopes and toric geometry: the 2×3 family, the square case, higher levels and stability, technical report, October 2026 (revised).

[2] T. Delzant, Hamiltoniens périodiques et images convexes de l'application moment, *Bull. Soc. Math. France* **116** (1988) 315–339.

[3] L. Pachter and B. Sturmfels (eds.), *Algebraic Statistics for Computational Biology*, Cambridge University Press, 2005.

[4] I. M. Gelfand, M. M. Kapranov and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, 1994.

[5] C.-H. Cho and Y.-G. Oh, Floer cohomology and disc instantons of Lagrangian torus fibers in Fano toric manifolds, *Asian J. Math.* **10** (2006) 773–814.

[6] E. Lerman and S. Tolman, Hamiltonian torus actions on symplectic orbifolds and toric varieties, *Trans. Amer. Math. Soc.* **349** (1997) 4201–4230.

[7] V. K. Awasthi, Shadows, fibres and secants of the posterior tower, technical report, October 2026.

[8] V. K. Awasthi, Partial gluing and level two: graded explanations, latent actors and paths through singularities, technical note, October 2026.

[9] J. R. Boyd, *A Discourse on Winning and Losing*, briefing (1987); Air University Press edition, 2018.

[10] F. H. Knight, *Risk, Uncertainty and Profit*, Houghton Mifflin, 1921.

[11] A. Wald, *Statistical Decision Functions*, Wiley, 1950.

[12] I. Gilboa and D. Schmeidler, Maxmin expected utility with non-unique prior, *J. Math. Econ.* **18** (1989) 141–153.

[13] Y. Ben-Haim, *Info-Gap Decision Theory*, 2nd ed., Academic Press, 2006.

[14] B. Schwartz, Y. Ben-Haim and C. Dacso, What makes a good decision? Robust satisficing as a normative standard of rational decision making, *J. Theory Soc. Behav.* **41** (2011) 209–227.

[15] D. Z. Long, M. Sim and M. Zhou, Robust satisficing, *Oper. Res.* **71** (2023) 61–82.

[16] R. A. Howard, Information value theory, *IEEE Trans. Syst. Sci. Cybern.* **2** (1966) 22–26.

[17] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953) 265–272.

[18] N. I. Al-Najjar and J. Weinstein, The ambiguity aversion literature: a critical assessment, *Econ. Philos.* **25** (2009) 249–284.

[19] L. J. Savage, The theory of statistical decision, *J. Amer. Statist. Assoc.* **46** (1951) 55–67.

[20] M. Siniscalchi, Dynamic choice under ambiguity, *Theor. Econ.* **6** (2011) 379–421.

[21] E. Kamenica and M. Gentzkow, Bayesian persuasion, *Amer. Econ. Rev.* **101** (2011) 2590–2615.

[22] P. R. Milgrom, Good news and bad news: representation theorems and applications, *Bell J. Econ.* **12** (1981) 380–391.

[23] S. J. Grossman, The informational role of warranties and private disclosure about product quality, *J. Law Econ.* **24** (1981) 461–483.

[24] E. Henry and M. Ottaviani, Research and the approval process: the organization of persuasion, *Amer. Econ. Rev.* **109** (2019) 911–955.

[25] A. Di Tillio, M. Ottaviani and P. N. Sørensen, Strategic sample selection, *Econometrica* **89** (2021) 911–953.

[26] J. Chen, Optimal rate of convergence for finite mixture models, *Ann. Statist.* **23** (1995) 221–233.

[27] L. P. Kaelbling, M. L. Littman and A. R. Cassandra, Planning and acting in partially observable stochastic domains, *Artif. Intell.* **101** (1998) 99–134.

[28] D. V. Lindley, On a measure of the information provided by an experiment, *Ann. Math. Statist.* **27** (1956) 986–1005.

[29] K. Chaloner and I. Verdinelli, Bayesian experimental design: a review, *Statist. Sci.* **10** (1995) 273–304.

[30] M. Shi, Y. Liang and N. B. Shroff, Near-optimal partially observable reinforcement learning with partial online state information, arXiv:2306.08762v4 (first version 2023, titled "Theoretical hardness and tractability of POMDPs in RL with partial online state information"; a preliminary version appeared at IEEE MILCOM 2024).

[31] W. Shen, J. Dong and X. Huan, Variational sequential optimal experimental design using reinforcement learning, arXiv:2306.10430 (2023).

[32] I. Goodfellow et al., Generative adversarial nets, *Advances in Neural Information Processing Systems* **27** (2014).

[33] L. Mescheder, A. Geiger and S. Nowozin, Which training methods for GANs do actually converge?, *Proc. ICML*, PMLR **80** (2018).

[34] S. Nowozin, B. Cseke and R. Tomioka, f-GAN: training generative neural samplers using variational divergence minimization, *Advances in Neural Information Processing Systems* **29** (2016).

[35] S. Sullivant, Toric fiber products, *J. Algebra* **316** (2007) 560–577.

[36] A. Engström, T. Kahle and S. Sullivant, Multigraded commutative algebra of graph decompositions, *J. Algebraic Combin.* **39** (2014) 335–372.

[37] D. Quillen, On the (co-)homology of commutative rings, in *Applications of Categorical Algebra*, Proc. Sympos. Pure Math. **17**, Amer. Math. Soc., 1970, 65–87.

[38] M. André, *Homologie des algèbres commutatives*, Springer, 1974.

[39] J. L. Horowitz and C. F. Manski, Identification and robustness with contaminated and corrupted data, *Econometrica* **63** (1995) 281–302.

[40] P. Blanchard, E. M. El Mhamdi, R. Guerraoui and J. Stainer, Machine learning with adversaries: Byzantine tolerant gradient descent, *NeurIPS* (2017).

[41] M. Baruch, G. Baruch and Y. Goldberg, A little is enough: circumventing defenses for distributed learning, *NeurIPS* (2019).

[42] D. Yin, Y. Chen, K. Ramchandran and P. Bartlett, Byzantine-robust distributed learning: towards optimal statistical rates, *ICML* (2018).

[43] T. Seidenfeld and L. Wasserman, Dilation for sets of probabilities, *Ann. Statist.* **21** (1993) 1139–1154.

[44] R. Radner and J. E. Stiglitz, A nonconcavity in the value of information, in M. Boyer and R. Kihlstrom (eds.), *Bayesian Models in Economic Theory*, North-Holland (1984) 33–52.

[45] L. G. Epstein and M. Schneider, Recursive multiple-priors, *J. Econ. Theory* **113** (2003) 1–31.

[46] F. Fournier-Facio and R. Willett, The local lifting property, property FD, and stability of approximate representations, arXiv:2603.18456 (2026).

[47] G. Li, Derived representation schemes with arbitrary coefficients and associative smoothness, arXiv:2607.19654 (2026).

[48] S. Abramsky and A. Brandenburger, The sheaf-theoretic structure of non-locality and contextuality, *New J. Phys.* **13** (2011) 113036.

[49] S.-Y. Chang, Universal obstruction cohomology of categorified spectral objects: from descent filtrations to derived duality, preprint hal-05711275 (2026).

[50] S. Abramsky, S. Mansfield and R. S. Barbosa, The cohomology of non-locality and contextuality, *EPTCS* **95** (2012) 1–14.

[51] J. Bénabou and J. Roubaud, Monades et descente, *C. R. Acad. Sci. Paris Sér. A* **270** (1970) 96–98.

[52] J. M. Beck, *Triples, Algebras and Cohomology*, PhD thesis, Columbia University (1967); Reprints in Theory Appl. Categ. **2** (2003).

[53] G. Janelidze and W. Tholen, Facets of descent, I, *Appl. Categ. Structures* **2** (1994) 245–281.

[54] S. Eilenberg and J. C. Moore, Adjoint functors and triples, *Illinois J. Math.* **9** (1965) 381–398.

[55] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer (1998).

[56] R. Godement, *Topologie algébrique et théorie des faisceaux*, Hermann, Paris (1958).

[57] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge University Press (1994).

[58] J. Giraud, *Cohomologie non abélienne*, Springer (1971).

[59] M. Giry, A categorical approach to probability theory, in *Categorical Aspects of Topology and Analysis*, Lecture Notes in Math. **915**, Springer (1982) 68–85.

[60] K. Ikeda, Quantum entanglement as a cohomological obstruction, arXiv:2511.04326v2 (2026).

[61] T. Fritz, A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics, *Adv. Math.* **370** (2020) 107239.

[62] T. Fritz, T. Gonda and P. Perrone, De Finetti's theorem in categorical probability, *J. Stoch. Anal.* **2**(4) (2021), Article 6; arXiv:2105.02639.

[63] L. Chen, T. Fritz, T. Gonda, A. Klingler and A. Lorenzin, The Aldous–Hoover theorem in categorical probability, *Algebr. Stat.* **16** (2025) 131–174; arXiv:2411.12840.

[64] T. Fritz, T. Gonda, P. Perrone and E. F. Rischel, Representable Markov categories and comparison of statistical experiments in categorical probability, arXiv:2010.07416.

[65] V. K. Awasthi, Coupling algebras: gluing, the square-zero cone and the ring of contingency tables, technical report, Johns Hopkins University, October 2026, draft 4.

[66] V. K. Awasthi, A tangent structure for injected data: feasible cones, universality at corners, and jets on the tetrad scheme, companion paper, 2026.

[67] J. R. B. Cockett and G. S. H. Cruttwell, Differential structure, tangent structure, and SDG, *Appl. Categ. Structures* **22** (2014) 331–417.

[68] F. Roumen, Cohomology of effect algebras, *EPTCS* **236** (2017) 174–201; arXiv:1602.00567.

[69] Y. Yin and J. Zhang, Markov categories, causal theories, and the do-calculus, *Studies in Logic* (2021); arXiv:2204.04821.

[70] E. Di Lavore, M. Román and P. Sobociński, Partial Markov categories, arXiv:2502.03477 (2025).

[71] C. F. Manski, Identification and statistical decision theory, *Econometric Theory* **41**(4) (2025) 977–993; arXiv:2204.11318.

[72] P. J. Huber, Robust estimation of a location parameter, *Ann. Math. Statist.* **35** (1964) 73–101.

[73] F. R. Hampel, The influence curve and its role in robust estimation, *J. Amer. Statist. Assoc.* **69** (1974) 383–393.

[74] B. Biggio, B. Nelson and P. Laskov, Poisoning attacks against support vector machines, *Proc. ICML* (2012), arXiv:1206.6389.

[75] J. Steinhardt, P. W. Koh and P. Liang, Certified defenses for data poisoning attacks, *Advances in Neural Information Processing Systems* **30** (2017), arXiv:1706.03691.

[76] I. Diakonikolas, G. Kamath, D. Kane, J. Li, A. Moitra and A. Stewart, Robust estimators in high dimensions without the computational intractability, *SIAM J. Comput.* **48** (2019) 742–864.

[77] M. Drton, B. Sturmfels and S. Sullivant, Algebraic factor analysis: tetrads, pentads and beyond, *Probab. Theory Related Fields* **138** (2007) 463–493, doi:10.1007/s00440-006-0033-2.

[78] C. F. Manski, Identification problems and decisions under ambiguity: empirical analysis of treatment response and normative analysis of treatment choice, *J. Econometrics* **95** (2000) 415–442.

[79] J. Stoye, Minimax regret treatment choice with finite samples, *J. Econometrics* **151** (2009) 70–81.

[80] R. Giacomini and T. Kitagawa, Robust Bayesian inference for set-identified models, *Econometrica* **89** (2021) 1519–1556.

[81] J. L. Montiel Olea, C. Qiu and J. Stoye, Decision theory for treatment choice problems with partial identification, arXiv:2312.17623.

[82] G. Duarte, N. Finkelstein, D. Knox, J. Mummolo and I. Shpitser, An automated approach to causal inference in discrete settings, *J. Amer. Statist. Assoc.* **119** (2024) 1778–1793.

[83] S. Watanabe, *Algebraic Geometry and Statistical Learning Theory*, Cambridge University Press, 2009.

[84] M. Drton and M. Plummer, A Bayesian information criterion for singular models, *J. R. Stat. Soc. Ser. B* **79** (2017) 323–380.